



TrackWise Training Program:

Crystal Reports Design for TrackWise

Manual Version – September 2015

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Sparta Systems, Inc.
2000 Waterview Drive
Hamilton, New Jersey 08690

Sparta Systems Europe, Ltd.
1 Hachoshlim Street
Herzliya, Israel 46724

(888) 261-5948
support@sparta-systems.com

972.9.970.6000
europe-support@sparta-systems.com

Sparta Systems Asia Pacific Ltd.
503 Silvercord Tower
230 Canton Road, TST, Hong Kong

+852.2376.9345
asia-sales@spartasystems.com

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Preface

The Objective

The objective of the Crystal Reports Design for TrackWise course is to provide the skills and knowledge necessary for effective reports design in the TrackWise environment. The report designer goes through a range of topics and challenging scenarios. In addition, the course focuses on the creation of professional looking reports. The student gains an understanding of table linking and is able to identify which TrackWise tables contain each specific piece of information.

The Course Workbook

This Course Workbook shall serve as a useful tool during the Crystal Reports design training course. Within this manual is a detailed curriculum of Workbook Exercises that instructs the student with a step by step experience through examples. In On-Your-Own Exercises, the students take recently learned skills and apply report requirements.

Workbook Exercises

Throughout this training course, you are asked to follow the instructor's example and training manual to prepare an example. Workbook Exercise notations indicate where you can pick up with the book to follow along. After the instructor has presented the example, use this guide to aid you in preparing the same example in your personal environment. Look for the following object throughout the manual where you are to re-do the instructors example:



On Your Own Exercises

At the end of each hands-on section, typically per Workbook Exercise set, there is an On-Your-Own Exercise section where the book defines user requirements that you must fulfill. During this time, feel free to ask the instructor for assistance during this process. Look for the following object throughout the manual:



1. Overview

1.1 Reporting Overview

Reports generation is almost an art by itself. However, when generating reports, remember reports are designed to transfer data to the user. Therefore, make them simple, clear to read and understandable.

The first step in generating reports is to obtain clear report requirements. The following questions are to be considered based on the type of reports requested.

Single Reports	Multi-Reports
Should the report include related records such as the parent data and/or siblings, based on the type of project report?	How should the data be grouped, based on a particular data field or specific date pertaining to the record?
Does each project have at least one single report to print out the record content?	Does the report need to filter out records that could come back, based on the fetched results?
Is there an audit trail report available for single report activity history?	Is there a need to include charts or graphs to illustrate the amount of returned data?

This course includes other examples based on the specific topics for report design. They are addressed in the other chapters. Once the requirements are known, draw a simple layout as a guide to design reports. Diving into complex formulas and details is counter-productive because it is very easy to lose sight of the target report agenda.

1.2 Integration with TrackWise Web

During the report design, reports are generated from Crystal Reports to test and verify. All of the records in the database are fetched and displayed in the report because there is neither query nor scope used. When running the same report from within TrackWise, the records displayed are based on current query and scope results. Crystal provides a way to filter for a sample of records which is addressed in the SelectExpert chapter. Reports are uploaded to two different folders based on report type. Single reports are uploaded to TWReports_1 and multi-reports are uploaded to TWReports. Both folders are accessible from the TrackWise server.

1.3 Database Connectivity

Crystal Report uses an ODBC connection to connect to the TrackWise database. Based on the database management system of Oracle or MSSQL Server, the appropriate driver is used. The data source name is defined using the ODBC Data Source Administrator that is found in administrative tools under the control panel. The following example is based on a SQL Server connection.



For classroom purposes, the DSN name **introcrdsn** is established. Refer to the following steps as a guide to setting up a data source name for future use.

1. Navigate to the ODBC Data Source Administrator that can be found in **Start – Programs – Administrative Tools – Data Sources (ODBC)** or from the Control Panel.
2. Select the **System DSN** tab:

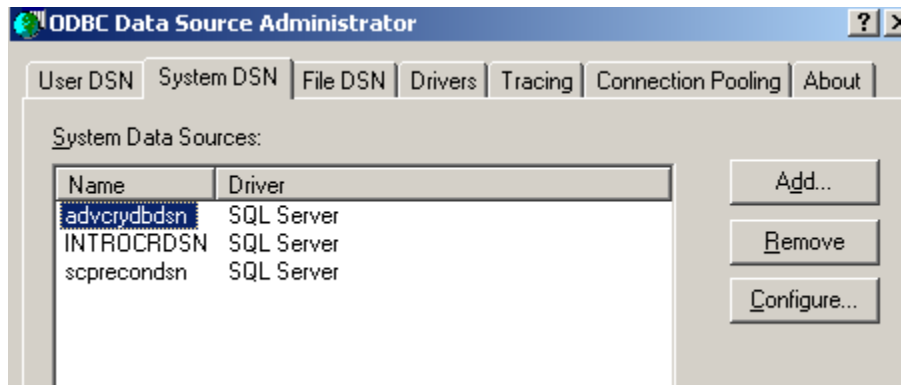


Figure 1-1 Example of System DSN tab on the ODBC Data Source Administrator window.

3. Select **Add** and refer to the **SQL Server** driver below :

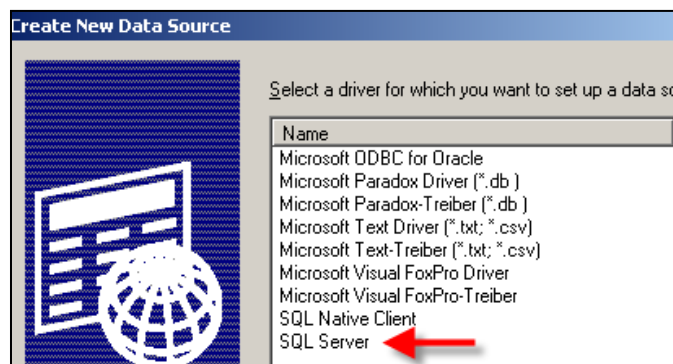


Figure 1-2 Select the SQL Server driver.

4. Click **Finish**.

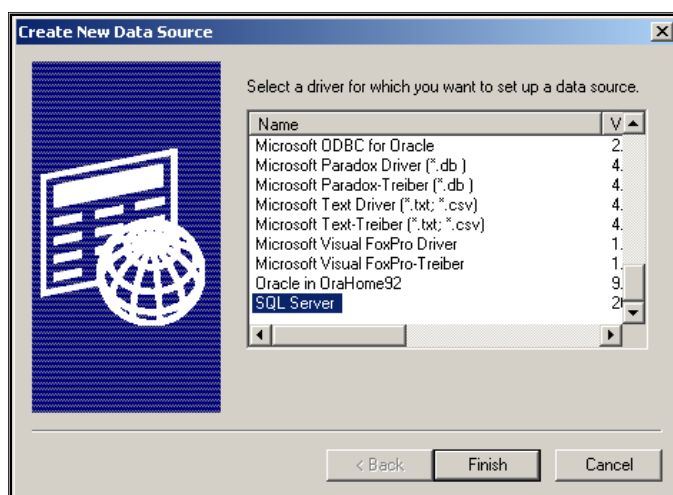


Figure 1-3 The Finish option is available after selecting the driver.

5. For practice, type in the ODBC **Name** of **devdb** and **Server** as **localhost**. Click **Next**.

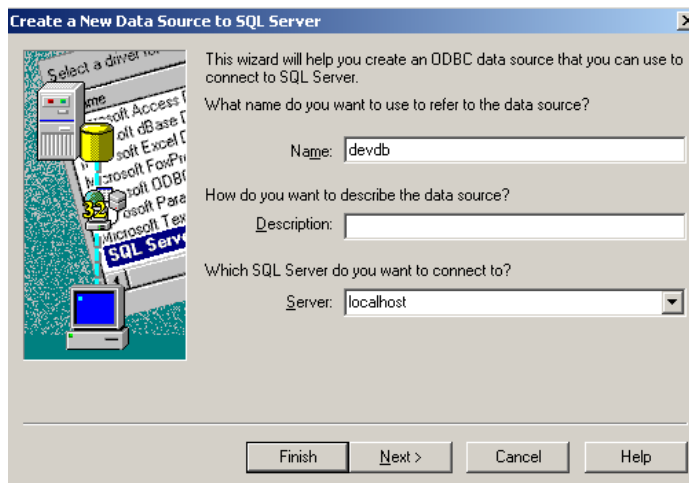


Figure 1-4 Provide a name for the data source and the server location.

6. Select SQL Server Authentication using a login ID and password of **sysadm** in lower case letters. This is the schema owner account to the database which owns the tables. Click **Next** to continue.

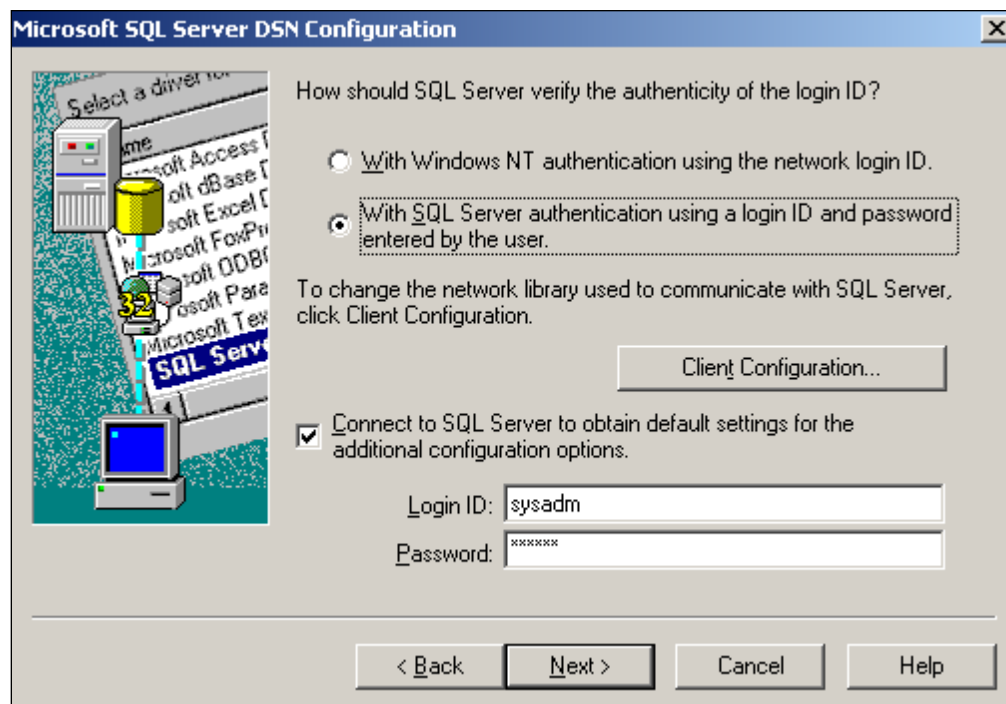


Figure 1-5 Select SQL Server Authentication and use the schema owner account to connect.

7. Select **Change the default database to** and select **introCrystal** from the list. Click **Next**.

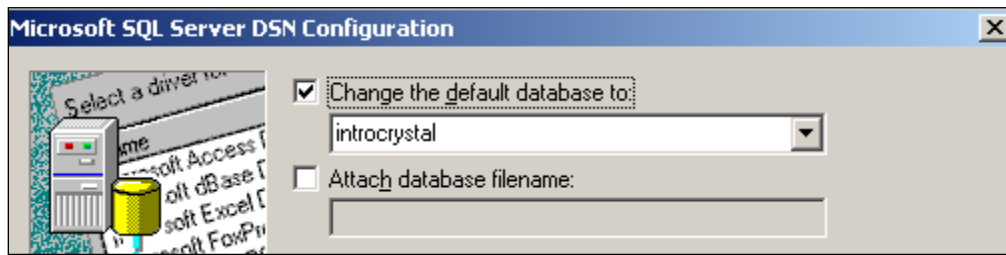


Figure 1-6 Use the configuration window to choose a default database.

8. Click **Finish**. A window appears to test the connection. Click **Test Data Source**. A successful network connection message appears.

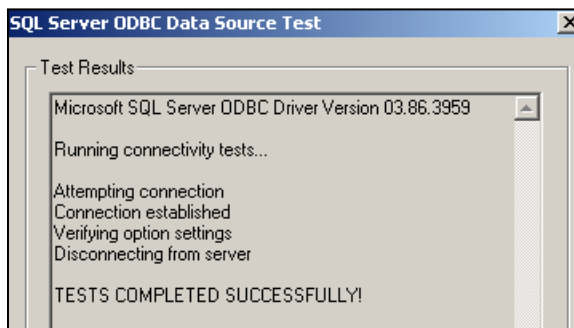


Figure 1-7 Example of a successful network connection.

9. Click **Finish** until the ODBC DSN box closes.

Conclusion:

A DSN name is needed on the computer which Crystal Reports Developer is installed in order to create reports with. In the next chapter a DSN name is used to connect to the database and begin report creation. Obtaining the username and password is available from the database administrator.



2. Table Linking

2.1 Definition and Explanation

The first step in designing a report is identifying which tables contain the required information for the report and the link between the tables involved. A correct table link is the foundation of each report and critical in order to display the correct data. This chapter includes the most often used data tables in every Crystal Report.

2.2 Changing Crystal Options

The following Crystal options are recommended when creating Crystal Reports. Before adding data tables to the report, refer to the following steps.



1. Double-click **Crystal Reports 2008** on the desktop. The screen opens to the start page.
2. Click **File – Options**.
3. Under the **Database** tab, ensure **Sort Fields Alphabetically** is selected. This is a useful feature when linking tables later.

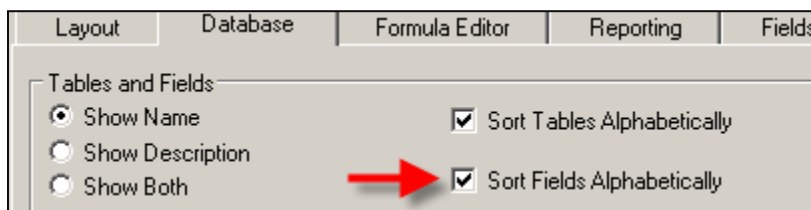


Figure 2-1 Sort Fields Alphabetically option on the Database tab.

4. Ensure **Automatic Smart Linking** is not selected. When linking tables, do not allow Crystal Reports to automatically link columns from one table to another. It is up to the Crystal Reports designer to link the tables.

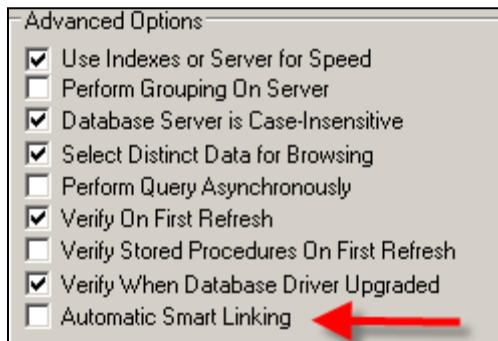


Figure 2-2 Under Advanced Options, ensure Automatic Smart Linking is not selected.

5. Navigate to the the **Fields** tab and select **String**.
6. Select **Can Grow**. That way the string fields in the report can wrap to the next line if needed and not be truncated. Click **OK**.

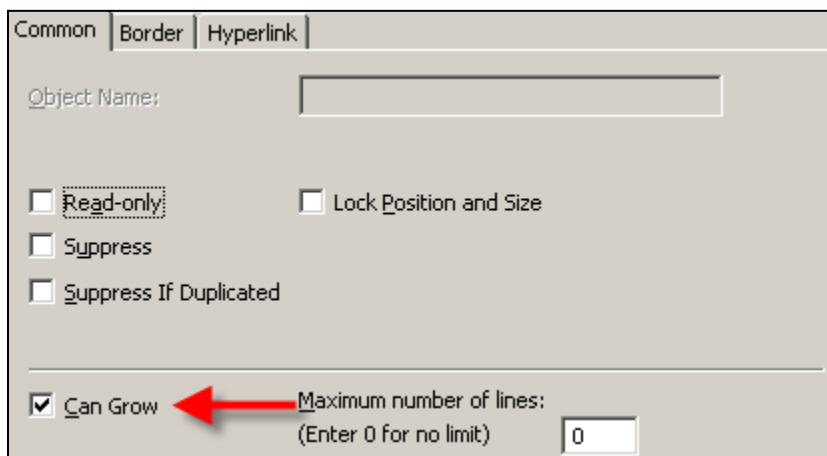


Figure 2-3 Select the Can Grow option on the Common Tab.

7. Refer to the **Reporting** tab. Select **Autosave Reports After** and use the default five minutes or modify this setting for sooner. This allows Crystal to save a copy of the report after a certain amount of time. When finished with the options, click **OK**.



Figure 2-4 Use the Autosave Reports After option to set a default time for Crystal to save a copy of the report.

8. The following option is used when a report is already opened in Crystal. On the main report design screen, click **File – Save Data with Report**. This deselects the option from being used. Saving data in reports causes conflicts in TrackWise , particularly with parameters as discussed in a future chapter.

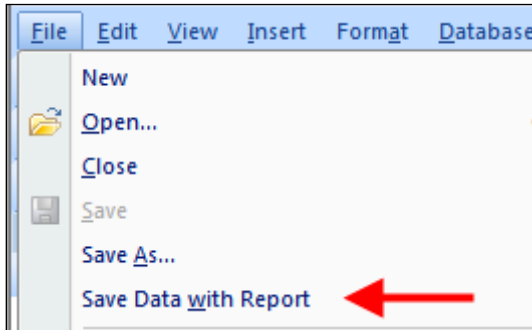


Figure 2-5 Example of Save Data with Report option deselected.

2.3 Table Linkage using Equal and Left Outer Joins

Crystal Reports prints data from the TrackWise database, based on query and scope results, from the TrackWise desktop. The data is stored within TrackWise database tables that are connected to one another. This process is implemented in order to resolve a numeric value to a name value. TrackWise data supports two kinds of table links which are left outer join and equal join. The following provides an explanation about the two types and when they are used. This is to prepare for the upcoming workbook exercise of linking tables together.

Equal join

Equal (Inner) join produces only the set of records that match both in Table A and Table B.

Example:

Table A		Table B	
id	name	id	name
1	Bell	1	Franklin
2	Galileo	2	Bell
3	Knight	3	Edison
4	Inventions	4	Knight

SQL Query:

```
SELECT * FROM TableA
INNER JOIN TableB
ON TableA.name = TableB.name
```

Results:

id	name	id	name
1	Bell	2	Bell
3	Knight	4	Knight

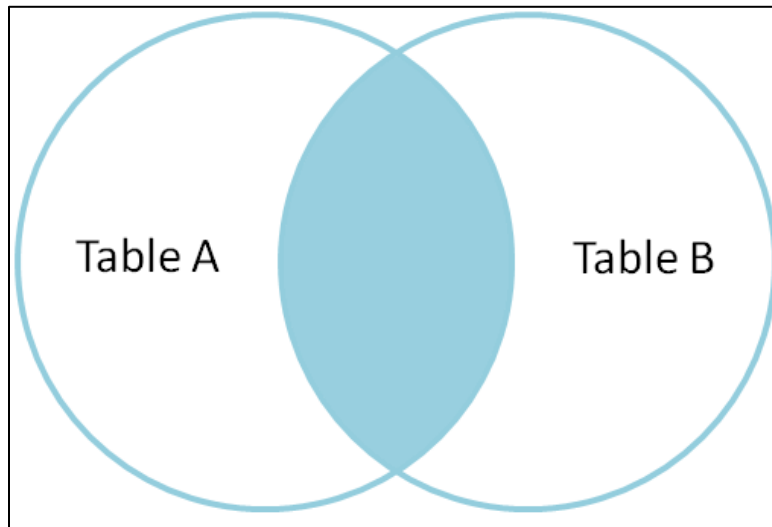


Figure 2-6 Visual example of Inner Join based on match in both Table A and Table B.

An equal join between two database tables expects an ID number to be sent over in order to be looked up. An example would be the PR table, which contains data about every record in TrackWise, and the Project table, which contains information about Projects. For every record in TrackWise, each one has a Project associated with it. An Equal Join is used because an ID number must always be available to be sent over from the PR to Project table.

Refer to the following example of an equal join between PR and Project.

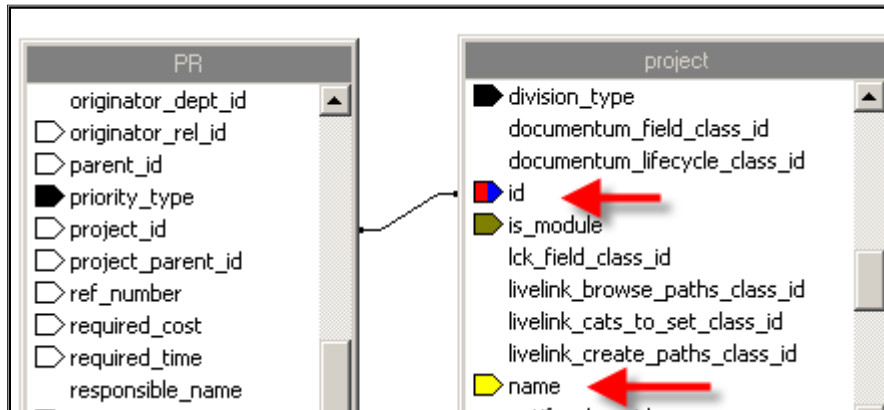


Figure 2-7 Equal join between the PR table and the Project table for the look of the ID. The Project table also has the name column which contains the project names.

The arrow pointing to ID in Project is where the look up ID appears in. The name column contains the values which are the Project names. The reason for an equal join is because there must always be an ID number to send over to be looked up (every record has a Project associated with it).

The following figure is a visual representation of the content inside of the tables. The PR table contains the record numbers under the ID column. For every record there is a project_id as well.

id	parent_id	project_id
-1	{null}	{null}
1001	{null}	7

Figure 2-8 Visual of content inside the tables. The Project id for PR 1001 in this example is 7.

In this example, PR#1001 has a project_id value of seven. In the next screen capture, project_id seven is sent over to the project table where it is looked up. From the project table perspective, PR#1001 is a Complaint record.

id	name
1	Action Item
2	Audit - Internal
3	Audit - Supplier
4	Change Control - Equipment
5	Change Control - Material
6	Change Control - Process
7	Complaint

Figure 2-9 The Project name for Project Id 7 is Complaint.

For a Crystal Report to print the word Complaint, there needs to be a proper link between PR and Project. If an equal join is used between two tables where an ID number is not available to be sent over, the whole record is removed from the report. Most links between tables in a report uses another join type called Left Outer.

Left outer join

Left outer join produces a complete set of records from Table A, with matching records (where available) in Table B. If there is no match, the right side will contain “null”.

Example:

Table A		Table B	
id	name	id	name
-----		-----	
1	Bell	1	Franklin
2	Galileo	2	Bell
3	Knight	3	Edison
4	Inventions	4	Knight

SQL Query:

```
SELECT * FROM TableA
LEFT OUTER JOIN TableB
ON TableA.name = TableB.name
```

Results:

id	name	id	name
-----		-----	
1	Bell	2	Bell
2	Galileo	null	null
3	Knight	4	Knight
4	Inventions	null	null

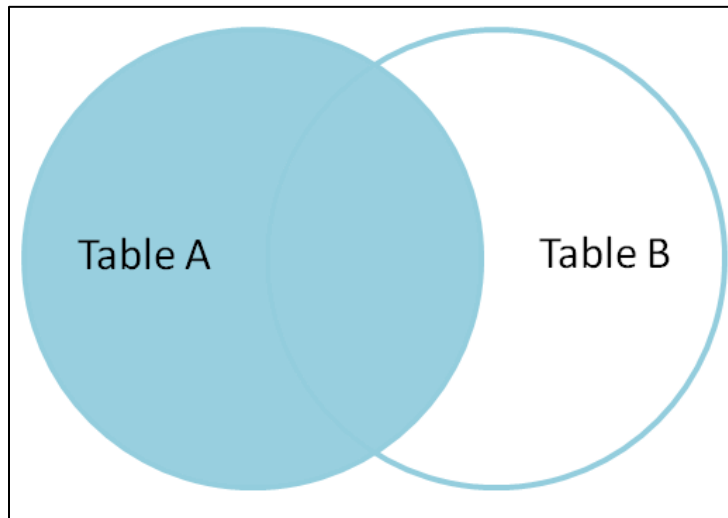


Figure 2-10 Visual example of producing a complete set of results from Table A.

Left outer joins is used for a majority of the links in a report. It provides a connection between two database tables but it is not guaranteed that an ID number is forwarded on to the other lookup table. The reason is because the information is not available yet or at all. When users run reports in TrackWise, they incorporate a query to fetch records with. The returned results have fields within the record that are not always filled out yet. To show other data fields that are available from the returned record, the left outer join does not penalize the record.

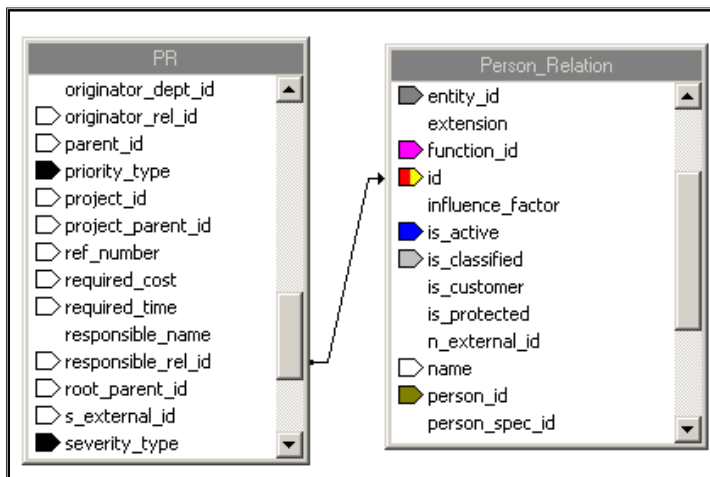


Figure 2-11 Left outer join of PR.responsible_rel_id to Person_Relation.id.

In this example, the Person Relation table is the look up table for person names. If there is not a person assigned to the record, responsible_rel_id would not send over a number to look up. Leaving the link as a left outer join allows the possibility that an ID could be available in another point of time.

The following figure from the PR table shows PR#1159 without a responsible_rel_id. This means the record is not assigned to anyone. Records 1006 and 1019 are assigned to the same person with the ID of two.

id	responsible_rel_id
1159	{null}
1006	2
1019	2

Figure 2-12 Example three records in the PR table. Two records that are assigned to the same id and one record not assigned.

To guarantee information to appear in Crystal Reports for record 1159 even though it is unassigned, the link between the PR table and Person_Relation is left out. The next screen capture shows who responsible_rel_id two is by linking to Person_Relation.

id	name
-1	{null}
1	Doe, Jane
2	Smith, John

Figure 2-13 Person_Relation table identifies Responsible_Rel_id two is Smith, John.

Based on this information, record 1006 and 1019 are both assigned to John Smith. Using left outer joins allows TrackWise records to appear in the report with or without an ID to look up.

2.4 Adding Tables to the Report

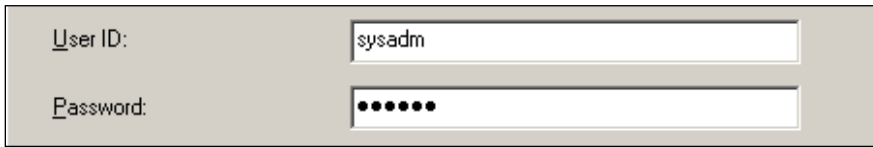
The following workbook exercise demonstrates how to create a new report, add data tables and link the tables for a functional TrackWise report.



Workbook Exercises

1. From the **File** menu, click **New – Blank Report**. Double-click **Create New Connection – ODBC RDO - INTROCRDSN**.

2. A window appears asking for the password. Enter **sysadm**. Click **Finish**.



The screenshot shows a login dialog box with two input fields. The first field is labeled 'User ID:' and contains the text 'sysadm'. The second field is labeled 'Password:' and contains a series of dots, indicating a masked password.

Figure 2-14 Enter database credentials.

3. The following window appears. Expand the **introcrystal** database.

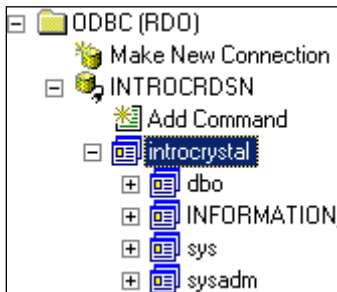


Figure 2-15 View of expanded introcrystal database.

4. Expand **sysadm** the database owner. Note the three options, two that are used in TrackWise reports.

- The table list contains all of the database system tables.
- The views contain all views that have been converted through the TrackWise update schema utility for previous versions or through Data Fields in version 8.4 and above.

Views must be created to appear in the expanded list. This is addressed in the update schema chapter.

5. Expand the **Tables** folder.

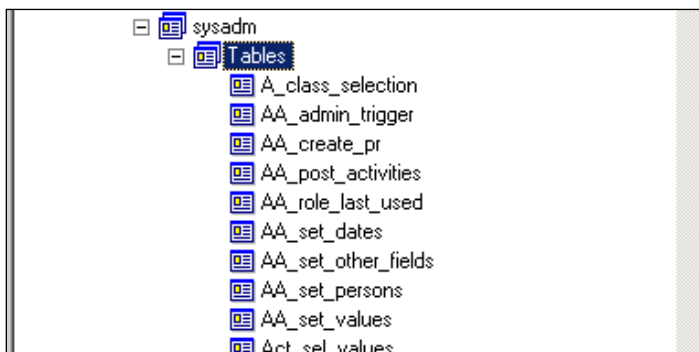


Figure 2-16 View of expanded Tables folder.

6. Select the following tables by either double-clicking the table or by highlighting the table and clicking the Right Arrow (>) button:

- **Division_Type**
- **Person_Relation**
- **PR**
- **Project**

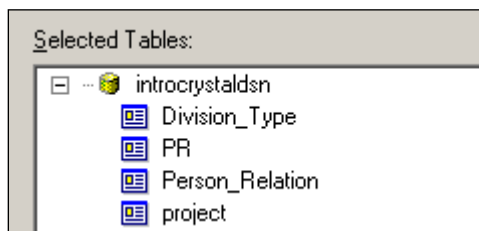


Figure 2-17 View of the four tables appearing under the Selected Tables panel.

7. Click **OK**.
8. The next screen to appear is the **Links** window. Drag and drop the tables in the following order as depicted in the following screen capture.

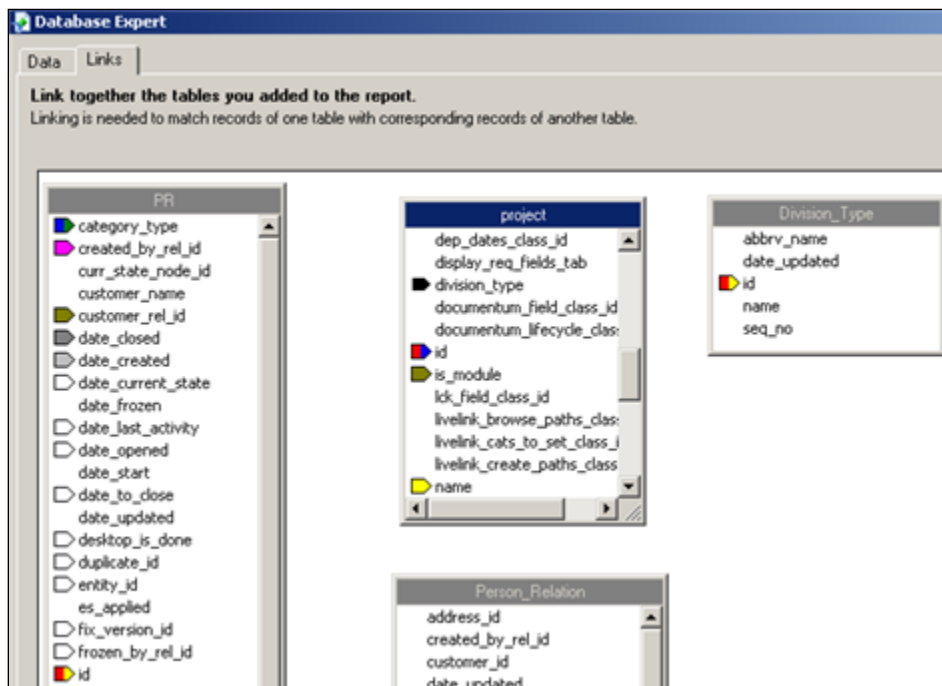


Figure 2-18 View of the Links tab with the four selected tables.

9. Link the tables together by using the following information. **PR.PROJECT_ID -> PROJECT.ID [Equal join]**. By default, Crystal Reports sets the link as an equal join. This join is to be equal because we know that every TrackWise record must have a Project.

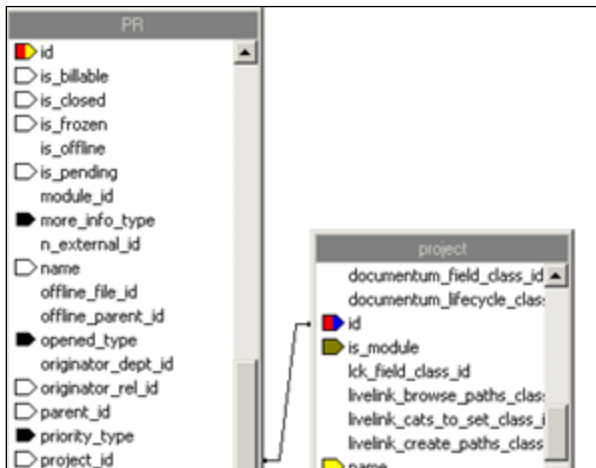


Figure 2-19 Link between the PR and project table.

10. The next join is from **Project** to **Division_Type**. **PROJECT.DIVISION_TYPE -> DIVISION_TYPE.ID** This join is equal because every TrackWise record must have a Division.

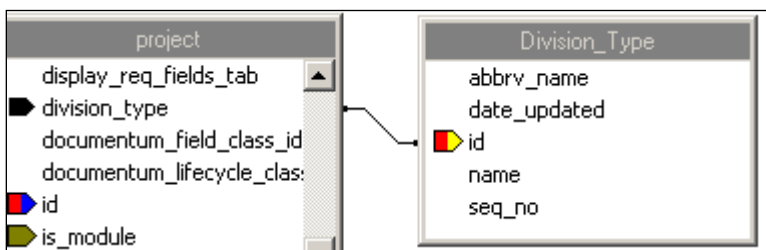


Figure 2-20 Link between the project and Division_Type table.

11. Finally, the last join is from the **PR** table to **Person_Relation**. **PR.RESPONSIBLE_REL_ID - PERSON_RELATION.ID [Left-outer join]**

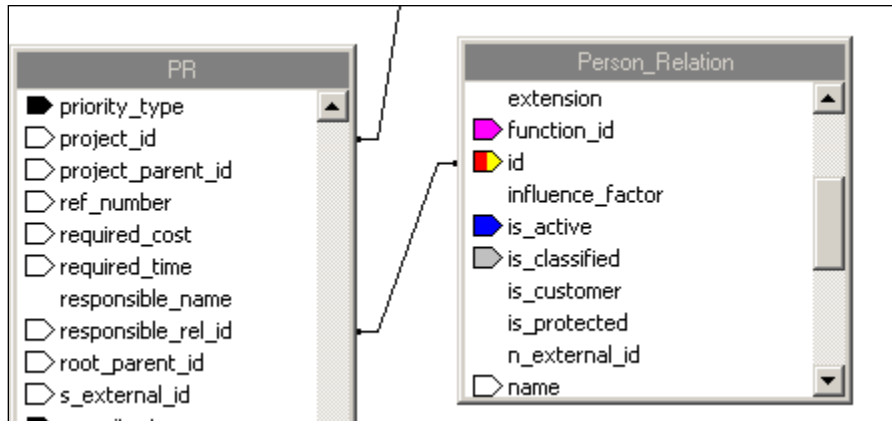


Figure 2-21 Link between PR and Person_Relation table.

12. This join is left-outer because it is not certain if this record has an Assigned To person associated with it. To make it left-outer, double click the link. A window appears with the option to change the link type.

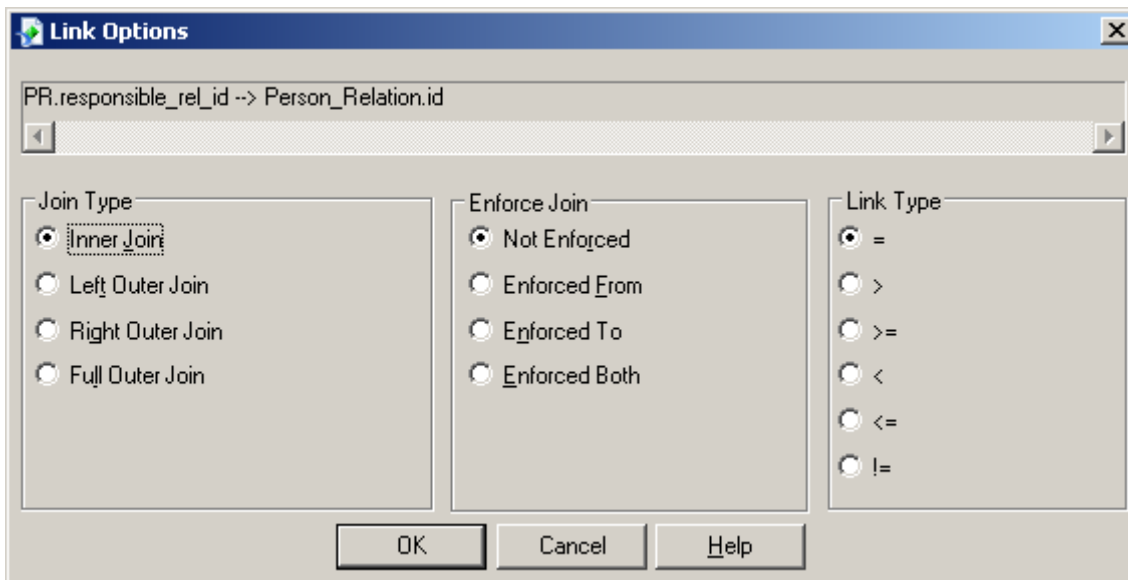


Figure 2-22 The Link Options window is available to change the link type.

13. Click **Left Outer Join** and **OK**.

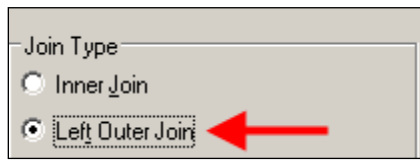


Figure 2-23 Link Options window with the Left Outer Join type selected.

14. The link from PR to Person_Relation now displays an arrow. This is based on the left outer link.

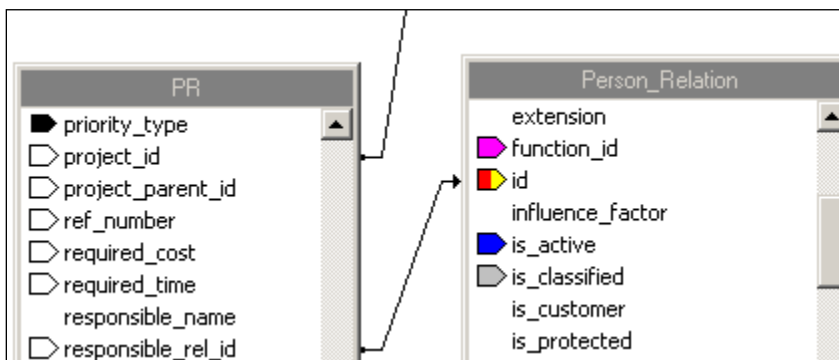


Figure 2-24 Link between PR and Person_Relation now with a Left Outer join type.

15. Click **OK** to exit the links screen.

16. Click **File - Save** and label this report as **Report1.rpt**. Save the report on the main Desktop screen.

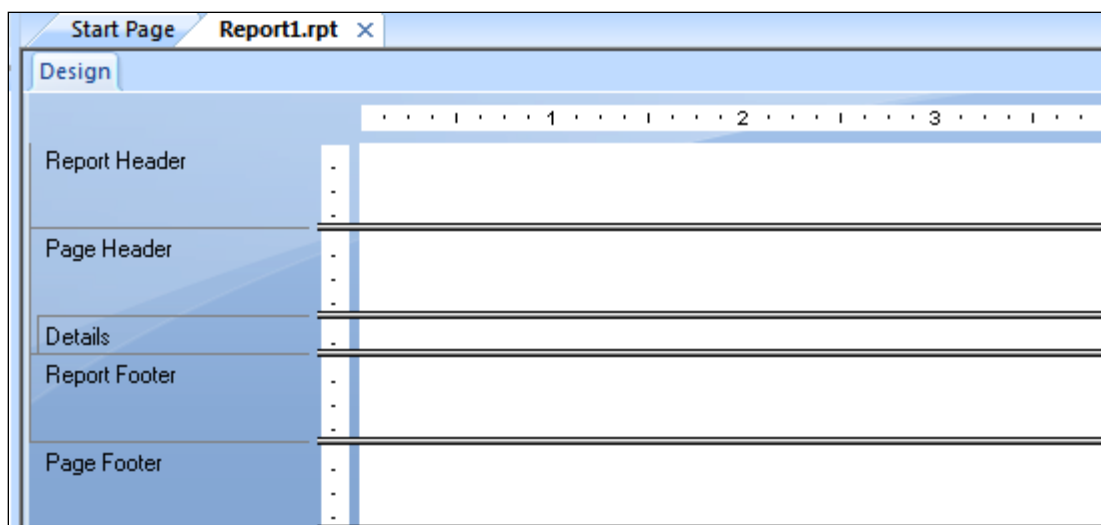


Figure 2-25 Design tab of Report1.

17. In addition to saving the report with a name, click **File – Summary Info**. A screen appears where the name of the report is available to be added along with additional information for background purposes.

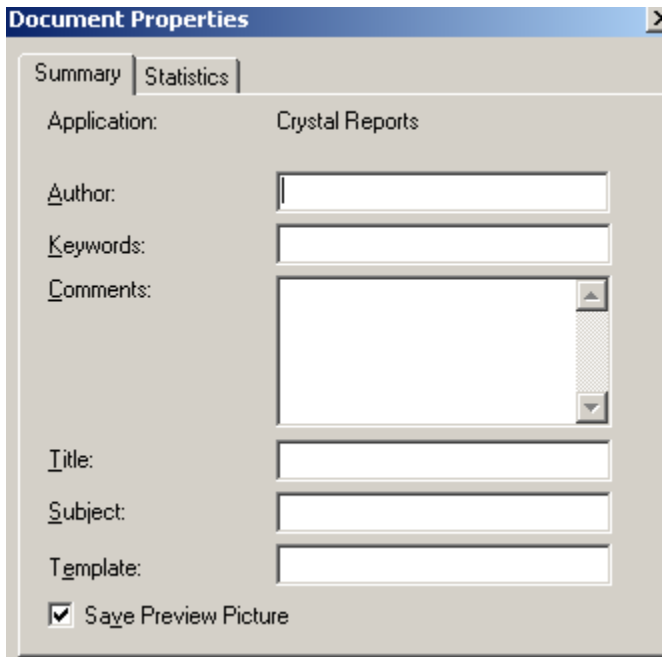
The image shows a screenshot of the 'Document Properties' dialog box, specifically the 'Summary' tab. The dialog has a title bar with 'Document Properties' and a close button. Inside, there are two tabs: 'Summary' and 'Statistics'. The 'Summary' tab is active. It contains several fields: 'Application:' with the value 'Crystal Reports'; 'Author:' with an empty text box; 'Keywords:' with an empty text box; 'Comments:' with a large empty text area and a vertical scrollbar; 'Title:' with an empty text box; 'Subject:' with an empty text box; and 'Template:' with an empty text box. At the bottom, there is a checkbox labeled 'Save Preview Picture' which is checked.

Figure 2-26 View of available document properties that can be edited.

Conclusion:

The basic design of a TrackWise Crystal Report uses the PR, Project, Division_Type and Person_Relation tables to resolve information in order for it to appear on the report. The common link between tables to use is left-outer due to the fact that not every returned record from the TrackWise query and scope includes records with all of the information available. To ensure that records fetched from TrackWise are included in the reports, make the links between tables left-outer for every table.



3. Multi-Record Report Design

3.1 Definition and Explanation

Multi-record reports are reports that display information from TrackWise records, based on query and scope results. Any record related field of information can be displayed on the report, including pieces of information that is not necessarily available on the TrackWise front end. The following workbook exercise is to design a report based on multiple records printed in a report.

3.2 Using the Field Explorer

The Field Explorer is on the left side of the screen. It presents to the report designer a list of all currently linked tables to select and place data columns onto the report. The Field Explorer can be placed anywhere on the design screen for referencing.

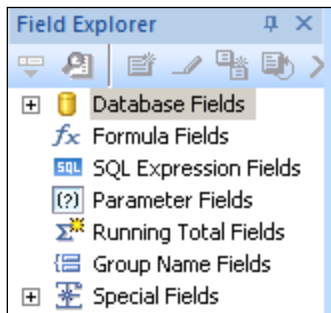


Figure 3-1 Field Explorer section on design tab.

Expand the **Database Fields** tree. Each table that was added to the report in the last chapter exists in this list. One can further expand the tree to see the fields that are contained within the particular table.

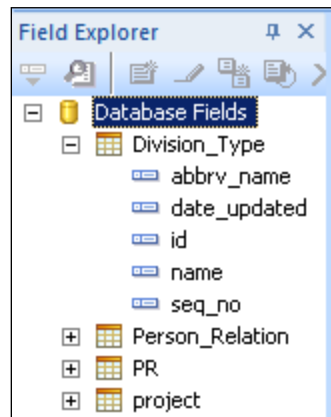


Figure 3-2 View of expanded Database Fields section in the Field Explorer.



- | Table and Column | Data Type |
|----------------------|---------------------------------------|
| DIVISION_TYPE.NAME | Displays the Division name |
| PROJECT.NAME | Displays the Project name |
| PR.ID | Displays the record ID (PR#) |
| PERSON_RELATION.NAME | Displays the name of the Assigned To. |

Figure 3-3 Example of design tab with column fields added to the Details section.

- | <u>name</u> | <u>name</u> | <u>id</u> | <u>name</u> |
|-------------|-------------|-----------|-----------------|
| Holmdel, NJ | Action Item | 1,015 | Bramley, Harrel |
| Holmdel, NJ | Action Item | 1,016 | Thomas, John |
| Holmdel, NJ | Action Item | 1,018 | Bramley, Harrel |
| Holmdel, NJ | Action Item | 1,019 | Smith, John |
| Holmdel, NJ | Action Item | 1,021 | Bramley, Harrel |
| Holmdel, NJ | Action Item | 1,022 | Bramley, Harrel |
| Holmdel, NJ | Action Item | 1,029 | Bramley, Harrel |
| Holmdel, NJ | Action Item | 1,060 | Bramley, Harrel |

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3. To return to design mode, click the **Design** tab.

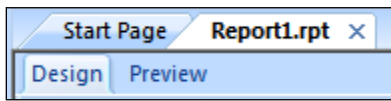


Figure 3-5 Example of Design tab selected.

4. Refer to the following chart for an explanation of each row in Design mode.

Row Name	Description
Report Header	Prints information on the top of the first page of the report only. This is used for the title of the report with company letterhead style.
Page Header	Prints information on top of every page in the report. In a multi-report this is used to display the name of the data fields on the report, much like how TrackWise does on the desktop layout.
Details	Prints the data from the dragged and dropped columns from the tables. This prints a row of information for every returned record.
Report Footer	Prints information on the bottom of the last page of the report. Can be used to display confidentiality information.
Page Footer	Prints information on the bottom of every page of the report. Can be used to display page numbers.

5. In the page header, double-click the first name column to edit the text. Name the first column **Division**.

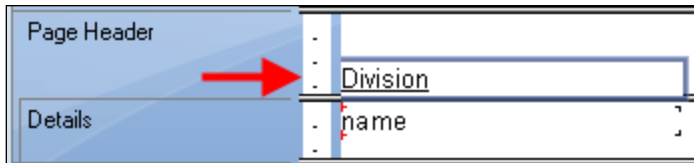


Figure 3-6 First name column renamed in Page Header to display as Division.

6. Provide a style change to the new label in page header. While highlighting the **Division** text, click **Bold**. De-select the underline option (**U**).

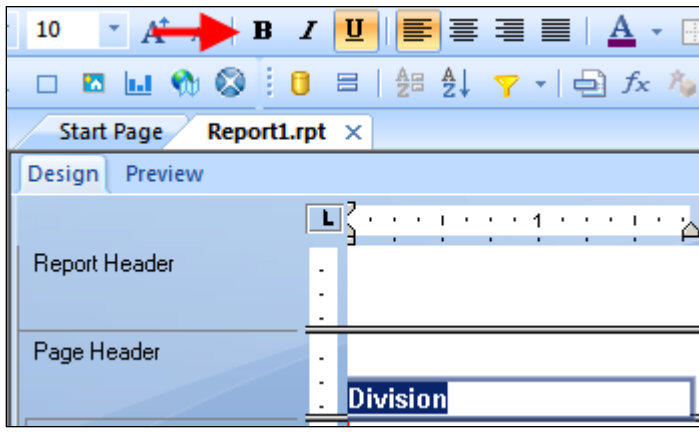


Figure 3-7 Highlight Division and click bold to change style.

7. Rename the second name column header to **Project** and apply the same style.
8. For the third column header, name it **PR#** and click left align.
9. For the last one, name it **Assigned To**.

Report Header				
Page Header	Division	Project	PR#	Assigned To
Details	name	name	id	name
Report Footer				
Page Footer				

Figure 3-8 Example of updated Page Header.

10. When finished, each column now has a name to describe what kind of data it is.

3.3 Adjusting the field size, page layout and location

Each field can be independently resized and moved on the report for best fit. In order to resize the field, select it first in either **Design** or **Preview** mode.



Workbook Exercises

1. Notice four anchor points on each side of the field. Place the cursor on top of an anchor point until the pointer shows up as a double-sided arrow, hold the left mouse button down and move the field in the necessary direction.



Figure 3-9 View of four anchor points on each side of the field.

2. Adjust the page layout to accommodate multiple columns of data fields. Reports can print in either Portrait or Landscape. Click **File – Page Setup** and choose **Landscape** for page orientation.

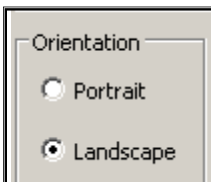


Figure 3-10 The available options for page layout.

3. Click **OK** and notice design mode acquiring additional space for fields.

3.4 Changing a number field format

The PR.ID field contains a comma. Since this format is not applicable for this field the format can be changed. In either **Design** mode or in **Preview** mode, right-click the **PR.ID** field and select **Format Field** from the list.

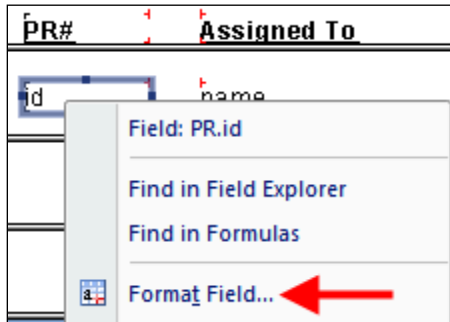


Figure 3-11 The Format Field option is available to change the format of fields.



Workbook Exercises

1. From the **Style** selection box, select **-1123** as the format. Click **OK**. This window allows you to select a pre-defined format or define your own format.

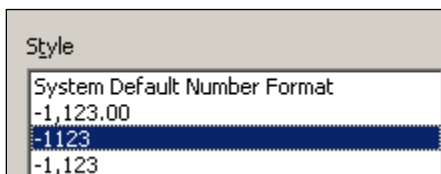


Figure 3-12 Select a pre-defined format or define a format.

2. Refresh the report and the PR# now shows up in its new format. Click **Save**.

Division	Project	PR#	Assigned To
Holmdel, NJ	Action Item	1015	Bramley, Harrel

Figure 3-13 Example of PR# formatted to display without a comma.

3.5 Field Alignment

Refer to the following steps to align the fields across.



Workbook Exercises

1. Highlight and select all four fields in Details by holding down the Shift key and selecting the four fields.

Page Header	Division	Project	PR#	Assigned To
Details	name	name	id	name

Figure 3-14 Example of multiple fields selected.

2. Left-click one of the fields, and with the button pressed, drag the mouse across. Notice all of the fields in Details and the corresponding Page Headers move together.
3. Click in-between Details in blue and the white column. This adds a line up marker where the fields in Details can line up against.

Page Header	Division
Details	name

Figure 3-15 Example of line up marker added to Details section.

4. A grid can also be used to line up data fields and labels with. This option is available from **View – Grid – Design**.

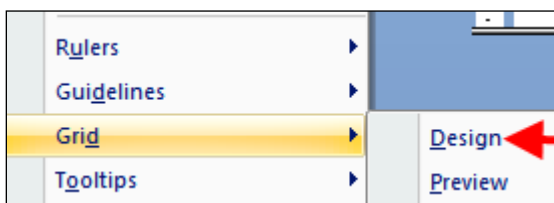


Figure 3-16 The grid option is available on the Design and/ or Preview tab.

5. Another option is to use guidelines from the top and side markers. This option is available from **View – Guidelines – Design**.



Figure 3-17 Example of Guidelines selected for the Design table

On Your Own Exercises

1. Add the **PR_STATUS_TYPE** table to the report to show the names of the record states. This is available from **Database – Database Expert** where additional tables can be added. Refer to the **introCrystal** database to find the table.

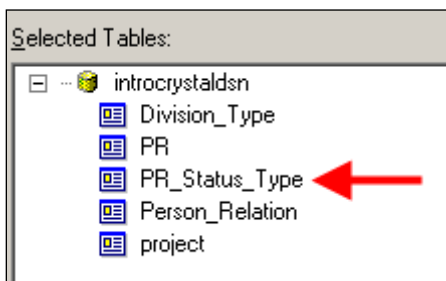


Figure 3-18 Example of PR_Status_Type added from the introcrystal database.

2. Link the table from **PR.STATUS_TYPE** to **PR_STATUS_TYPE.ID** as an equal join. Every record has a status.

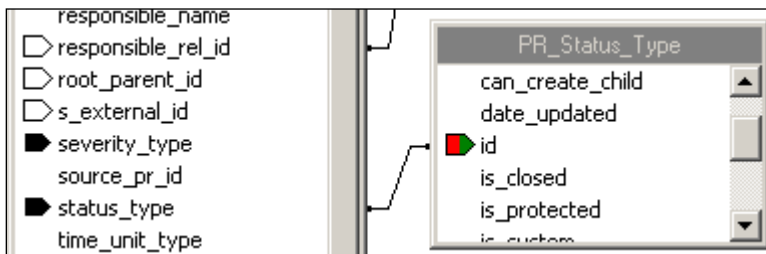


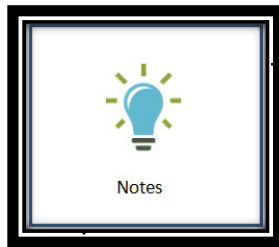
Figure 3-19 Link between PR and PR_Status_Type tables.

3. On the reports details screen, drag and drop **PR_STATUS_TYPE.NAME** to the Details section. Resize the fields so all fields fit nicely on the report.

4. Run the report in Crystal. Notice the new column displays the name of the state each record currently is in.

Conclusion:

The Details row of the report prints information based on the records returned in TrackWise. In Crystal Reports all of the records are returned for testing purposes. A multi-report is designed to show many records in one report as a preview of the actual content within the record. In the next chapter, the ability to group and sort the returned data is discussed and demonstrated.



4. Grouping and Sorting

4.1 Definition and Explanation

Grouping and sorting is a fundamental reporting concept, which allows joining records together with common attributes. Breaking records into groups can give a report a more professional look and make reading the report clear and concise. Some examples of grouping include group by Assigned To, Project, Division and product, just to name a few. Grouping places records into associated buckets of information. Multi-reports should contain at least one group or more in order to organize the data.

4.2 Grouping

A Crystal Report can contain one or more break groups, depending on the report requirements. Each time a break group is added, two new sections appear: Group Header and Group Footer. A Group Header contains the name of the group. Think of this as the label for the bucket of records. A Group Footer contains summary information such as the count of records in that group. Think of this as the bottom of the bucket of records.

4.3 Inserting a Break Group

Open the report that was just saved in the previous section. Select **File – Save As** and save this report as **report2.rpt**.



Workbook Exercises

1. Click **Insert - Group**. Select **DIVISION_TYPE.NAME**. Click **OK**.

Page Header	-	Division	Project
Group Header #1:	-	Group #1 Name	
Details	-	name	name
Group Footer #1:	-		

Figure 4-1 Example of Design tab with a Group inserted.

2. Click the **Preview** tab. The records are now grouped together by the Division.

PH	Division	Project	PR#	Assigned To	State
GH1	Canada				
D	Canada	Complaints	1140	Dillon, Corey	Opened
D	Canada	Complaints	1141	Bramley, Harrel	Opened
D	Canada	Complaints	1142	Jovanis, Mike	Opened
D	Canada	Complaints	1153	Dillon, Corey	Opened
D	Canada	Complaints	1155	Jovanis, Mike	Opened
D	Canada	Complaints	1154	Bramley, Harrel	Supervisor Review
GF1	Holmdel, NJ				
GH1					
D	Holmdel, NJ	Training - Employee	1160	Bramley, Harrel	Active

Figure 4-2 Preview tab with data grouped by Division.

3. On the left side is the **Group** tree. Based on the field type to group, a tree-like navigation structure appears in order to quickly navigate to a particular section of the report. With four different Divisions in this database, each one is presented here to access.

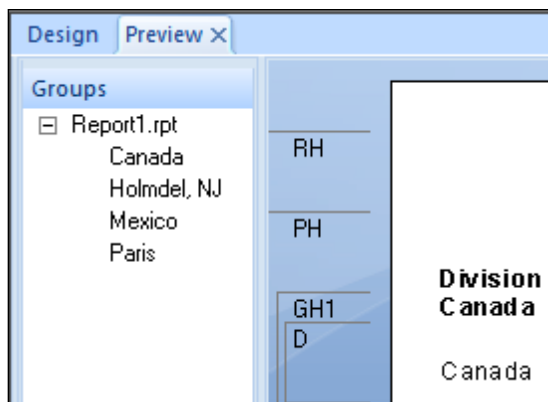


Figure 4-3 Display of Group tree based on inserted group by Division.

4. To navigate through the next several pages, use the arrows at the top right of the screen. Click **Save**.

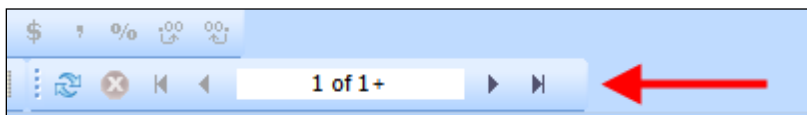


Figure 4-4 Use arrows to navigate between the preview pages.

4.4 Repeat Group Header on Each Page

On the next several pages of the report, a group begins on one page and continues to the next. However, when looking on the next page, there is no way to tell what group the records belong to since there is no heading. Crystal Reports allows the repeating of the Group Header on each page for situations like this.



1. In the design mode, right-click **Group Header #1** and select **Change Group** from the menu.
2. Select the **Options** tab and enable **Repeat Group Header on Each Page**. Click **OK**.

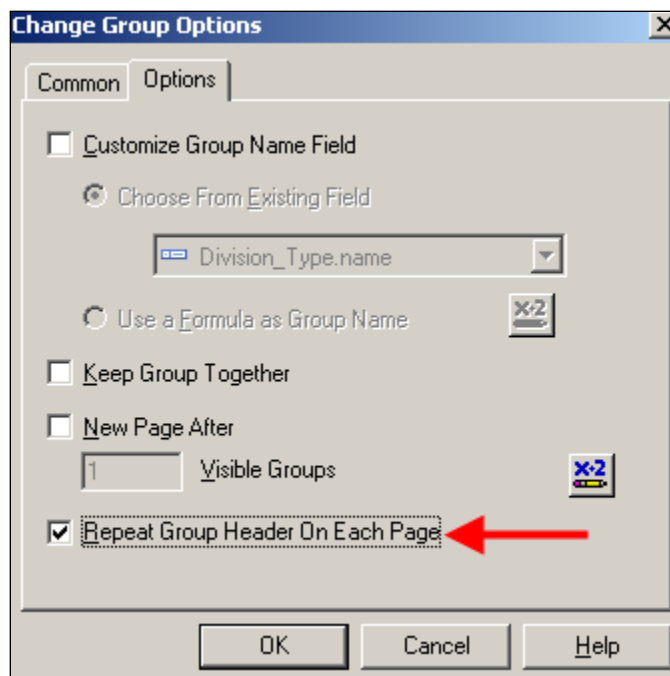


Figure 4-5 Repeat Group Header on Each Page option selected to enable.

- Preview the report and notice how the Break Group header repeats when the records go to a second page.

PH					
GH1	Division	Project	PR#	Assigned To	State
D	Holmdel, NJ	Complaint	1081		Closed - Cancelled
D	Holmdel, NJ	Change Control - Process	1124	Broderson, Michael	Closed - Cancelled

Figure 4-6 Example of repeated Break Group header.

4.5 Start each Group on a New Page

Notice that on some of the pages in the report, multiple groups are on the same page and different groups start on different locations of the same page. A common request is to start each group at the top of a new page for the purpose of report simplicity.



Workbook Exercises

- In design mode, right-click **Group Header #1** and select **Change Group** from the list.



Figure 4-7 The Change Group option.

-
- Click **Options**. Select **New Page After** and confirm number one for **Visible Groups**. This allows a new page to start for each group of records after the first group appears. This also prevents a blank page one and instead uses page one with the first incoming group.

Change Group Options

Common Options

☒ Customize Group Name Field

☒ Choose From Existing Field

Division_Type.name

☐ Use a Formula as Group Name

☐ KeeP Group Together

☒ New Page After

1 Visible Groups

☒ Repeat Group Header On Each Page

Figure 4-8 Example of Options tab of the Change Group Options window with New Page After selected.

- Click **OK** and return to the design screen. Click **Save** and run the report. Notice that each group based on division name starts on a new page.

Groups						
Report1.rpt Canada Holmdel, NJ Mexico Paris	RH					
	PH					
	GH1	Division	Project	PR#	Assigned To	<u>State</u>
	D	Canada	Complaints	1140	Dillon, Corey	Opened
	D	Canada	Complaints	1141	Bramley, Harrel	Opened

Figure 4-9 Preview of new page based on Division.

- Go back to the **Design** tab. Remove the first column from Details and Page Header for Division. Since the grouping is based on Division, it is not necessary to keep the column as well.

Page Header				
	Project	PR#	Assigned To	State
Group Header #1:	Group #1 Name			
Details	name	id	name	name

Figure 4-10 Example of Design tab with Division removed from Page Header and Details sections.

- Highlight all four data fields from **Details** with the mouse and drag them over to be slightly closer to the left side of the screen.

Page Header				
	Project	PR#	Assigned To	State
Group Header #1:	Group #1 Name			
Details	name	id	name	name

Figure 4-11 Example of highlighted fields.

6. Save the report and preview again. Notice the extra space is put to good use.

Groups					
Report1.rpt					
Canada	RH				
Holmdel, NJ					
Mexico	PH				
Paris					
	GHI				
D					
	D				

	Project	PR#	Assigned To	State
Canada				
	Complaints	1140	Dillon, Corey	Opened
	Complaints	1141	Bramley, Harrel	Opened

Figure 4-12 Preview of report with modifications.

7. Finally, to put the name of the break group on the center top of the page, right-click the page header and select **Insert Section Below**.

Page Header

Group Header #1

Details

Group Footer #1:

Report Footer

Page Footer

Page Header

Suppress (No Drill-Down)

Section Expert...

Show Short Section Names

Insert Line

Delete Last Line

Arrange Lines

Fit Section

Insert Section Below

Figure 4-13 The Insert Section Below is available with a right-click on a section.

8. **Page Header b** appears. Move this new Page Header up in order to bring the first page header down. Left-click **Page Header b** and hold the mouse button while moving the mouse up.

Page Header a	-				
Page Header b	-				
Group Header #1:	-	Project	PR#	Assigned To	State
	-	Group #1 Name			

Figure 4-14 Example of a second Page Header.

9. Page Header b contains the names of the fields and is closer to Details. Move **Group#1 Name** to the blank Page Header. Center the name and change the font size to stand out. Make sure to give it enough room based on the anchor points for resizing.

Page Header a	-				
Page Header b	-				
Group Header #1:	-	Project	PR#	Assigned To	State
Details	-	name	id	name	name

Figure 4-15 Move the Group #1 Name object to the blank Page Header.

10. Save the report and preview. Notice the name appears on the top of every page based on Page Header a and the field headers on Page Header b.

RH				
PHa				Canada
PHb				
GH1	Project	PR#	Assigned To	<u>State</u>
D	Complaints	1140	Dillon, Corey	Opened
D	Complaints	1141	Bramley, Harrel	Opened

Figure 4-16 Preview of report with Division now displayed at the top of each page.

4.6 Sorting Information

Sorting refers to how the records are displayed in the most inner break group on a report. A report can have multiple break groups based on Division, Project and Status. However, after all the records have been grouped by Division, Project and Status, they can be sorted by the Date Opened and by the Assigned To. The following workbook exercise shows how data within a break group can be sorted by a particular field type.



Workbook Exercises

1. Click **Record Sort Expert** or refer to **Report – Record Sort Expert** to open the configuration window.
2. Under **Available Fields**, highlight **Person_Relation.name** and bring it over to the **Sort Fields** window.

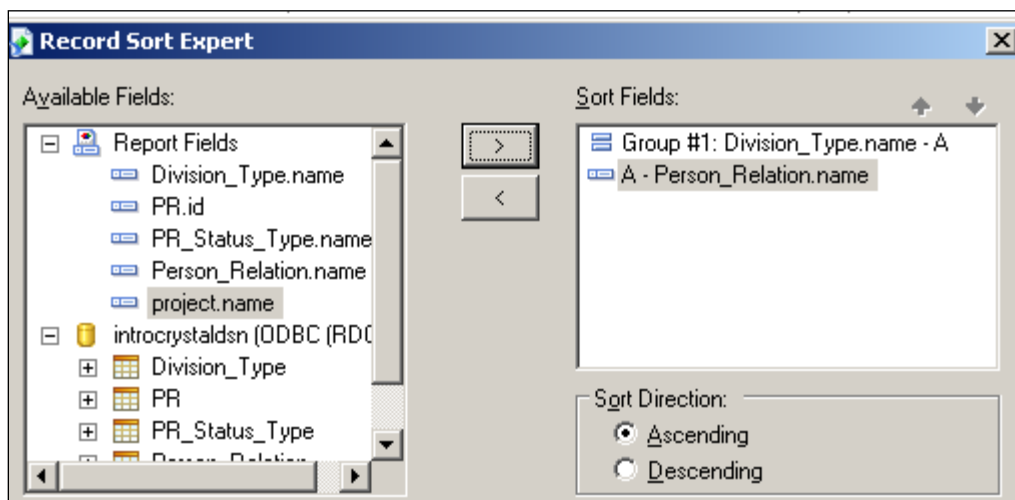


Figure 4-17 Record Sort Expert window with Person_Relation added to Sort Fields section.

3. Keep the **Sort Direction** as **Ascending** so the names appear in alphabetical order. Click **OK**.

4. Save the report and click the **Preview** tab. Notice in each group of records, the order is alphabetically.

RH				
PHa				Canada
PHb				
GH1	Project	PR#	Assigned To	State
D	Complaints	1141	Bramley, Harrel	Opened
D	Complaints	1154	Bramley, Harrel	Supervisor Review
D	Complaints	1140	Dillon, Corey	Opened
D	Complaints	1153	Dillon, Corey	Opened
D	Complaints	1142	Jovanis, Mike	Opened
D	Complaints	1155	Jovanis, Mike	Opened

Figure 4-18 Preview of Assigned To sorted ascending.

5. Click the **Design** tab. To add another sorting option, refer back to the **Record Sort Expert**. From the **Available Fields** screen, highlight and move **Project.name** over to **Sort Fields**. Click **OK** and preview the report again.

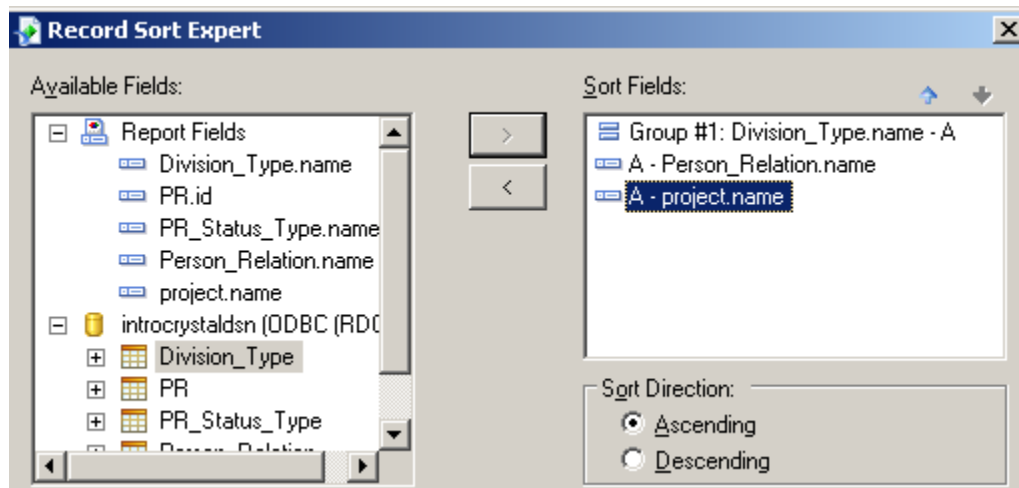


Figure 4-19 Record Sort Expert with Project added to Sort Fields section.

6. When previewing a report, notice the records for Holmdel NJ are sorted first by Assigned To (person_relation.name) and then by Project (Project.name).

PHa	Holmdel, NJ			
PHb				
GH1	Project	PR#	Assigned To	State
D	Audit - Internal	1052	Bramley, Harrel	Pending Response
D	Change Control - Process	1059	Bramley, Harrel	Pending SME Post-Approval
D	Change Control - Process	1078	Bramley, Harrel	Pending SME Pre-Approval
D	Complaint	1004	Bramley, Harrel	Closed - Done
D	Complaint	1095	Bramley, Harrel	Closed - Done

Figure 4-20 Preview of sort by Assigned To and then by Project.

7. When grouping or sorting with a field that is blank, those records appear first followed by records with the data field filled out.

4.7 Multiple Break Groups

In addition to grouping on division name, a second group is added based on the state of the record.



1. In Design mode, click **Insert – Group**. Add **PR_Status_Type.name** as the second data field to group by. Click **OK**.

- The next break group appears on the report under Group Header #2.

Page Header a	-				Group #1 Name
Page Header b	-				
	-	Project	PR#	Assigned To	State
Group Header #1:	-				
Group Header #2:	-	Group #2 Name			
Details	-	name	id	name	name

Figure 4-21 Design tab with second group inserted.

- Remove **State** from the report in **Details** because it is now used to group the records together. Save the report and click **Preview**.

PHa			Canada
PHb			
GH1	Project	PR#	Assigned To
GH2	Opened		
D	Complaints	11 41	Bramley, Harrel
D	Complaints	11 40	Dillon, Corey
D	Complaints	11 53	Dillon, Corey
D	Complaints	11 42	Jovanis, Mike
D	Complaints	11 55	Jovanis, Mike
GF2	Supervisor Review		
GH2			
D	Complaints	11 54	Bramley, Harrel

Figure 4-22 Preview of report with second grouping by State.

4. While previewing the report, refer to the tree menu on the left side. As a result of adding two break groups, the tree contains both Division and States names. This provides an easy way to locate specific information on the report.

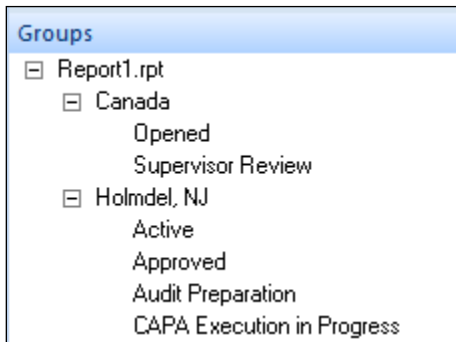


Figure 4-23 Group Tree menu with both Division and States.

The following chart suggests a variety of group options in reports.

Break Group Suggestions	Description
Assigned To	Grouping on Assigned To or any person-related field allows the report to print which records are associated with whom. Ideal for manager reports of open records from colleagues.
Product	Grouping on Product Names allows the report to group on returned product records that are past due or high priority.
Division	Grouping on Division allows the report to organize the returned records based on origin of occurrence.
Disposition	Grouping on Disposition allows the report to group on records that need to be held, released or destroyed based on the value inside the record details.

4.8 Upload Reports to TrackWise

To run the reports in TrackWise, refer to the following workbook exercise where the file is copied and pasted into the appropriate directory. This is a step that a network administrator needs to do, depending on permissions on the TrackWise server.

There are two folder directories to upload reports to. One is called TWReports which contains multi-reports. These reports print data based on the user's query and scope settings.

The other folder is called TWReports_1. This folder contains single reports which are covered later in class. These reports are based on a particular highlighted record on the TrackWise desktop.



Workbook Exercises

1. Minimize Crystal Reports developer. Click **My Computer** and locate **C:\TrackWiseWeb\intro_TrackWise \TWReports**
2. Clear all reports within this folder.
3. Copy **Report 2.rpt** from the Desktop over to this folder.
4. Close the folder. Click **Start – Programs –Administrative Tools – Services**. Restart the **Apache Tomcat 7.0 Tomcat7_TrackWise** service.
5. Log in to **intro_WebAdmin**.
6. Navigate to **System Configuration – Projects – Report Classes**.
7. Click on **All Divisions/Projects** and add the report to the **Multiple Reports** tab. Then click on **Save**.
8. Restart the **Apache Tomcat 7.0 Tomcat7_TrackWise** service.
9. Log in to TrackWise by clicking **Start – Programs – TrackWiseWeb – intro_TrackWise**.
10. Click the **Analytics** tab.
11. On the left had side of the page, expand **Report Templates**.

12. From the Report list, select **Report 2.rpt**.
13. From the Query list, select **All records including closed (pub)**
14. From the Scope list, select **All Projects (pub)**
15. Click **Run** to see the report. Try a variety of Query and Scope combinations to view additional results.

Conclusion:

Break Groups organize multiple amounts of records into a cohesive order. The addition of break groups also creates a group tree to easily navigate the report for specific kinds of data. By placing records into organized buckets, sorting arranges the data within the break group.



5. Running Totals

5.1 Definition and Explanation

Running Totals work in conjunction with Break Groups to provide powerful summaries and statistics, based on returned records. Running totals can be as simple as counting the records in each break group or the total number of returned records in the report. Running Totals only work within group footers or report footers.

Running Totals created in one report can be duplicated and copied over to work in another report. When using a template report, running totals are able to be copied over to the report itself.

5.2 Defining a Running Total

With **Report2.rpt** opened, click **File – Save As** and name it **Report3.rpt**. In the next workbook exercise, a running total is created to count the number of records per state.



1. In Design mode, refer to **Field Explorer** and right-click **Running Total Fields**. Click **New**.

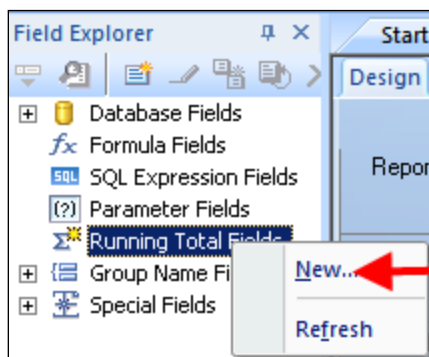


Figure 5-1 the New option appears with a right-click on Running Total Fields form the Field Explorer.

2. Provide a name for the running total to describe its functionality. In this example, name it **Record Count Per State**.

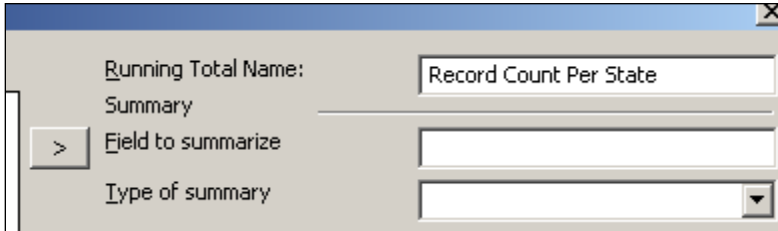


Figure 5-2 Name added to Running Total Name field.

3. Select **PR.ID** from the **Available Table and Fields** and click the upper-most right arrow. This changes the value in the **Field to Summarize** to **PR.ID**. Crystal incorrectly assumes that you want to sum the **PR.ID**. From the **Type of Summary** list, select **Count**.

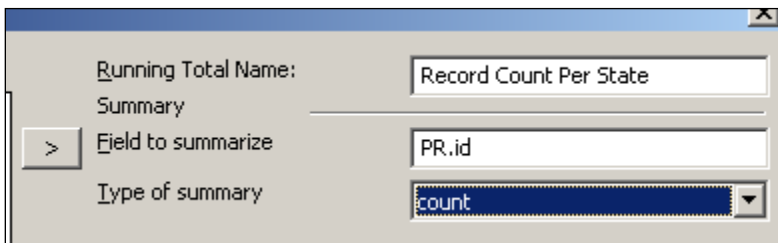


Figure 5-3 PR.id added to Field to summarize and count selected for Type of summary.

4. In the **Evaluate** section, select **For Each Record** to count every record in the group footer.

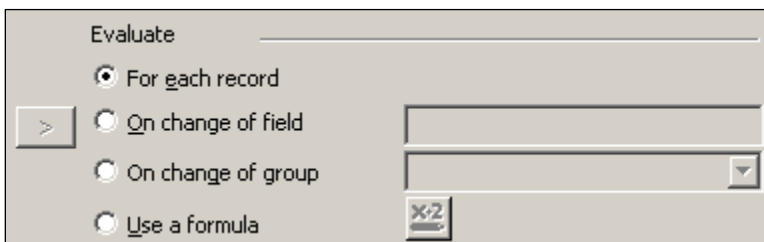


Figure 5-4 The For each record option is selected under Evaluate.

5. In the Reset section, select the **On Change of Group** option. From the corresponding list, select **Group #2 PR_STATUS_TYPE.NAME**. This guarantees that for each status, the counter resets. This option is based on the location of the running total. If it is counting within group #2, the reset occurs in this group so the next batch of records is counted.

If it is counting all records in the report within report footer, **Reset** is set to **Never**.

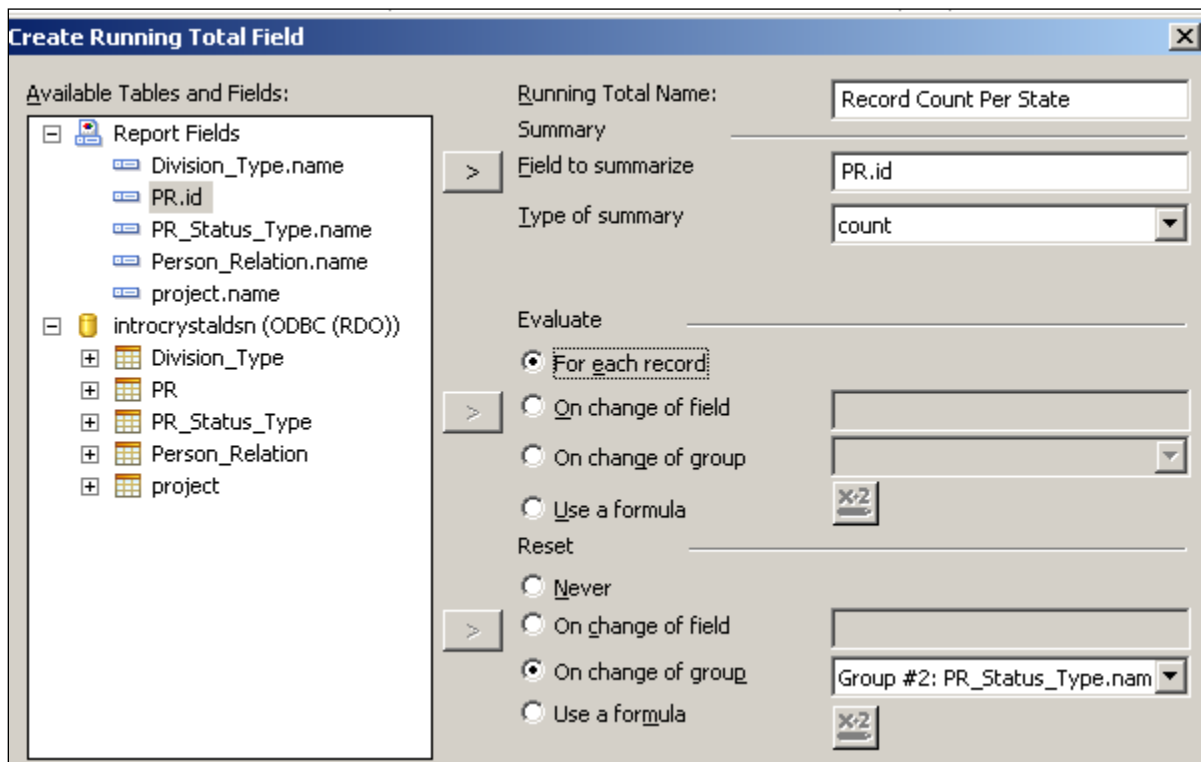


Figure 5-5 On change of group option and Group #2 selected under the Reset section.

6. Click **OK**. The Running Total appears in the **Field Explorer**.

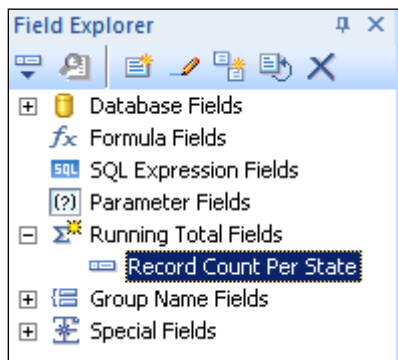


Figure 5-6 The new Running Total appears in the Field Explorer.

- In Design mode, open the **Field Explorer** and select the running total. Drag and drop it into **Group Footer #2** for **PR_STATUS_TYPE.NAME**.

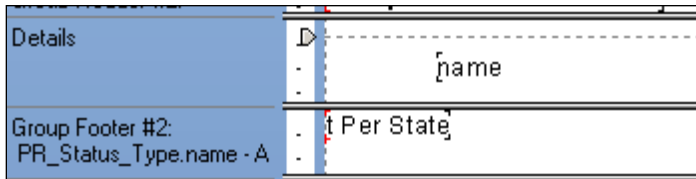


Figure 5-7 Example of running total field added to Group Footer #2.

- Refresh and preview the report and notice that after the change for each Status, the total number of records in group footer two appears.

GH2	Approved		
D	Action Item	1022	Bramley, Harrel
D	Corrective Action	1071	Bramley, Harrel
D	Action Item	1019	Smith, John
GF2	3		

Figure 5-8 Preview of count based on the defined running total.

5.3 Embedding a Running Total into a Text Object

To better display the running total value in a clear and professional manner, it is common to embed the value into a text object concatenated with a narrative.



- In Design mode, delete the running total from **Group Footer #2** section (not from the Field Explorer window).

2. Click **Text Object** on the standard toolbar.

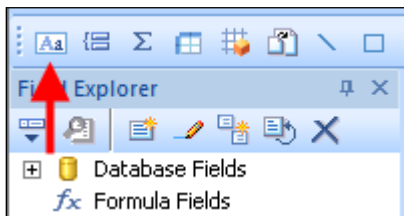


Figure 5-9 Location of Text Object tool.

3. The cursor appears as a **plus** symbol used to draw a text object. Place the symbol in Group Footer #2 and hold the left mouse button down. Move the mouse across the screen to the other side of the footer. Let go of the mouse button and a text box appears.

Group Header #1:				
Group Header #2:	Group #2 Name			
Details	{name}	{id}	{name}	
Group Footer #2:				
PR Status Type name - Δ				

Figure 5-10 Example of Text box added to Group Footer #2.

4. Place the new text object in the **Group Footer #2** section. After placing the text object here, open it up by expanding from the right anchor.
5. Double-click the new text object to initiate edit mode and type: **The Total Number of records for this Status:** (Insert a space after the colon)
6. From the **Field Explorer** window, drag the **PR_STATUS_TYPE.NAME** field into the text object.

Group Header #2:	Group #2 Name			
Details	{name}	{id}	{name}	
Group Footer #2:	The Total Number of records for this Status: {name}			
PR Status Type name - Δ				

Figure 5-11 Example of text object embedded with running total field.

7. After the **PR_STATUS_TYPE.NAME** field, enter a single space and then type the word **is**.
8. From the **Field Explorer** window, drag the running total **Record Count per Status** into the text object.

9. Save and preview the report. Notice the new summary line.

PHb	Project	PR#	Assigned To
GH1	Active		
GH2			
D	Training - Employee	1160	Bramley, Harrel
D	Training - Employee	1164	Latos, Ingrid
GF2	The Total Number of records for this Status: Active is 2.		

Figure 5-12 Preview of embedded running total in a text object.

10. Text objects can contain static and dynamic information. Typical use for text objects are for field labels and report titles. Define a new running total that counts the number of different types of Assigned To names using **(PERSON_RELATION.NAME)** for each Division. Name this new running total **Different Assigned To Users**.
11. Set **Type of Summary** to **distinct count**. This option counts a name only once even if it appears more than once in the group of records. To determine how many different people are involved with records, **distinct count** is used.

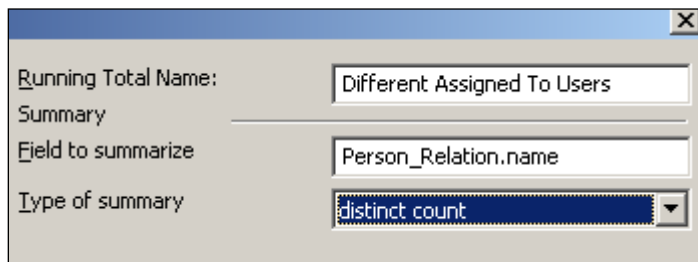


Figure 5-13 Distinct count selected for the Type of summary field.

12. For **Evaluate**, set **For each record**.
13. Reset **On change of group** to **Group#2: PR_Status_Type.name**

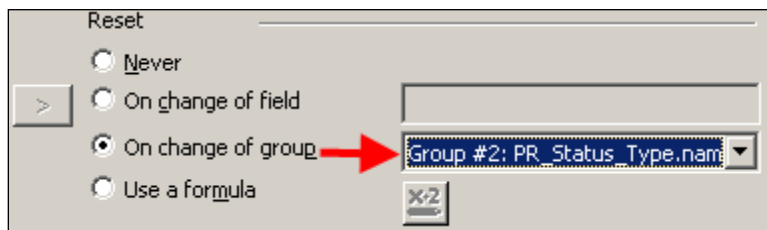


Figure 5-14 On change of group and Group #2 selected under the Reset section.

14. Click **OK**.

15. Create another text object in Group Footer#2 with the following: **The Total Number of Different Assignees for the State : <PR_Status_Type.Name> is <Running Total: Different Assigned To Users>.**

Group Footer #2:	-	The Total Number of records for this Status: {name} is {#Record Count Per State}.
PR_Status_Type.name - A	-	The Total Number of Different Assignees for the State: {name} is {#Different Assigned To Users}.

Figure 5-15 Example of additional running total embedded in text object.

16. Click **Save** and then **Preview**. In the following screen capture for the state Approved, three records appear with two different Assigned To names.

GH2	Approved		
D	Action Item	1022	Bramley, Harrel
D	Corrective Action	1071	Bramley, Harrel
D	Action Item	1019	Smith, John
GF2	The Total Number of records for this Status: Approved is 3.		
	The Total Number of Different Assignees for the State: Approved is 2. 		

Figure 5-16 Preview of text and running total field.

5.4 Counting Closed Records Only

Another use of running totals is to only count closed records. Instead of counting all records, the evaluate section is used to only include records that are closed. In the PR table, the IS_CLOSED column indicates which records are closed. If this equals one, the record is closed. If it equals zero, the record is still opened and should not be counted. In the next exercise, a running total is created to count how many records in a Division are closed by placing this into Group Footer #1.



1. Create another running total. Enter **Closed Records** in **Running Total Name**. For **Field to summarize**, use **PR.ID** For **Type of summary**, set it to **count**.

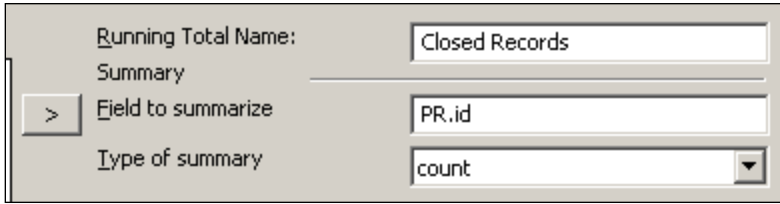


Figure 5-17 Running Total Name and Summary section completed.

2. For Evaluate, set the button to **Use a formula** and click **X+2**. Formulas are discussed in greater depth later in a future chapter. The X+2 button is used to access the formula workshop screen.

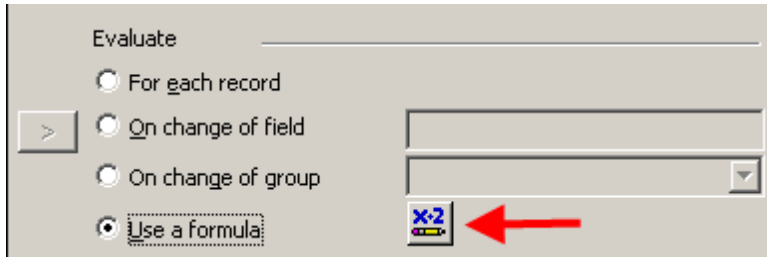


Figure 5-18 Location of X+2 button to access the formula workshop.

3. On the **Formula Workshop** screen, locate the PR table from **introCrystalsn ODBC**. Open the PR table to show the columns of information.

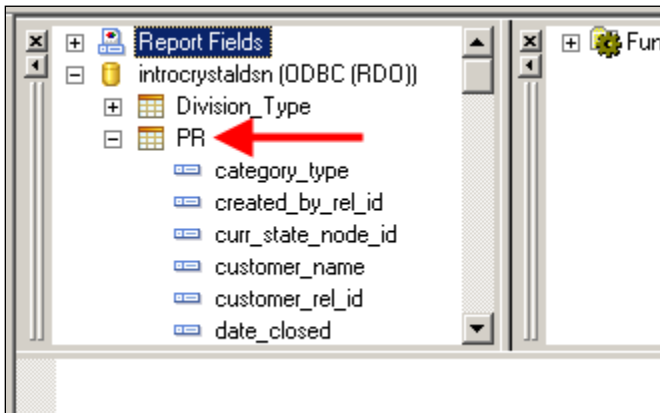


Figure 5-19 Expanded PR table to show the columns of information.

4. Double-click the item **is_closed**. It appears on the main Workshop screen.

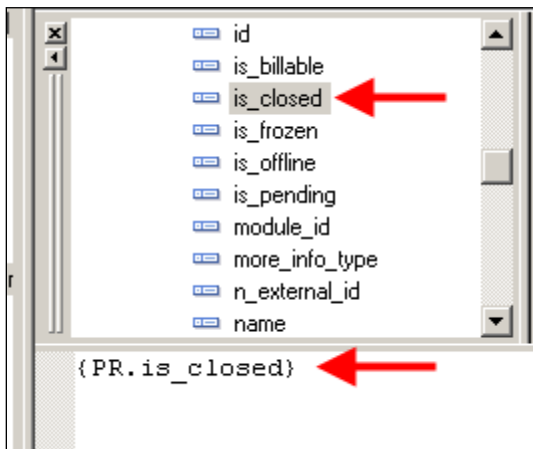


Figure 5-20 **is_closed** column added to main workshop screen.

5. After the bracket of information type an equal sign followed by a one. This indicates that if the record is closed; count it as part of the running total. If it is not closed, do not count it.

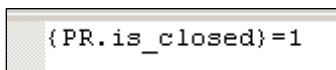


Figure 5-21 Formula identifies closed records.

6. Click **X+2** on the top left side of the screen to check the formula. A window appears indicating no errors found.

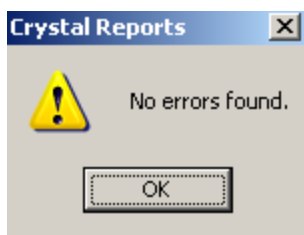


Figure 5-22 No errors found message.

7. Click **Save and Close**.

8. Reset **On Change of group** to **Group#1: Division_Type.name**. Click **OK**.

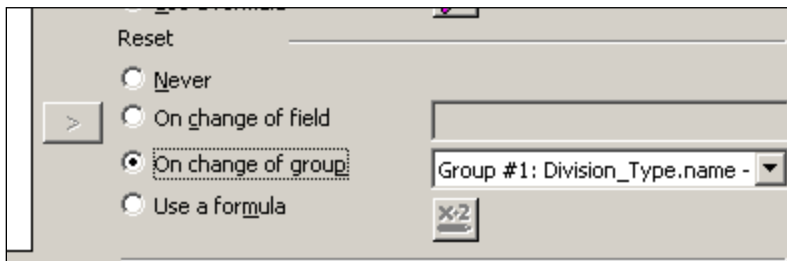


Figure 5-23 On change of group and Group #1 selected under Reset section.

9. On the Design screen, create a text box in Group Footer #1 and enter the following text.
The Total Number of Closed records for the Division <Division_Type.name> is <Running Total: Closed Records>.

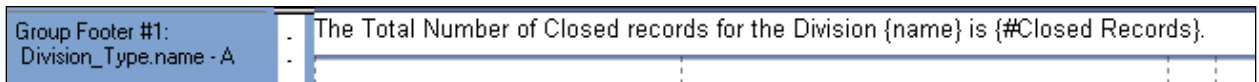


Figure 5-24 Running Total embedded in text object on design tab.

10. Click **Save** and preview the report. Refer to the last page for Holmdel NJ records. Notice a number appears to indicate how many are closed.

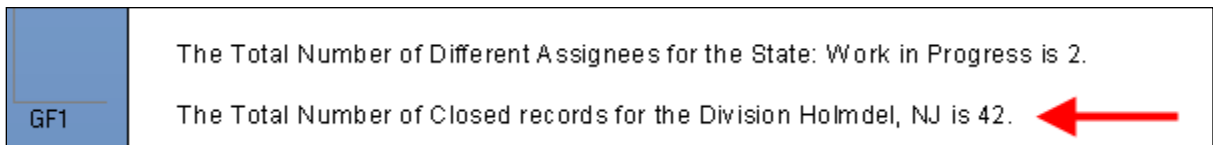


Figure 5-25 Preview of text object with the embedded running total.

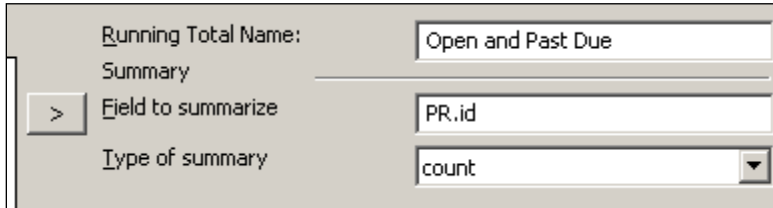
5.5 Counting Open and Past Due Records

Running totals can also count the number of records still open but past due. The PR table contains a column called **date_to_close**. This is the Date Due field seen in most records. The next workbook exercise uses a formula to evaluate records that are still opened with **date_to_close** set to yesterday or earlier.



Workbook Exercises

1. Create another running total. In the **Running Total Name** field, enter **Open and Past Due**. Select **PR.ID** in **Field to summarize** and select **count** for **Type of Summary**.

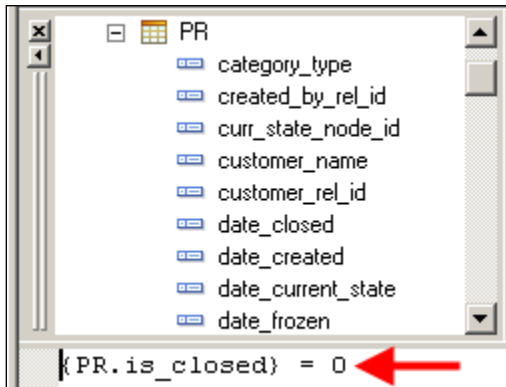


The dialog box shows the following fields:

- Running Total Name:** Open and Past Due
- Summary:** (empty)
- Field to summarize:** PR.id
- Type of summary:** count

Figure 5-26 Running Total Name and Summary completed.

2. Use a formula to evaluate which records to count. Click **X+2**. On the formula workshop screen refer to the PR table. Double-click **PR.is_closed**. Add an **equal sign** followed by a **zero**.



The Formula Workshop dialog box shows the following fields:

- PR** (selected table)
- category_type**
- created_by_rel_id**
- curr_state_node_id**
- customer_name**
- customer_rel_id**
- date_closed**
- date_created**
- date_current_state**
- date_frozen**

The formula field contains: `{PR.is_closed} = 0` with a red arrow pointing to the equals sign.

Figure 5-27 Formula indicates open records.

3. Click **Enter** and include the text **and**. Click **Enter**.

4. For the next line double-click **date_to_close**. This is the date due of the record. To only count records where the date is in the past to show late records, refer to the other list on top called **Functions – Date Ranges**.

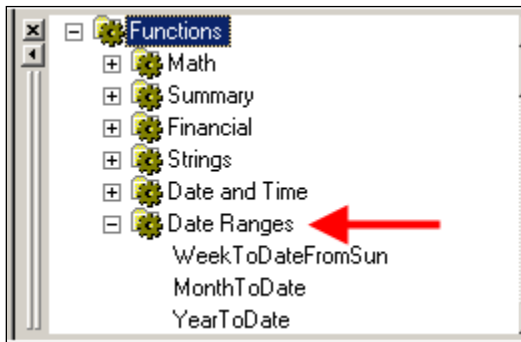


Figure 5-28 From the Functions panel, expand the Date Ranges section.

5. Find the date range **AllDatesToYesterday**. Double-click this function to display within the formula workshop.

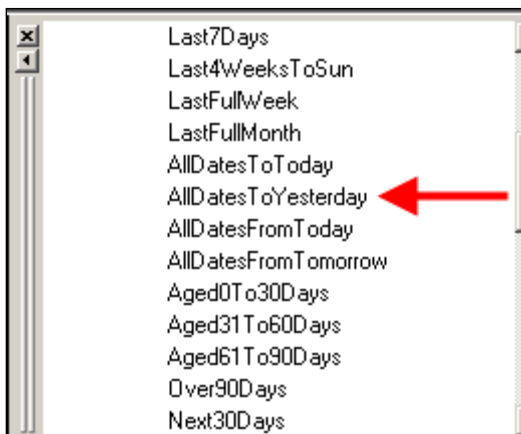


Figure 5-29 Locate the AllDatesToYesterday function.

6. Before checking and saving the formula, refer to the following screen capture.

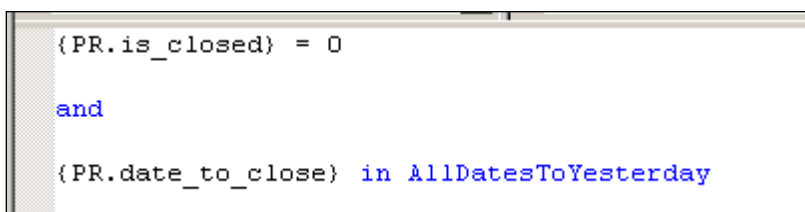


Figure 5-30 Formula for open and past due records.

7. Click **Save and Close**. On the running total screen, set **On change of group** to **Group #1:Division_Type.name**. For each Division, there is a running total to count how many records with a date due are overdue.

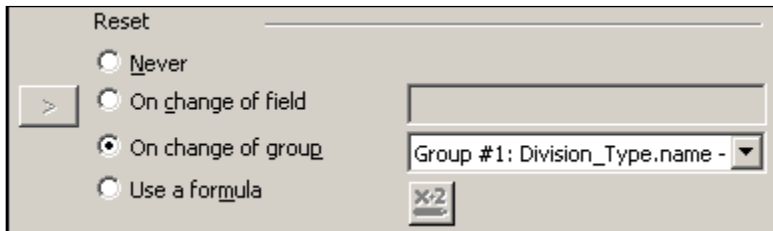


Figure 5-31 On change of group and Group #1 selected under the Reset section.

8. Click **OK**.
9. On the Design screen add another text box within Group Footer#1 with the following information:

The Total Number of Open records that are Past Due in Division (Division_Type.name) is (Running Total – Open and Past Due).

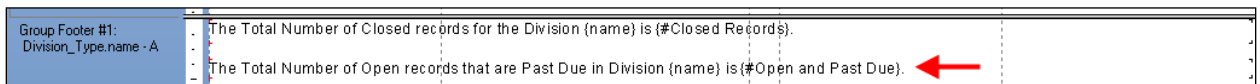


Figure 5-32 Running Total field embedded in text object.

10. Save the report and preview. At the end of every Division, a break group number appears to indicate how many of the open returned records are past due. Check the last page for Holmdel NJ records.

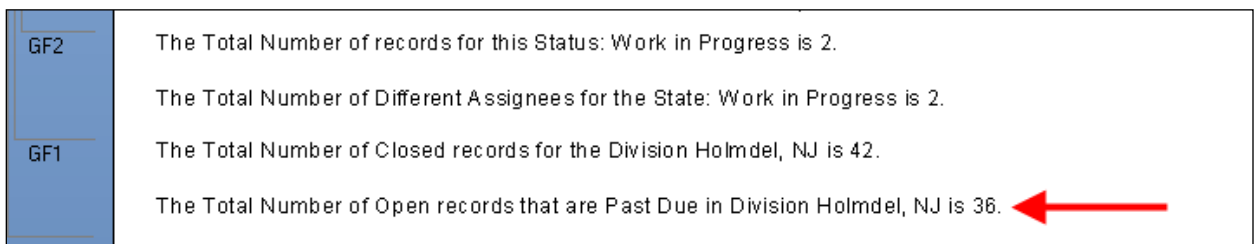


Figure 5-33 Preview of the text object and running total.



On Your Own Exercises

1. Add a running total into the Report Footer to count all PR.ID records and not reset. Create a text box with narration to describe what the number is.
2. Add another running total into the Report Footer to count the number of unique assignees on the report and not reset. Create another text box with narration to describe what the number is.
3. Save the report and preview.

Conclusion:

Running Totals count the number of records and other particulars within a group or report footer. They can be used to count open records that are overdue, unique number of objects within the returned results and how many records from the returned results are closed. The placement of the running total is crucial as different numbered results appear depending on the location of the count.



6. Parameters

6.1 Definition and Explanation

Parameters are a way for Crystal Reports to collect information from TrackWise, before generating the report. It is a question that Crystal is asking TrackWise before the report generates.

- Which Query was used to generate this report?
- Which Scope was used to generate this report?
- What is the name of the user who ran this report?

TrackWise provides Crystal Reports with each parameter value. Users are never prompted for this information. During report design in Crystal, the parameters appear when the report is previewed. They can mostly be ignored and the report still runs.

6.2 Defining a Parameter

1. Open the report used in the previous section and save it as **report4.rpt**.
2. At the top of the report, right-click **Report Header** and click **Insert Section Below**. This created another Report Header for the upcoming parameters.

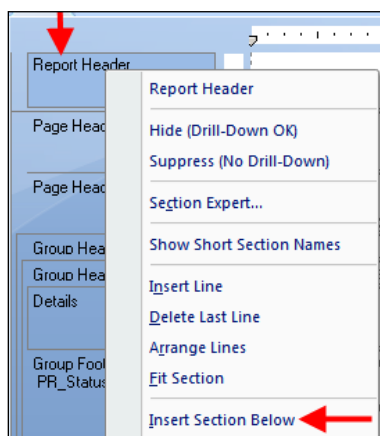


Figure 6-1 The Insert Section Below option is available with a right-click on Report Header.



Workbook Exercises

1. Repeat this step for a total of six new Report Header rows, labeled A through F.

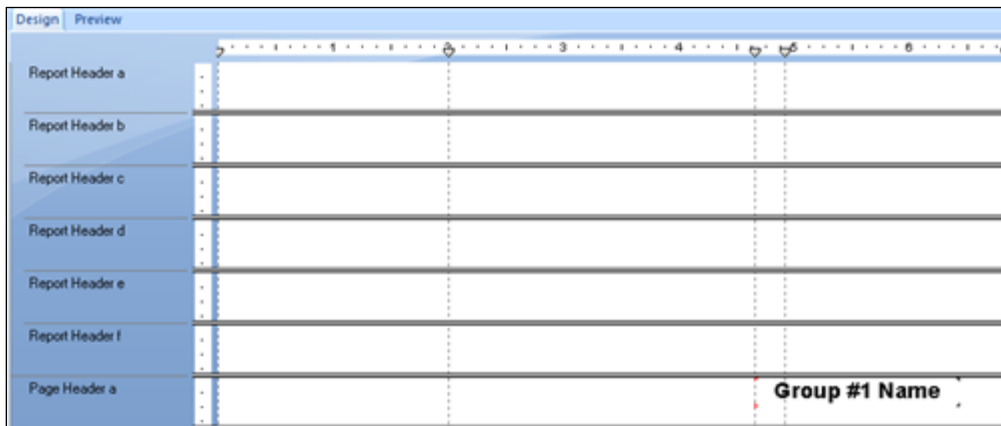


Figure 6-2 Additional six Report Header rows added to Design.

2. The first parameter, **SPTimeZone**, is used with a formula for an upcoming chapter. It asks TrackWise which time zone is the user, who is running the report, is in. Date Time fields in records reflect local time to a user in TrackWise. The information needs to appear the same way in the report as with TrackWise.
3. Under **Field Explorer** , right-click **Parameters Fields** and select **New**.

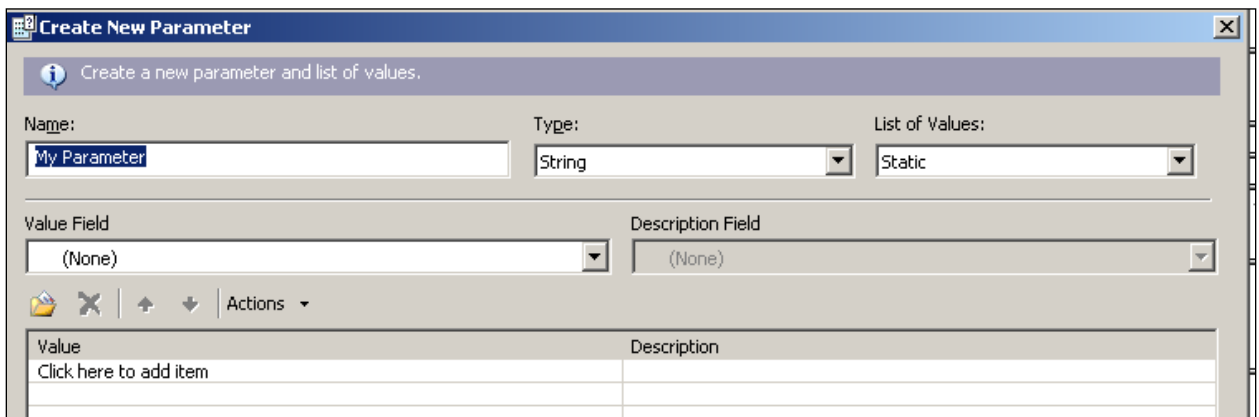


Figure 6-3 Create New Parameter window.

4. Enter **SPTimeZone** in the name field. Leave all the other settings as is.

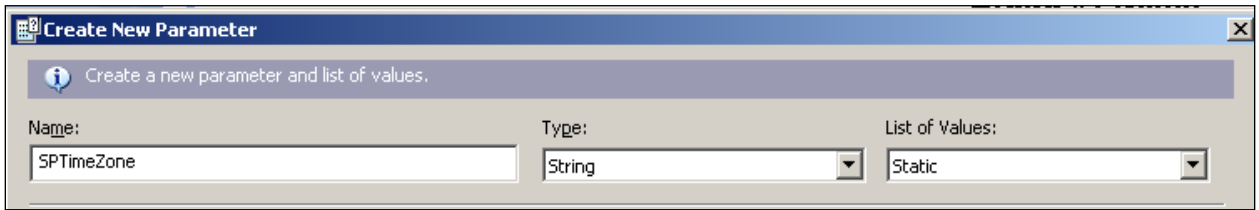


Figure 6-4 SPTimeZone parameter settings.

5. Click **OK**. The parameter appears under **Field Explorer**.

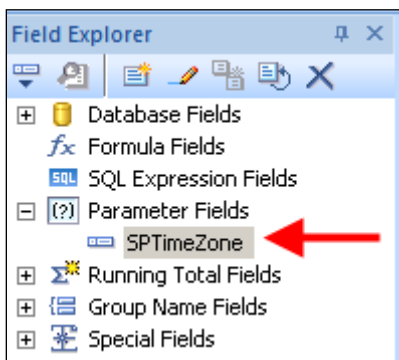


Figure 6-5 SPTimeZone added to Field Explorer.

6.3 SPDivisionsProjects

SPDivisionsProjects [String] is used to show all the Division/Projects in the user's current scope, at the time of running the report.



1. Define the parameter **SPDivisionsProjects** as a string. Click **OK**.

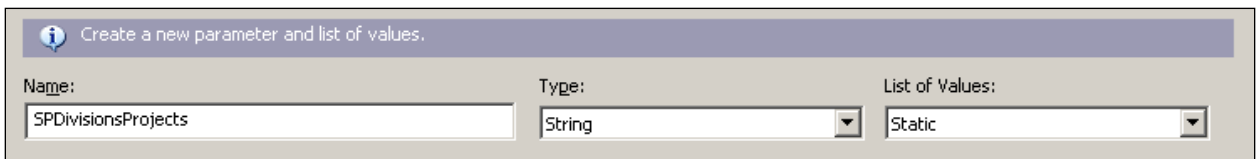


Figure 6-6 SPDivisionsProjects parameter settings.

2. Place a text object in Report Header A and enter the following message with SPDivisionsProjects to follow.

Division and Projects in Current Scope:

3. Add **SPDivisionsProjects** into the text object and save the report:

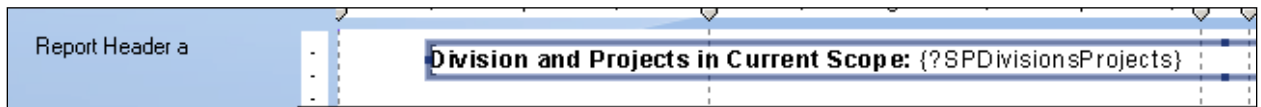


Figure 6-7 SPDivisionsProjects paramter embedded in text object.

6.4 SPProjects

SPProjects [String] shows a list of all Projects in the user's current scope at the time of running the report.



Workbook Exercises

1. Define the parameter **SPProjects** as a string. Click **OK**.

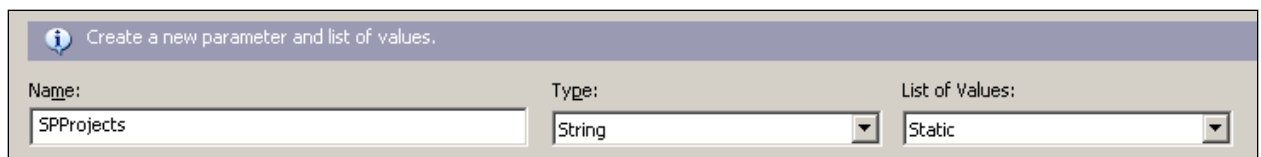


Figure 6-8 SPProjects paramter settings.

2. Place a text object in Report Header B and type the following message with SPProjects to follow.

Project(s) in Current Scope:

3. Add **SPPProjects** into the text object and save the report:

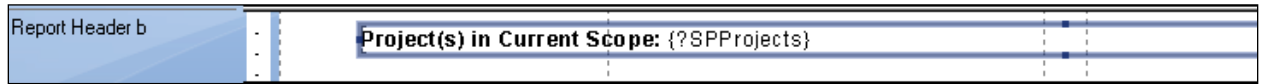


Figure 6-9 SPPProjects paramter embedded in text object.

6.5 SPDivisions

SPDivisions [String] shows a list of all Divisions in the user's current scope at the time of running the report. Create the parameter and add a text object to incorporate it within.

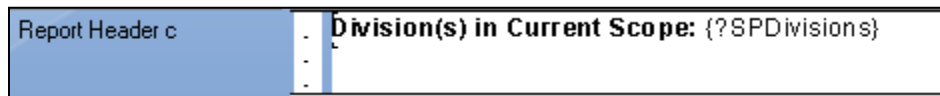


Figure 6-10 SPDivisions parameter embedded in text object.

6.6 SPScope

SPScope [String] shows the name of the user's saved scope at the time of running the report. Create the parameter and add a text object to incorporate it within.

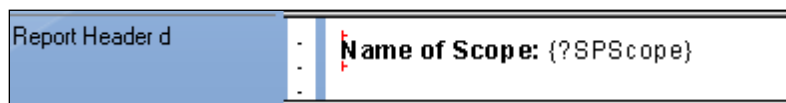


Figure 6-11 SPScope paramter embedded in text object.

6.7 SPQueryHeading

SPQueryHeading [String] shows the query description at the time of running the report. Create the parameter and add a text object to incorporate it within.

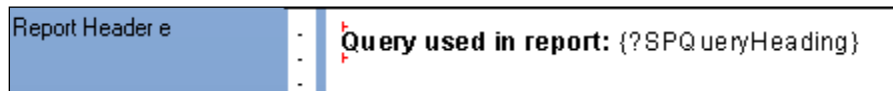


Figure 6-12 SPQueryHeading embedded in text object.

6.8 SPUser

SPUser [String] shows the name of the user currently running the report. Create the parameter and add a text object to incorporate it within.



Figure 6-13 SPUser embedded in text object.

6.9 nUserRelId

nUserRelId [Number] shows the **Person_Relation.id** of the person currently running the report. It can display the user's address, email, phone number and other details about the user. This is used in the Subreport chapter. Create the parameter for now with the type value set to **Number**.

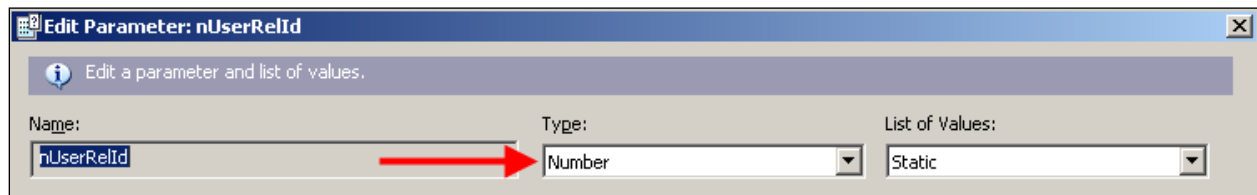


Figure 6-14 nUserRelId parameter settings.

6.10 SPAdminScopeId

SPAdminScopeId [Number] shows the Divisions and Projects of the Coordinator's Admin Scope used when running the report. This parameter is used with a subreport to display the information.

6.11 SPQueryId

SPQueryId [Number] shows the Coordinator's Admin Query used in running the report. This is used with a subreport containing the query description.

6.12 LanguageName

LanguageName[String] relays the language ID based on the translated dialect TrackWise is using. For TrackWise systems using MultiLanguage, this parameter generates the returned data into the specified language.

Translating Crystal Reports requires a report technique called **sub reporting**. This is covered later in the course along with the instructions on how to use this parameter if required.



1. Copy the report template into the **TWReports** folder in the **TrackWiseWeb** folder. The directory is the following:

C:\TrackWiseWeb\intro_TrackWise\TWReports

2. Close the folder. Click **Start – Programs – Administrative Tools – Services**. Restart the **Apache Tomcat 7.0 Tomcat7_TrackWise** service.
3. Log in to **intro_WebAdmin**.
4. Navigate to **System Configuration – Projects – Report Classes**.
5. Click on **All Divisions/Projects** and add the report to the **Multiple Reports** tab. Then click on **Save**.
6. **Restart** the **Apache Tomcat 7.0 Tomcat7_TrackWise** service.
7. Log in to TrackWise by clicking **Start – Programs – TrackWiseWeb – intro_TrackWise**.
8. Run the report from TrackWise using different queries and scopes. Notice each of the six parameters are filled out based on the way the report was generated.

9. If the parameters are not populated with data, return to Crystal and deselect **Save Data With Report**. This is found under **File**. Deselect this option, save the report and reload the same file to TWReports again. Saving data with the report prevents the parameters from asking TrackWise questions that it needs to answer.

Conclusion:

Parameters are questions for TrackWise to answer upon running the multi-report. A report of this kind needs to indicate its origins by providing the scope, query and name of the user who ran the report. The parameters in this chapter were placed in the report header to show this information on the first page based on report header. These parameters come with TrackWise and are activated by typing the names exactly as listed in the book. They are case sensitive.



7. Formulas

7.1 Definition and Explanation

Formulas are a powerful feature of Crystal Reports by providing an internal programming language, allowing the designer to fully control the report. They provide the ability to convert date time fields, conditionally display information within the report and others. Due to the abundance of Crystal functionality, refer to the help menu to learn more about the formula syntax available to design with.

In the following workbook exercise, a formula is used to convert date time fields from GMT to local time, based on the user's regional settings on the computer. The formula uses SPTimeZone the parameter from the last chapter to find out from TrackWise the correct time zone.

7.2 Date Time Conversion

1. Open the report saved in the previous step and save it as **report5.rpt**.
2. From the **Field Explorer** window, right-click **Formula** and select **New**. The **Formula Name** window opens.

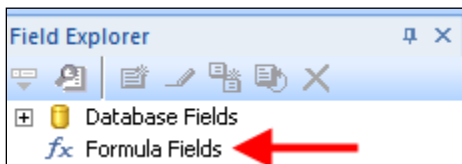


Figure 7-1 Location of Formula Fields option under the Field Explorer section.



Workbook Exercises

1. Provide a meaningful **Name** for the formula. In this example, call it **Date Opened**. Click **OK**.

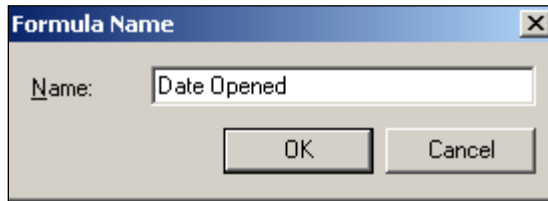


Figure 7-2 Use the Name field to provide a name for the formula.

2. The **Formula Workshop** screen opens. Here is an overview of the Options.
 - a. Save saves the formula.
 - b. **X+2 with a checkmark** is a spell check function for formulas. It does not guarantee that the formula works as expected based on the intended logic. Instead it verifies the settings are correct without extra parenthesis or commas.

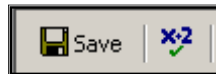


Figure 7-3 Location of X+2 with a checkmark button.

3. An important and fundamental feature of TrackWise is Universal Time Code. Each date time value that a user enters into a record is converted from the user's local time zone to Greenwich Mean Time (or GMT). This same value is converted from GMT back to the user's local time zone when viewing the information in TrackWise and while running reports. This information is stored in the database as GMT and it is the responsibility of the report designer to apply a formula of this type for each date time field displayed on the report.

4. Click **Functions - Additional Functions- gmt (u2lgmt.dll)**

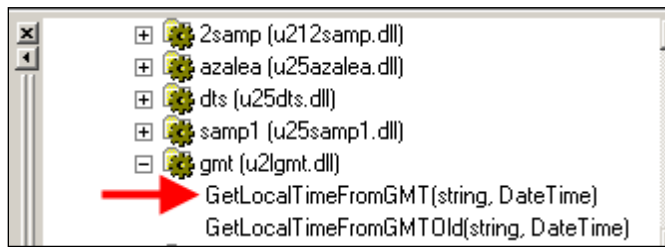


Figure 7-4 Expand gmt under Additional Functions.

5. Double-click **GetLocalTimeFromGMT(String, Date/Time)**function.

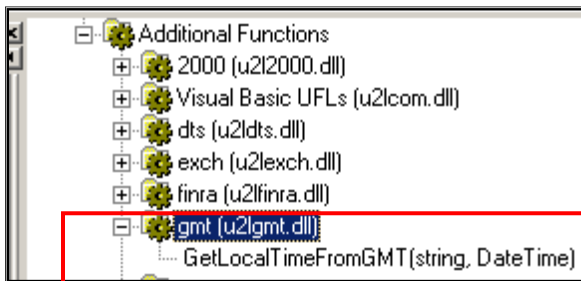


Figure 7-5 The GetLocalTimeFromGMT function is located under the gmt section.

6. The text appears on the main formula design screen. Place the cursor between the parentheses and to the left of the comma.

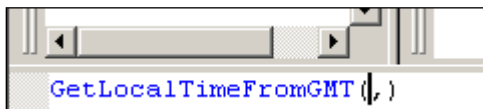


Figure 7-6 Example of formula added to main formula design screen.

7. From the **Report Fields** tree, double click the **SPTimeZone** parameter.



Figure 7-7 Add the SPTimeZone paramter to the formula.

8. It appears between the parentheses and to the left of the comma. This is the parameter to ask the time zone question of the user.

```
GetLocalTimeFromGMT({?SPTimeZone})
```

Figure 7-8 Placement of the SPTimeZone paramter in the formula.

9. Place the cursor to the right of the comma. At the top of the screen, open **introCrystalsdn** to see the list of tables.
10. Locate the PR table. Find the column called **date_opened**. Double-click **date_opened** for it to appear to the right of the comma within the parenthesis.

```
GetLocalTimeFromGMT({?SPTimeZone},{PR.date_opened})
```

Figure 7-9 Add the date/time field to convert to the formula.

11. Check for errors and then click **Save and Close**. Return to design mode. Drag **Date Opened** into the details section of the report. Provide a label in the page header.

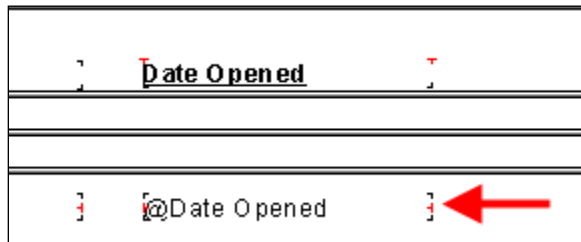


Figure 7-10 Example of Date Opened formula added to details section.

12. Save the report and copy it into the TrackWiseWeb TWReports folder. Run the report from TrackWise and compare the date opened on the report to the actual date opened in the record itself.

Similar formulas have to be defined for every Date Time field to be displayed on the report. Other examples from the PR table include the following.

Table / Column name	Description
PR.date_created	Date / Time field column for Date Created .
PR.date_closed	Date / Time field column for Date Closed .

PR.date_current_state	Date / Time field column for Since. This appears on the top right side of the form under the label In State for longer than one year.
-----------------------	--

7.3 Swapping First Name and Last Name

The Assigned To field displays in the format of Last Name, First Name on the report. There may be a requirement to show the first name followed by the last name. In this case, the formula consists of a few functions. The comma is used as a delimiter to separate the name. Find the location of the comma in the string, find the total number of characters in the string and then how many characters is the first name and last name.

This formula is used for **System Person** fields such as **Assigned To, Originator, Performed By** and **Posted By**. User Defined Person Fields show a person's name correctly (First Name, Last Name) when using views, a topic discussed in a future chapter and therefore this formula is not required.

Create a new formula, with a meaningful name that describes which person field is being converted, such as **Assigned To**.



1. In the formula, enter the following:

```
right({Person_Relation.name},length({Person_Relation.name}) -
instr({Person_Relation.name},"",-1) & " " & left({Person_Relation.name},
instr({Person_Relation.name},"",-1)
```

```
right({Person_Relation.name},length({Person_Relation.name}) -
instr({Person_Relation.name},"",-1) & " " & left({Person_Relation.name},
instr({Person_Relation.name},"",-1)
```

Figure 7-11 Formula to display first name followed by the last name.

- Without person_relation.name, the formula appears as the following. The parentheses would contain a different person field based on table and column name.

**Right ({Name}, Length({Name }) - instr({Name },",") - 1) & " " & left({Name },
instr({Name },",")-1)**

- There are a few functions used in this formula.
 - Right (str, length)** - Extracts the given number of text characters from the right side of the specified string. Example: Right("abcdefg", 3) returns "efg".
 - Length (str)** - Length returns the number of characters in a text string. Example: Length("Account")returns 7.
 - InStr (str1, str2)** - The InStr function returns the position of the first occurrence of one string within another. This position is a 1 based index of the characters in str1. If str2 is not found in str1, the InStr function returns 0. Example: InStr("abcdefg", "bcd") returns 2.
 - Left (str, length)** - Extracts the specified number of characters from the left side of the string. Left("abcdefg", 4) returns "abcd".

7.4 Calculating Number of Days Open

Another formula that is used often is a formula that calculates the number of days open per record. If the record is closed, subtract Date Opened from Date Closed. If the record is still opened, subtract Date Opened from the current date and time.

Another important aspect is that Date Opened is GMT and current date is not. Therefore, you must perform the GMT conversion before applying the calculation.



Workbook Exercises

- Create another formula called **Number of Days Opened**.

2. Enter the following formula.

**if {PR.is_closed} = 1 then date({PR.date_closed}) - date({PR.date_opened}) else
CurrentDate - date ({@Date Opened})**

3. Check for type errors. When finished, click **Save**. Drag the formula into the report and create a new column header for it.

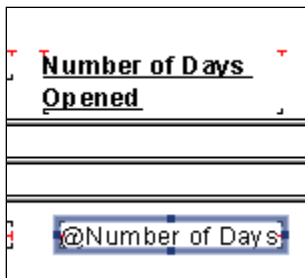


Figure 7-12 Formula added to details section.

4. Save the report. Run the report and notice the number of days opened vary based on open records in the report.

7.5 Conditional Display

In the previous section, we calculated and displayed the number of days open for each record. A more visual alert for late records is possible in Crystal Reports. For example, the TrackWise record ID can appear in red and bold each time a record is open for more than 90 days.



1. Open the report from the previous step, and go to **Design** mode.
2. Right-click the **PR.ID** field, Select **Format Field** in the menu and then the **Font** tab.

3. Click **X+2** to correspond with the **Color** field to open the **Formula Editor** window.

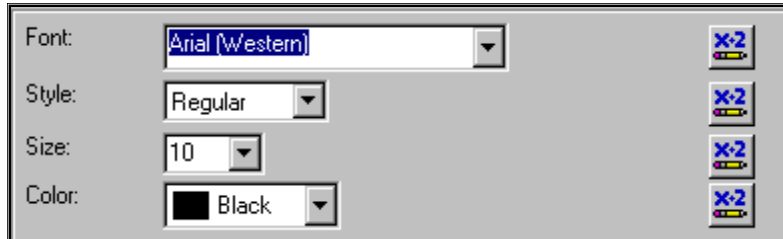


Figure 7-13 Font tab from the Format Field

4. In the formula text section, enter the following:

if {@Number Days Open} > 90 then crRed else crBlack
5. One formula can be used into another formula. Save the formula to return to the **Font Attributes** screen.
6. Copy the report into the TrackWise folder and run the report. Observe how the **PR.ID** field becomes red when records are opened for more than 90 days.

7.6 Formula Based Break Group

In a previous chapter, it was explained how to insert a Break Group. The break group was based on a database field such as Division, Project, and Status etc. While this is most common, it is also possible to base a break group on the results of a formula. In this exercise a new break group is added based on the formula that calculates the number of days open.



Workbook Exercises

1. In **Design** mode, select the **Group** option from the **Insert** menu. This opens the **Insert Group** window.

2. In addition to the database fields, notice all the formulas currently defined in the report. Select **Number of Days Opened** formula and click **OK**.



Figure 7-14 Example of formula as an option for Insert Group.

3. Notice the new group available on the report. Indent the group name to the right to make the report more readable, and delete from the report the field that shows the number of days open for each record.
4. Save the report and run it from TrackWise and observe how the records are grouped by the number of days opened.

PHb	<u>Project</u>	<u>PR#</u>	<u>Assigned To</u>	<u>Date Opened</u>
	Opened			
GH1	88.00			
GH2				
GH3				
D	Complaints	1,153	Dillon, Corey	10/22/2010 8:35:00PM

Figure 7-15 Preview of records grouped by number of days opened formula.

5. In TrackWise, notice that the format for **Number of Days Opened** in the group tree and on the report both contain two zeroes after the decimal.

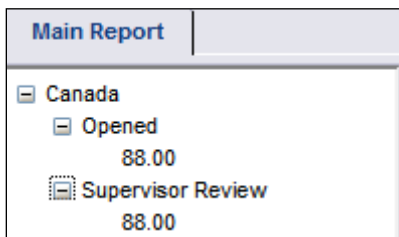


Figure 7-16 Group tree with the number of days opened formula.

6. To show an integer, as opposed to this default format, return to **Design** mode. Locate the break group based on the formula **Number of Days Opened**.



Figure 7-17 Location of Group Header # 3 break group.

7. Right-click the group and select **Change Group** from the list to open the **Break Group Definition** window.
8. In the **Group Options** section, select the **Options** tab and enable the **Customize Group Name Field** option and also select the **Use a Formula Group Name** option.

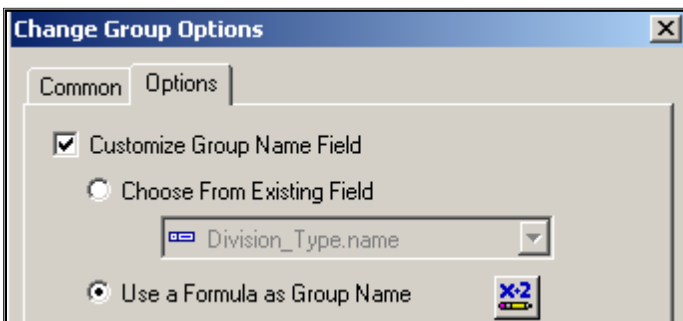


Figure 7-18 Options tab from the Change Group Options window.

9. Click **X+2** corresponding to the **Use a Formula as a Group Name** option to open the **Formula Editor**.
10. In the **Formula Editor** window, enter the following in the formula text:

totext ({@# Days Open}, '##')
11. The command **totext** converts date opened as a text value in order to remove the zeroes after the decimal. Save the report and run it again from TrackWise. Observe how the format changed into an integer.



Figure 7-19 Group tree displaying numbers formatted to integers.

7.7 Blank Record Values

Sometimes, a report runs against a record which does not display data because the field simply does not have the data filled in yet. For example, **Assigned To** (PR.Responsible_Rel_ID > Person_Relation.ID) could be filled out later in a workflow but at the time the report generated, this field was blank. The following formula allows a specific message or body of text to appear within a data field box if data is missing.

if isnull({@Assigned To}) then 'N/A' else {@Assigned To}



Create the formula to display 'N/A' where the Assigned To field is blank:

1. Create another formula called **Assigned To or Blank**. Use the syntax from the previous page to complete the formula.
2. When finished, replace the existing **Assigned To** field based on person_relation.name with this formula instead.
3. Save the report and run it in Crystal Reports. Notice that records that are unassigned have the letters N/A in them.

Complaint	1,081	N/A
-----------	-------	-----

Figure 7-20 N/A displayed on report.

Add the Originator Field to the report:

1. Refer back first to **Database – Database Expert**. Click **Person_Relation** under Selected Tables and press **F2**. Give this table the alias **Assigned_To**.

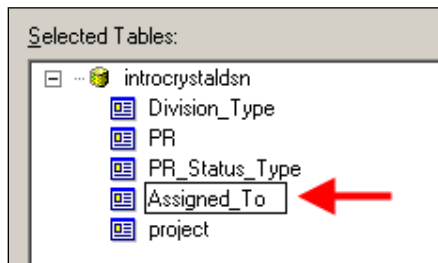


Figure 7-21 Person_Relation table renamed to Assigned_To.

2. Add **Person_Relation** to **Selected Tables**. Highlight **Person_Relation** and click **F2**. Rename the table **Originator**.

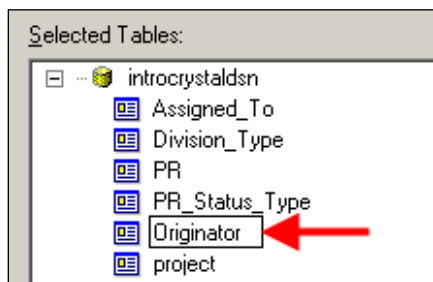


Figure 7-22 Second Person_Relation table renamed to Originator.

3. Click the **Links** tab. Link the PR table to Originator using the following equal join link.
PR.originator_rel_id to **Originator.id**

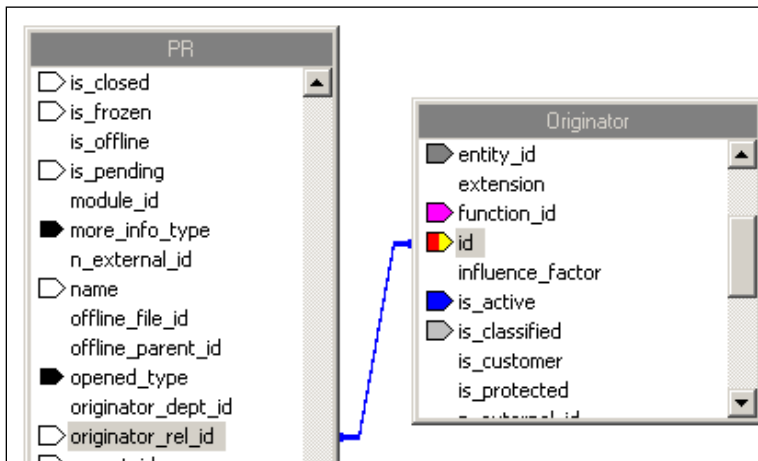


Figure 7-23 Link between PR and Originator tables.

4. Refer back to the Design screen. Add the **Originator.name** column to the report within **Details** and provide a page header label.
5. Save the report and run it. Notice the name of the Originator appears on the report. A name conversion formula is required to show first name last name as with the Assigned To field.

Conclusion:

Formulas provide the chance to dynamically change the appearance of the report based on returned data. They are also used to alter the appearance of fields such as Assigned To and convert date/time fields such as Date Opened. Features not available out of the box from Crystal are attainable based on particular types of formulas. Formulas are also used to determine the amount of time a record has been opened for based on when it was closed or if it still opened.



8. Headers, Footers, Typical GxP Compliance Concerns

8.1 Definition and Explanation

When generating reports in a GMP environment, there are numerous required components to meet compliance requirements. The following presents all the necessary components.

Each report has a Page Header and a Page Footer (not be confused with Report Header and Report Footer). Information on the Report Header shows on the first page only while information on the Page Header shows on the top of each page. Similarly, information on the Report footer shows on the last page only, while information on the Page footer shows on the bottom of each page.

8.2 Page Number

Open the report used in the previous section and save it as **report6.rpt**.



Workbook Exercises

1. While in design mode, open **Field Explorer** and expand the **Special Fields** folder.

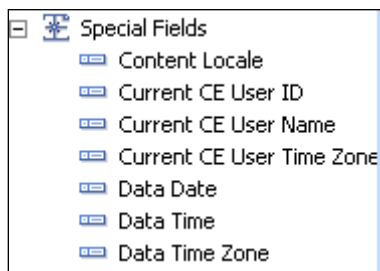


Figure 8-1 Expanded Special Fields folder under the Field Explorer.

2. Drag-and-drop **Page N of M** field to the top of the **Page Footer** section.
3. Save and preview the report. Notice that sequential page numbers at the bottom of each page.

8.3 Print Date vs. Data Date

An important component in the report is the date and time when the report was executed. However, there is common confusion between two sets of Crystal Reports Special Fields. The first component is Print Date and Print Time. This component reflects the date and time that the report is sent to print.

Data Date and Data Time reflects the exact date and time that the information was queried from the database. The second method is more accurate and therefore recommended to use with compliant reports.



Workbook Exercises

1. Refer to report **header F**. Add the word **on** after **SPUser**.
2. Drag **Data Date** to report header F.
3. Add the word **at** after data date.
4. Drag **Data Time** to report header F.
5. Save the report and run it in TrackWise. Notice the information is automatically filled out with the date and time information.

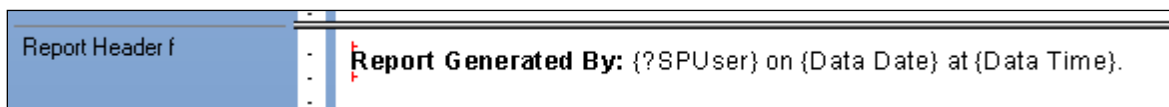


Figure 8-2 Special Fields added to the Report Header.

8.4 Time Zone of the User who Generated the Report

Through the use of a formula, you can also print the time zone of the user who ran the report. Parameter SPTimeZone is used to ask TrackWise which time zone the user logged in as.

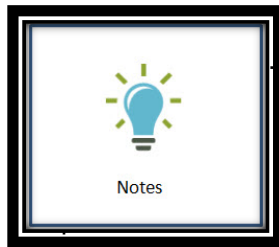


Workbook Exercises

1. Create the formula **Local Time Zone**. Enter the following information.
**if {?SPTimeZone}="America/New_York" then "Eastern Standard Time" else
if {?SPTimeZone}="America/Chicago" then "Central Standard Time" else
if {?SPTimeZone}="America/Denver" then "Mountain Standard Time" else
if {?SPTimeZone}="America/Los_Angeles" then "Pacific Standard Time"**
2. Check the formula and then save it. Drag the formula into report header F. Run the report in TrackWise.

Conclusion:

Simple but often overlooked add-ons to the report make a big difference in terms of appearance and functionality. Whether it is numbering pages, or presenting the date and time the data was queried from the database, typical compliance to a standard such as these is essential.



2. Right-click **Details** and select **Suppress (No Drill-Down)**. Details are not required in this example.

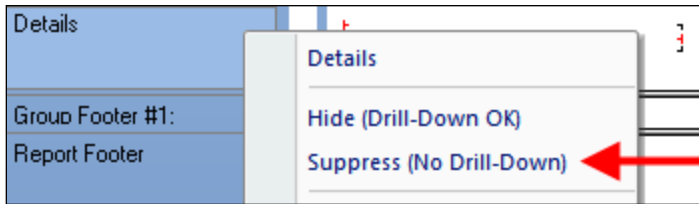


Figure 9-2 Location of Suppress (No Drill-Down) option.

3. To insert a new graph, click **Insert – Chart**. Place the outlined chart onto the report within the Division Group Footer. The **Chart Expert** window opens.
4. Select the **Type** tab. From the **Chart type** list, select the **Pie** option and select **Use depth effect**.

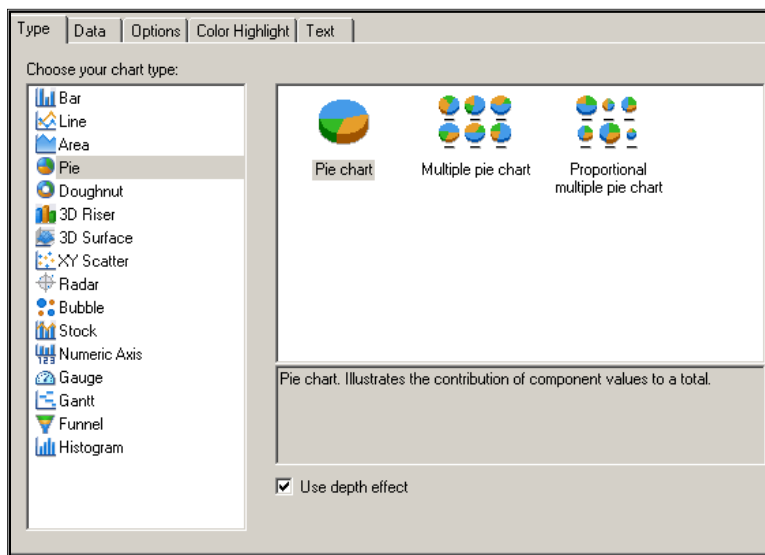


Figure 9-3 Example of Type tab with pie chart and use depth effect selected.

5. On the **Data** tab, select the **Assigned To** formula from the **Available fields**. Click the right arrow to move this into the **On change of** box. This box addresses what kind of information is to appear as a pie slice. Other examples would include product names, record status, priority and others.
6. Select **PR.ID** from the **Available fields** and click the lower, right arrow to move this into the **Show value(s)** box. With the **Sum of PR.ID** field highlighted, click **Set Summary Operation**. This setting is used to count the number of returned records.

7. Select **Count** from the Set **Summary Operation** field and click **OK** when complete. With these settings, a pie slice is drawn for every assignee indicated within the returned records.

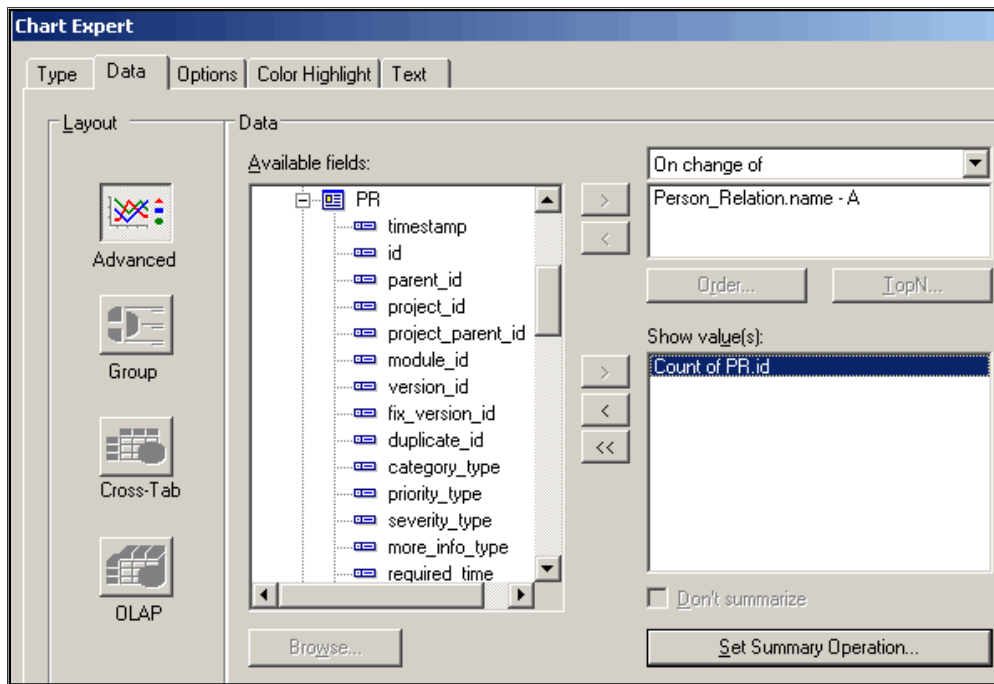


Figure 9-4 Summary Operation set to Count.

8. Click **OK**. Place the chart in the **Group Footer** section. Resize this object and preview.

9.3 Changing Chart Attributes

To adjust the chart attributes, right-click the newly created chart and select **Chart Expert**. Click the **Options** tab. On this screen the appearance and functionality of the chart is adjustable.



Workbook Exercises

1. In the **Data Points** section, select the **Show Value** and **Show Label** option. For the legend, set **Layout** to **Amount**.
2. Click the **Text** tab. Deselect **Auto-Text** for the **Title** field. Remove the current text and enter **Assigned To Distribution**.

3. Click **OK** when complete.
4. Right-click the chart again and select **Format Chart**.
5. In the **Format Editor** window, select **Border**. Enable the **Drop Shadow** option.

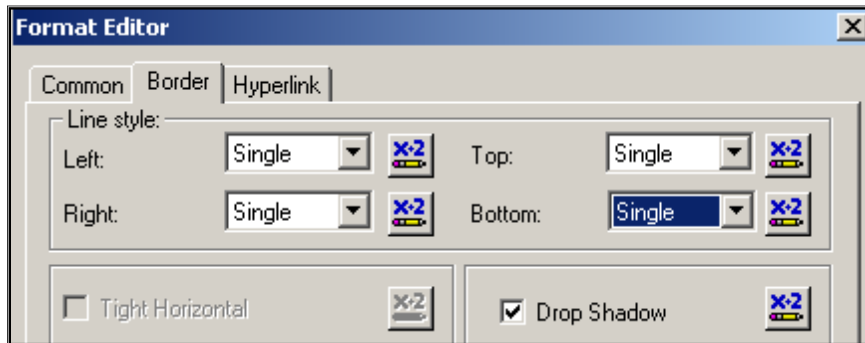


Figure 9-5 Border tab with Drop Shadow selected.

6. In the **Line Style** section, select **Single** for the Left, Right, Top and Bottom fields. Click **OK** when complete.
7. Each pie slice represents the number of records each assignee has. A pie chart can contain a pie slice for product types, statuses or priorities based on the variety of available values.

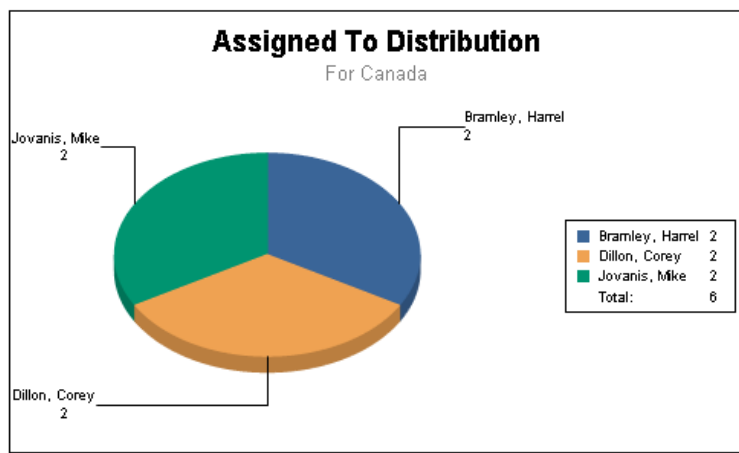


Figure 9-6 Example of pie chart representing the number of records per assignee.

9.4 Inserting a Bar Graph

The next graph displays the distribution of **Status** across **Projects**.



1. Open the report from the previous section. Save it as **report8.rpt**.
2. Delete the pie chart. Change the **Break Group** to group on **Project.name** instead of **Division_Type.name** by right-clicking the break group and selecting **Change Group**.
3. Select **Chart** from the **Insert** menu. Place the outline of the chart into the **Project.name Group Footer**.
4. The **Chart Expert** opens. Ensure **Bar** is selected, and then click the **Data** tab.
5. Select the **PR_STATUS_TYPE.NAME** field, and click the upper, right arrow to move this into the **On change of** section.
6. Select the **PR.ID** field and click the lower, right arrow to move this into the **Show values** section. Change **Sum of PR.ID** to be a count instead.
7. Click the **Options** tab. From the **Data Points** section, select **Show Value**.
8. Click the **Text** tab. Deselect the **Auto-Text** option for the **Title** field. Remove the current text, and type in **Status Distribution**.
9. Click the **OK** button when complete.

For any chart added to a report, right-click and choose **Auto Arrange Chart**. Crystal formats the chart to display the information with minimum overlapping.

9.5 Descending Bar Graph

In a standard bar chart, the data series are arranged in alphabetical order by default. One could change this to show the data series in ascending, descending or custom order.



Workbook Exercises

1. Right-click the graph and select **Chart Expert**.
2. Click the **Data** tab.
3. Highlight the **PR_STATUS_TYPE.NAME**, and click **TopN**. This opens the **Group Sort Expert**.

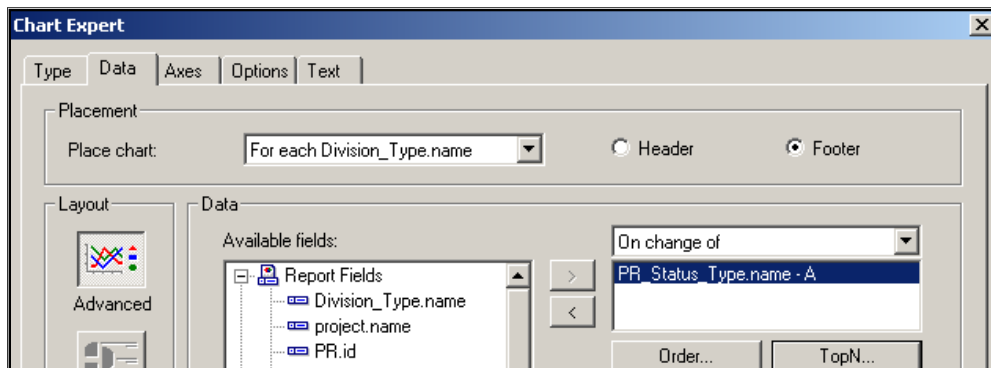


Figure 9-7

4. Select **All** from the **Group Sort** list. This highlights the **Count of PR.ID** in the **Based on** list. Select the **Descending** option at the bottom of this window.
5. Click the **OK** button when complete and exit the **Chart Expert** settings window. Preview the report.

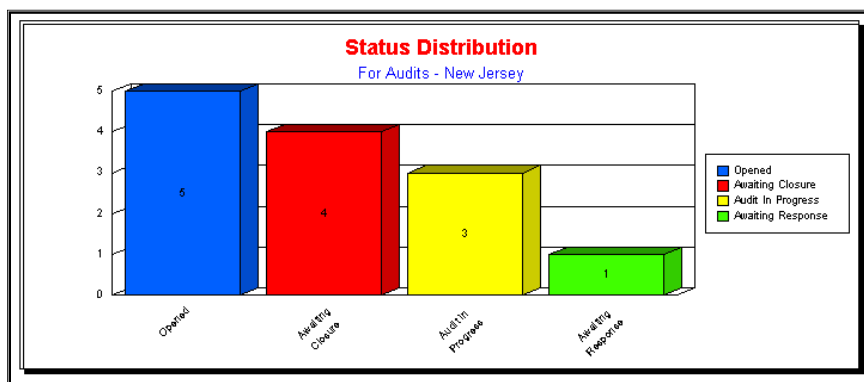


Figure 9-8 Descending bar chart.

9.6 Formula Based Graph

Each chart previously explained used database fields. This is the most common, however, you can use a formula as a basis of a chart. The next chart is an area type chart showing the number of records per days open.



1. Open the report used in the previous section. Remove the graph and save it as **report9.rpt**.
2. Select **Chart** from the **Insert** menu. Place the chart in the Report Footer. The **Chart Expert** opens.
3. From the **Chart Type**, select **Area** and select **Stacked area chart**.

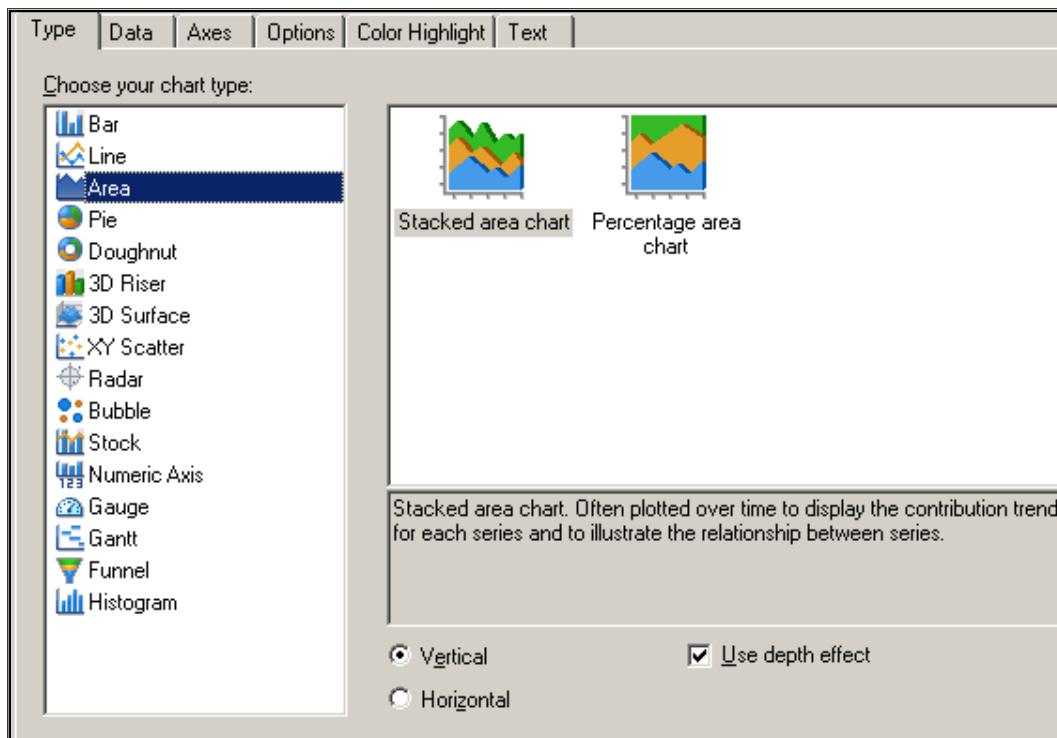


Figure 9-9 Stacked area chart type selected.

4. Click the **Data** tab. From **Available Fields**, select the **Number of Days Open** formula and click the upper right-arrow to move this into the **On Change of** box. This formula counted the number of days that were or are open.
5. From **Available Fields**, select the **PR.ID** field and click the lower, right-arrow to move this into the **Show values** box. Change the **Sum of PR.ID** to be a count.
6. Click **OK** when complete. Resize this object and preview.
7. Click the **Options** tab. Select **show value** and deselect **show legend**. Click **OK** and then **Preview**.

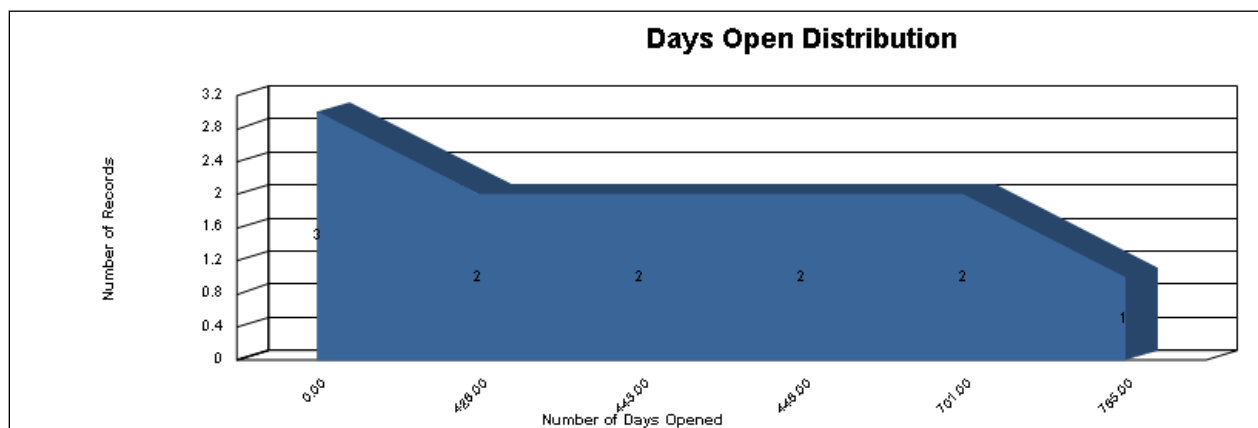


Figure 9-10 Example of stacked area chart.

9.7 Stacked Bar Graph

A stacked bar graph displays data across three axis and stacks the data in a single bar. The next report displays a count of records per Project, however within each project (data series), it is further broken down by **Assigned To** person. This chart is placed on the Division footer.



Workbook Exercises

1. Open the report from the previous section and save it as **report10.rpt**. Delete the current chart. Change the break group from **Project.name** back to **Division_Type.name**.

2. Select **Chart** from the **Insert** menu. Place the chart in the Division Group Footer on the report. The **Chart Expert** opens.
3. From **Chart Type**, select **Bar** and then select **Stacked bar chart**.

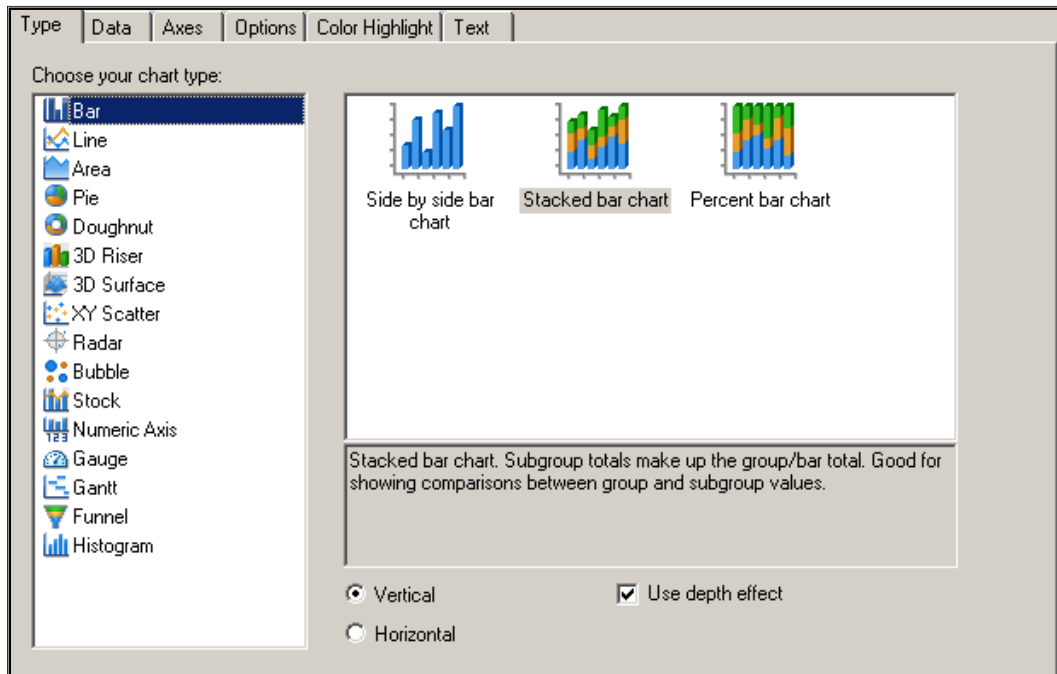


Figure 9-11 Stacked bar chart selected.

4. Click the **Data** tab. From **Available Fields**, select **PROJECT.NAME** and click the upper, right arrow to move this into the **On Change of** box. In addition, select **@Assigned To or Blank** formula.

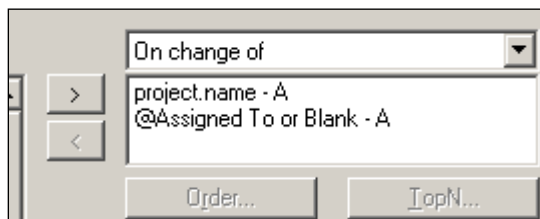


Figure 9-12

5. From **Available Fields**, select the **PR.ID** field and click the bottom, right arrow to move this into the **Show Values** box.
6. Click **OK** when complete.

7. Finally, right-click the chart for **Chart Options**. Click the **Data Labels** tab and select **Absolute Value** so each segmented section on the bar is no longer cumulative. Save the report and click **Preview**.

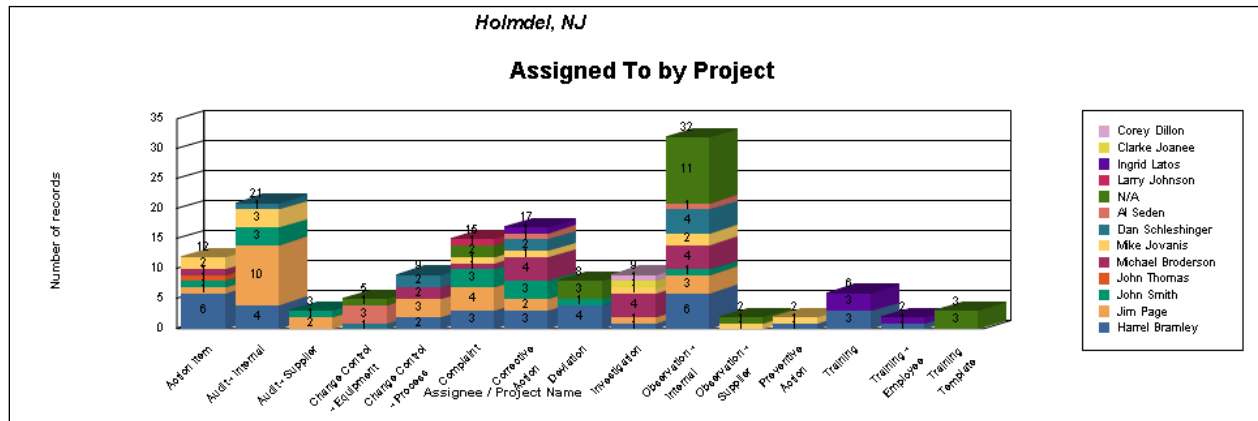


Figure 9-13 Example of stacked bar chart.

On Your Own Exercises

1. Open the report saved in the last section and remove the existing chart. Save the report as **report11.rpt**.
2. Create a bar graph in the **Division.Name** group footer of the report that shows the number of records created for each Division by month. Use the **Date Opened** formula for the chart's **On Change Of** option.
3. Click **Order**. Select **For Each Month** for the option to print a bar for each month records were opened.
4. Add the necessary titles to the report and any background, shadow and borders.
5. At the end in the report footer, add a **Pie Chart** that shows the number of records for each Division.
6. Run the report from TrackWise using different queries and different scopes to observe the results.

Conclusion:

Charts and graphs represent a visual approach for data returned from TrackWise. A pie chart shows the distribution of records amongst many assignees while a bar graph shows the number of records in each state. The information is based on the break groups which arrange the records in a particular order. In addition to the charts and graphs, columns of detailed record data can be included as well, to provide a high level and detailed listing of information.



10. Cross Tabs

10.1 Definition and Explanation

Cross tabs allow displaying information or calculation in a matrix format. The cross tab functionality in Crystal allows the report designer to create, in a short period of time, clear and comprehensive reports. In addition, cross tabs reports can be used as a basis for charts and graphs.

10.2 Defining a Cross Tab

Open the report saved in the previous chapter, remove all the charts and save it as **report12.rpt**.



1. In design mode, locate **Cross-Tab** and Drag it to the **Division Group Footer**. Afterwards, right-click **Cross Tab** and select **Cross-Tab Expert**.
2. From **Available Fields**, add **Project.name** into the **Columns** window.
3. From **Available Fields**, add **PR.ID** field into the **Summarized Fields** box.
4. From **Summarized Fields**, highlight the **Sum of PR.ID** field and click **Change Summary**. Select **Count** from the **Summary field calculates** option.

5. From **Available Fields**, add **Person_Relation.name** or if available, the **Assigned To** or **Blank** formula into the **Rows** field.

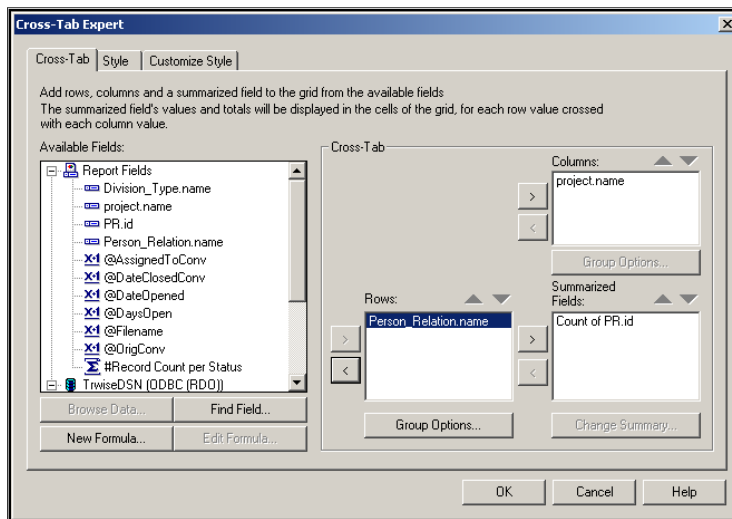


Figure 10-1 Cross Tab Expert with Column, Row and Summarized Field selections.

6. Click **OK** and preview to view the report.
7. Refer back to the **Cross Tab Expert**. Under the **Customized Style** tab, select **Repeat Row Labels**.

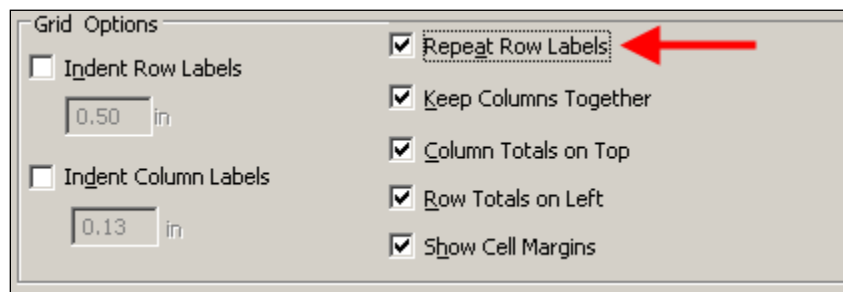


Figure 10-2 Customized Style tab.

8. Change the field alignment in order to give the report a better look. Copy the report into the TrackWise Web folder and run it from TrackWise to view the results.

10.3 Multi Axis Cross Tab

More than one row against one column can be displayed. It is possible to have multi rows break down against the same column.



Workbook Exercises

1. Open the report from the previous section. Remove the cross tab and save it as **report13.rpt**.
2. In Database Expert, add the **TW_V_PRIORITY** view to the report. Link the PR.ID column to the view's TW_V_PRIORITY.PR_ID column as a left outer join. Views are covered extensively in the Single Report chapter.
3. Select **Cross-Tab** from the **Insert** menu. Place the **Cross Tab** into the **Report Footer**. In the following example, the priority values are displayed on a per Site (Division) basis.
4. From **Available Fields**, add **TW_V_PRIORITY.S_VALUE** field into **Columns**.
5. From **Available Fields**, add **Division_Type.name** and **Project.name** into **Rows**.
6. From the **Available Fields**, add **PR.ID** into the **Summarized Fields** box. Change the **Sum of PR.ID** to a count.

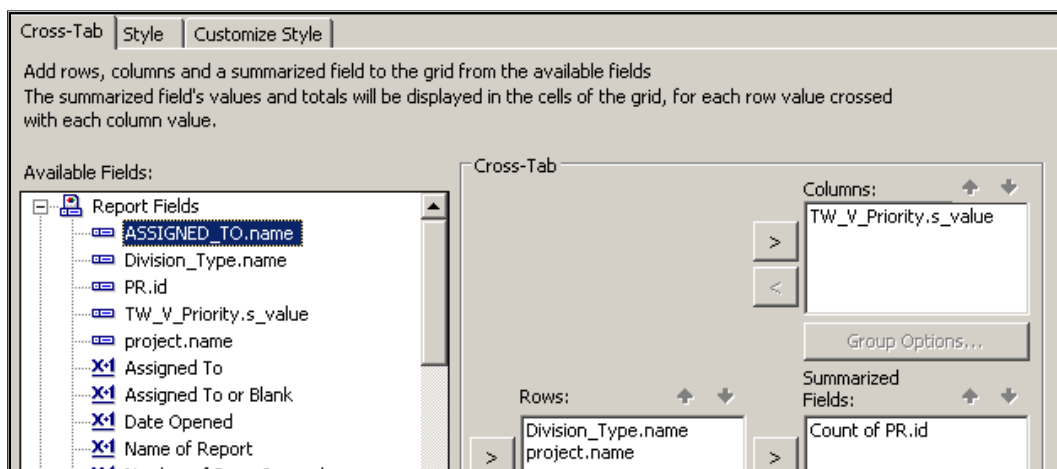


Figure 10-3 Cross Tab expert with values added to Columns, Rows and Summarized Fields.

7. Click **OK**. The cross-tab should appear in the report footer.
8. Click **Preview** to view the results. Resize the fields to make sure that fields are not truncated.
9. Return to **Design** mode. Right-click the cross tab and select **Cross-tab Expert** from the list. Click **Customize Style**.
10. Select **Grand Total** in **Columns** and select **Silver** from the **Background** color field.
11. Select the **Grand Total** field in the **Rows** section and select **Silver** from the **Background** color field. Click **OK** when complete.
12. In **Design** mode, select the **Row #1 Name** inside the cross-tab object and click the Control **B** button on the formatting tool bar to make it bold. Select the **Row #1 Name** inside the cross-tab, and click the **I** button to make it italic.
13. Select the **Grand Total** fields and increase the font size to **14**.
14. Save the settings and run the report from TrackWise to view the results.

10.4 Formula Based Cross Tab

Formulas can also be used to design a cross tab which provides a whole new dimension to the cross tab design.



1. Open the report from the previous section. Remove the cross tab and save it as **Report14.rpt**.
2. In Design mode, drag a **Cross-Tab** to the **Division Group Footer**.
3. Insert the **Project.name** in the **Rows** field and the **Number of Days Open** formula, in the **Columns** field. Set the **Summary Field** to **Count of PR.ID**.
4. Click **OK** and place the cross tab in the Division footer. Save and preview the report.

10.5 Cross Tab Based Chart

Cross-Tabs can be used as a basis for charts and graphs. Both charts can appear on the report together in order to show the information in two different styles. Open the report from the last section and save it as report **report15.rpt**. Leave the cross tab on the report.



1. Open the report in Design mode. Right-click the previously created cross tab and select **Insert Chart**.
2. A bar graph appears. To modify the graph, right-click the chart and click **Chart Expert**.

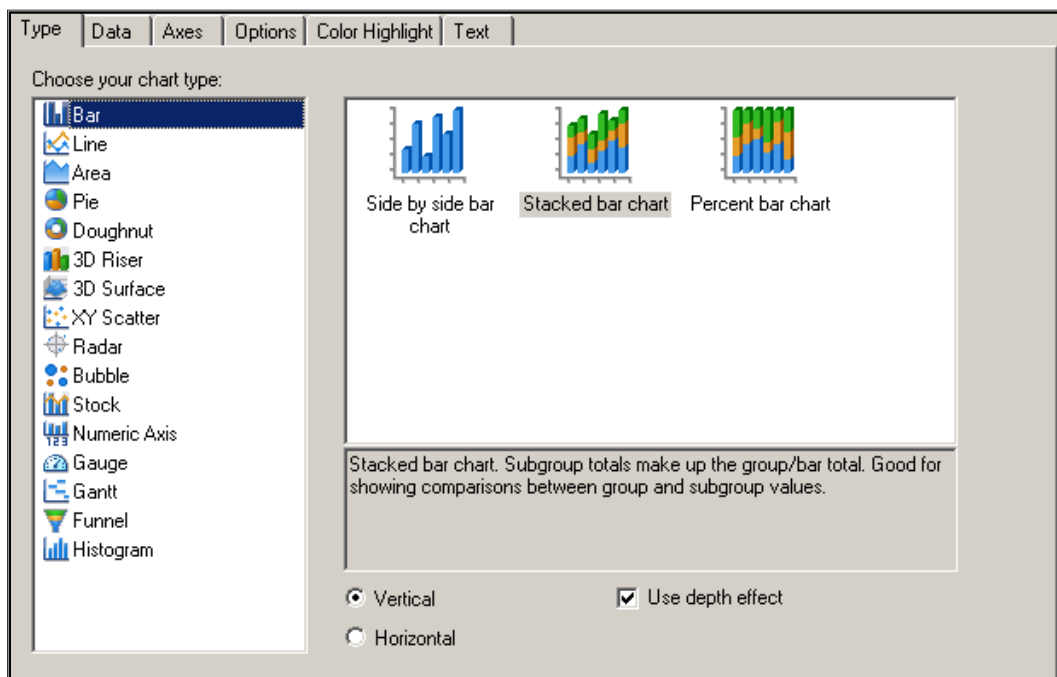


Figure 10-4 Stacked bar chart selected.

3. Click the **Data** tab. In the **Data** section, verify **Project.name** is set for the Subdivided by field.
4. Click **OK**. Verify that the chart version of the report is located in the report footer above the crosstab. Preview the report to view the results.

5. After running the report, adjust the chart properties for the text titles. Copy the report to TrackWise and test it.



1. Open the report saved in the last section and save the report as **report16.rpt**.
2. Remove the crosstab and chart from the previous exercise.
3. Insert a cross-tab in the report footer that shows **TW_V_PRIORITY.S_VALUE (column)** and **Division_Type.name (row)**.
4. Change the default cross tab attributes to make sure all fields fit and not truncated.
5. Change both **Grand Totals** (Column and Row) to have a silver background.
6. Insert a chart based on this cross tab. The chart should be a stacked bar chart showing a bar for each **Division_Type.name** stacked by **TW_V_PRIORITY.S_VALUE**.
7. Run the report in TrackWise using different queries and scopes. Observe the results.

Conclusion:

Cross tabs use rows and columns to meet with a calculation to represent the two intersecting values. This provides a numeric outcome based on the information provided. Examples provided here included assigned to name with project, to provide information representing the number of records the user has in that project.

Notes:

11. Select Expert

11.1 Definition and Explanation

Whenever a user runs a report in TrackWise, the report displays the records from the user's current desktop; based on the user's current scope and current query. However, there are scenarios in which the report requires an additional level of record filtering. Using the Select Expert, Crystal Reports allows the report designer to define which records to be further included or excluded from the report. Some examples include limiting the number of returned records to only be created 30 days ago or sooner. Another example would be to only show records due in the next ten days and exclude all other records due 25 days from now.

Whenever a Select Expert rule is in place, make sure to add a text box disclaimer indicating that the report is designed to show specific types of records.

11.2 Using the Select Expert with a Formula.

Open **report16.rpt** Click **File-Save As** and name it **report17.rpt**.



1. Open the report in Design mode. Select **Report - Select Expert – Record** from the **Report** menu. This opens the **Choose Field** window.

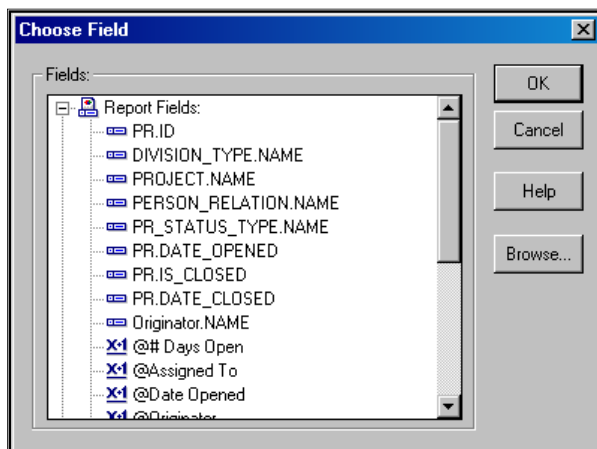


Figure 11-1 Select Expert-Record field selection window.

2. From the **Report Fields** folder, select **Number of Days Opened**. Select **is less than** with the value of **80**.

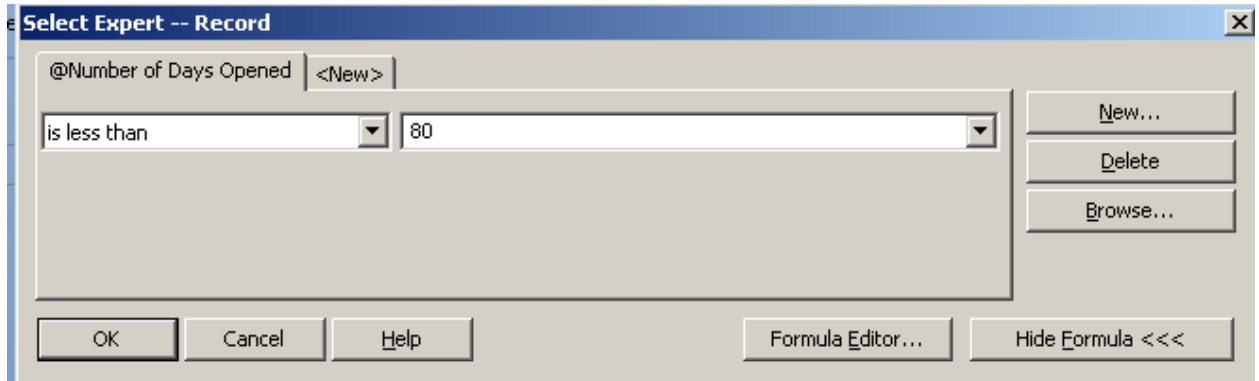


Figure 11-2 Select Expert with Number of Days Opened formula.

3. Click **OK** and run the report. Notice that only records opened for 80 days or less appear on the report. This limits the number of returned records regardless of how many records a user would fetch.
4. Copy the report to TrackWise. Use a Query and Scope to bring all records back and notice only a few records appear in the report.

11.3 Select Expert with Selection Values.

In the next example Priority is used to filter for high priority records only.



1. Open the report from the previous section, remove the existing select expert and save it as **report18.rpt**.
2. Open the report in Design mode and select **Report - Select Expert – Record**.

3. From the **Report Fields** folder, select **TW_V_PRIORITY.s_value**. Set the filter to **is equal to** with the value of **High**. Click **OK**.

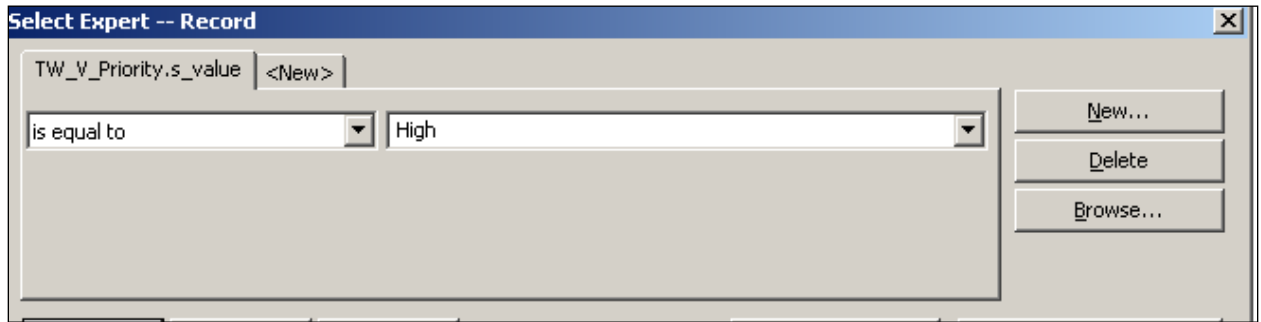
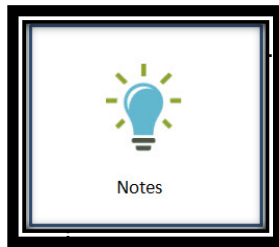


Figure 11-3 Select Expert with Priority view.

4. Run the report in TrackWise. Only high priority records appear in the report regardless of Query and Scope's initial returned records. The select expert has the final say as to which records returned print within the report.

Conclusion:

The select expert provides an internal filter within the report to specifically include or exclude records. Although query and scope in TrackWise populates the report with records, select expert has the final say as to what appears. A disclaimer on the report is recommended to inform users of an embedded filter in use.



12. Multi-Report Review

12.1 Definition and Explanation

Based on the previous chapters, which incorporated multi-report design, refer to the following exercise to build a report based on customer requirements.



On Your Own Exercises

1. Take **Report 18.rpt** and click **File – Save As** to name it **Desktop Report.rpt**
2. Remove all of the report headers except for the default one. Remove the contents inside.
3. Remove all page headers except for the default one. Clear out the contents inside.
4. Leave one group header to be edited but delete the others.
5. Clear out all of the data fields within details. Delete all select expert filters.
6. Remove sorting options from the record sort expert.
7. Ensure that there are no additional group footers or more than one report footer. Clear out the contents inside of report footer.
8. When finished, the report appears as it did in the beginning.
9. Using the existing tables, formulas and parameters, design a report based on the following requirements.
 - The report header indicates the scope and query used in TrackWise for the report.
 - The report header indicates the name of the report and the person who ran the report in TrackWise.
 - The report is grouped with the formula, Assigned To or Blank, to provide a tree structure named after each assignee.

10. The details and page header of the report contains the following.

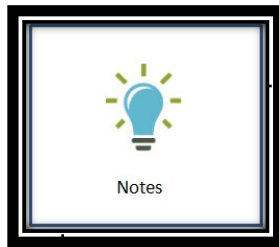
Page Header	PR#	Short Desc.	State	Date Opened	Date Due	Number of Days Opened
Details	PR.id	PR.name	PR_Status_Type.name	Date Opened Formula	PR.date_to_close	Number of Days Opened Formula

11. In the group footer, add a running total to count the total number of records for each person.

12. In the report footer, add a bar graph to display the total number of records each user is assigned to.

13. Provide formatting options, such as a line to separate each row of returned records.
When finished, save the report and run it in TrackWise.

Conclusion: Based on the previous chapters of multi-report design, this exercise enables the Crystal designer to incorporate settings and features by creating a new report. Although the field requirements are mentioned, often the users need to provide the exact field types to use for report design. This includes which grouping options to use and any applicable charts and running totals.



13. Single Record Report

13.1 Definition and Explanation

Single Reports are intended to be as detailed as possible, with all of the data fields in a single record. All Projects have a single report in TrackWise because this provides a way to pull data from one record, any accessible record, from the system and print this information into a report. The following workbook exercise instructs the report designer on how to find the fields in the database and the best way to get them to generate data onto the report.

One of the first steps is to create a new record in TrackWise in order to fill out the fields. This is to see the work in progress with all of the fields populated with information. Screenshots are recommended in order to follow a layout for the fields to be added to the report.

Single reports use one of two possible techniques to properly display data onto the report. These are based on using views or direct table linking.

13.2 Getting Started

By taking a recent report and saving it as a template, it can be used to generate other reports with. A template is a report with commonly used tables and formulas already in place as a springboard for report design.

1. With the last report worked upon, remove all break group headers, footers, any additional details and page header/footer sections.
 - a. Tip: You can select all items on a section by right clicking in the section area on the left and click **Select All Section Objects**.

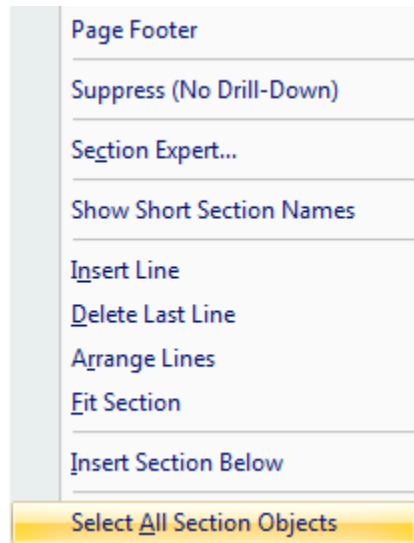


Figure 13-1 Location of the Select All Section Objects option.

2. Get the report to look like how it did when first logged in to Crystal Reports. Change the report back to portrait mode and ensure that the select expert does not have anything included.
3. When finished, save this report as **template.rpt** Use this report to create any other report with, either multi-reports or single reports.
4. Take the **template.rpt** report and select **File-Save As**. Name the report **SingleReport1.rpt** Insert a grouping based on **PR.ID** since only one record returns with this single report. The reason for the break group is to incorporate one of the two possible techniques to pinpoint data fields onto the report. This is called direct table linking and is addressed shortly.

5. Keep Details suppressed. In a single report, it prints rows of extra space if not utilized. Right-click **Details** and select **Suppress (No Drill Down)**. Details can be used to display record data if only using views to display data. This is addressed shortly.

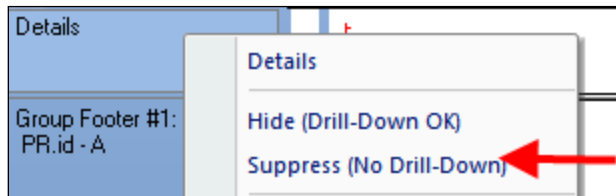


Figure 13-2 Suppress (No Drill-Down) option.

6. The report template appears as the following.

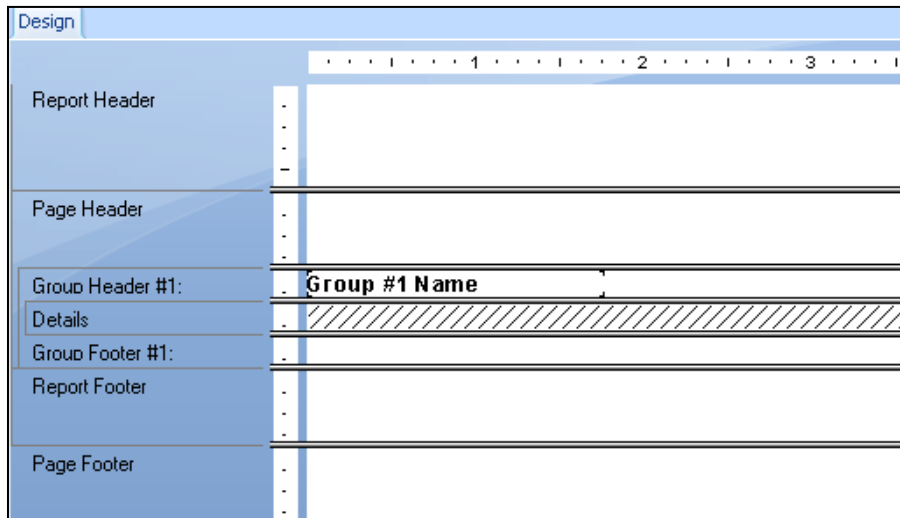


Figure 13-3 Example of report template.

7. In this example, Group Header#1 is not required since the data fields go into the Group Footer. The reason is because one of two techniques used to get the fields to display in the report relies on running totals to direct table link. Since running totals only work in footers, all data fields are added here so they match the pattern and arrangement of fields.

The Report Header includes the title of the report as seen in the following example. Use the text object option to include the following title.

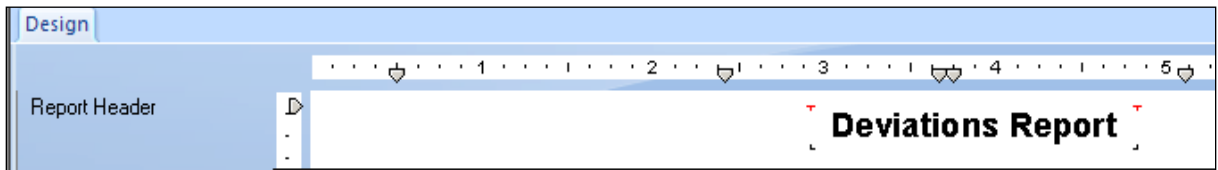


Figure 13-4 Example of title added to the Report Header.

8. The page header replicates what is seen within the details of the record. Refer to the following example in TrackWise, PR#1156 for the record to base the report on.

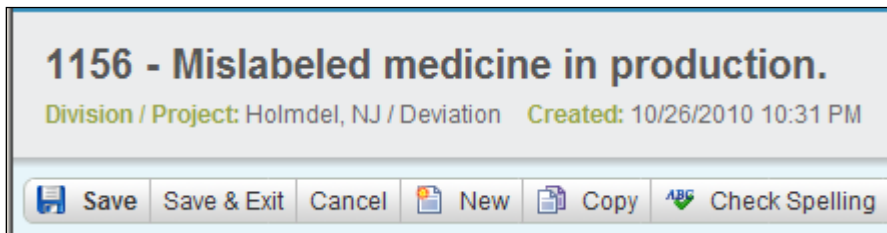


Figure 13-5 Header section of PR# 1156 in TrackWise.



In TrackWise, create a new record and complete the following fields that are currently blank but require data in order to ensure the report is designed correctly. **Save** when finished.

- Related Documents
- Date Due

- Original Date Due
- Quality Contact
- Assigned To
- Building
- Floor
- Location Type
- Equipment ID
- Equipment Description
- Timeliness Justification

Set **Select Expert** to filter PR.ID to equal the newly created record for reporting purposes.

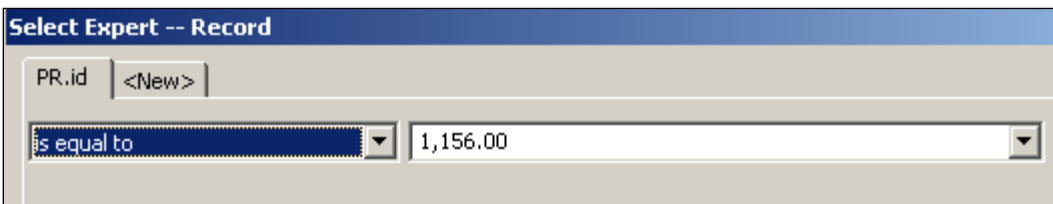


Figure 13-6 Example of Select Expert based on PR.id.

In the page header of the report, add the following fields. This is consistent information that appears on the top of every page of the report.

- **PR ID :** Use PR.ID . Change the format of ID to leave out commas and decimals.
- **Short Desc:** Use PR.Name to display this information.

Report Header	D	Deviations Report									
Page Header	D	id	-	name	+						

Figure 13-7 PR ID and Short Desc added to the Page Header.

Right-click **Page Header** and select **Insert Section Below**. Add another Page Header so short description does not run into the next set of fields.

In Page Header b, add the following fields.

- **Division / Project:** Use Division_Type.name / Project.name
- **Date Created:** Use a formula to convert PR.Date_Created based on the same way Date Opened was converted in a previous chapter.

GetLocalTimeFromGMT({?SPTimeZone},{PR.date_created})

The name of the current state and the number of days in the state are included on top of the screen. Add the next two additional fields in Page Header b.

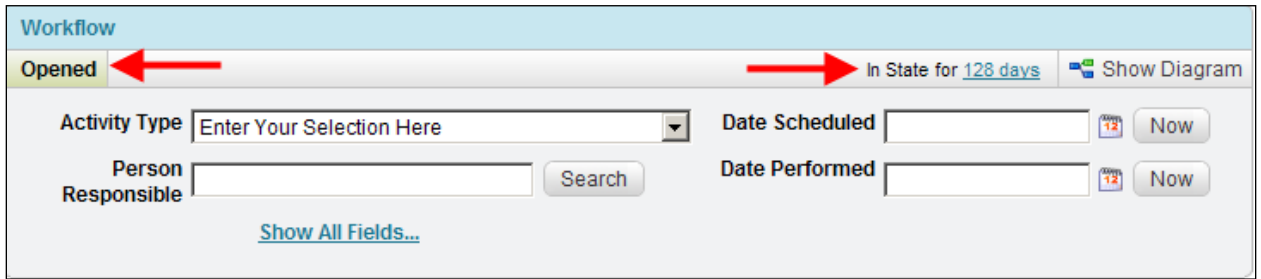


Figure 13-8 Location of current state and number of days in state in the PR record details.

- **PR State :** Use PR_STATUS_TYPE.NAME
- **In State for:** As with the formula for Date Opened, use a formula to convert PR.Date_Current_State to indicate the exact date and time the record entered the current state.

GetLocalTimeFromGMT({?SPTimeZone},{PR.date_current_state})

Click **Save**. Run the report and refer to the following screen capture, based on the fields added.

RH PHa PHb	Deviations Report	
	1156 - Mislabeled medicine in production.	
	Division / Project: Holmdel, NJ / Deviation	Date Created: October 26, 2010 10:31 pm
	PR State: Opened	In State Since: October 26, 2010 10:31 pm

Figure 13-9 Preview of report with header detail added.

Draw a line under the page header. Back in Design mode, click **Insert – Line**. With the mouse, left-click the starting point for the line and hold the mouse button down to draw the line across the whole page header section. Right-click the line and select **Format Line**. Select **Move to Bottom of Section when Printing**. Click **OK**.

Run the report in Crystal. The line appears below the page header.

RH	Deviations Report	
PHa	1156 - Mislabeled medicine in production.	
PHb	Division / Project: Holmdel, NJ / Deviation	Date Created: October 26, 2010 10:31 pm
RF	PR State: Opened	In State Since: October 26, 2010 10:31 pm

Figure 13-10 Line added under the page header using Insert – Line.

13.3 Using Views for Data Fields

There are two methods to use when adding data fields into a report. Both techniques are presented here to choose the one preferred.

Before adding data fields, first determine which fields came with TrackWise and which ones were created for the system by Administrators. Fields included with TrackWise are called System Fields. Typical examples of these kinds of fields are Assigned To, Date Due, Short Description and Originator.

Fields created for TrackWise are called User Defined Fields. By taking a look at the records, it is not obvious as to which fields are which. Run a Data Fields report in Web Administrator which is a way of knowing which fields are system-based or user defined.

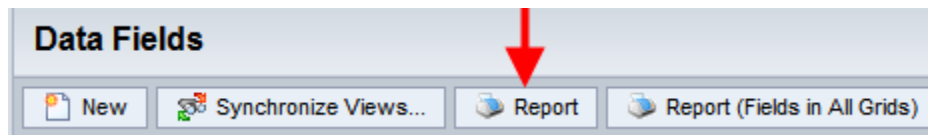


Figure 13-11 Location of Data Fields Report button in Web Administrator.

The column with the view names refer to user defined fields.

<u>Field Name - Customized</u> ▲	<u>Associated With Initial Group</u>	<u>Field Category</u>	<u>Field Type</u>	<u>View For Reports</u>
<u>Close Event Only By</u>	Yes	PR (User Defined)	Person	TW_V_CLOSE_EVENT_ONLY_BY
<u>Close Event Only On</u>	Yes	PR (User Defined)	Date/Time	TW_V_CLOSE_EVENT_ONLY_ON
<u>Close Requirements</u>	Yes	PR (User Defined)	Single Selection	TW_V_CLOSE_REQUIREMENTS
<u>Closed By</u>	Yes	PR (User Defined)	Person	TW_V_CLOSED_BY
<u>Closed On</u>	Yes	PR (User Defined)	Date/Time	TW_V_CLOSED_ON
<u>Combination Product</u>	Yes	PR (User Defined)	Single Selection	TW_V_COMBINATION_PRODUCT
<u>Comments</u>	Yes	PR (FlexField)	FlexField	TW_V_COMMENTS

User Defined Fields (UDF) can be added to a report in two ways. Here is an overview of each technique followed by the steps to add the fields first with views.

Views are a quick and easy way to bring these types of fields over to a report. They are mini queries that are searching for data that a record would have in order to properly display it. They do require additional work later on when moving the report from Development to Production by making sure they each point to a new data source. Crystal provides a way to switch out each object to a new data source. If this step is not implemented, then each view and table on the report is modified one at a time to point to the intended data source.

If a field is multi-based (Ex. Multi-text, Multi-Selection, File Attachment) a Crystal functionality called Subreport is used. Otherwise, refer to the following example on how to add a field this way.



The Deviation configuration form contains the following fields under General Information.

General Information	Impact Analysis	Investigation & Root Cause	Effectiveness Planning	Reference Info / Comments	E-Signatures	Admin Only (Hidden Fields)
General Information						
Originator John Smith Info		Date Opened 10/26/2010 10:31 PM Now				
Short Desc.  Mislabeled medicine in production.						
Related Documents C:\Documents and Settings\Administrator\My Documents\resume.txt View Remove Browse		Original Date Due is set upon initial quality review.				
Date Due 03/31/2011 		Original Date Due 03/23/2011 Now				
Supervisor/Manager Jane Doe Info		Quality Contact Harrel Bramley Info				
Assign the Investigator for the Deviation record.						
Assigned To Ingrid Latos Info						

Figure 13-12 Example of Deviation record details.

- Right-click **Group Header#1** and select **Suppress (No Drill Down)**. In the Group Footer, add the following system fields through the use of existing formulas and tables. Use a text box to create a label for each of these fields.

- Originator
- Date Opened
- Short Desc.

Group Header #1:	.	Group #1 Name	.
Details	.		.
Group Footer #1: PR.id - A	.	Originator: {@Originator} Date Opened: {@Date Opened} Short Desc. {name}	.

Figure 13-13 Fields added to Group Footer #1.

- The following fields are user defined by administrators. One of the two techniques to display the fields onto the report is through views. A view is a mini-query searching the

database to see if the record has a value implemented. Click **Database – Database Expert**. Expand the database name, schema owner and Views list.

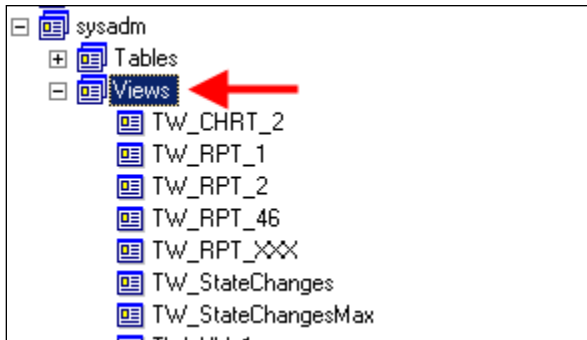


Figure 13-14 Expanded Views folder from the Database Expert.

3. All views start with the letters TW_V followed by the field name. There is a character limit when displaying the view name through the database expert. Scroll through the list and double-click the following views to use in the report.

- TW_V_Priority
- TW_V_Date_Occurred
- TW_V_Original_Date_Due
- TW_V_Type
- TW_V_Building
- TW_V_Floor
- TW_V_Location_Type
- TW_V_Supervisor_Manager
- TW_V_Quality_Contact

4. Each view appears on the right side of the screen under **Selected Tables**.

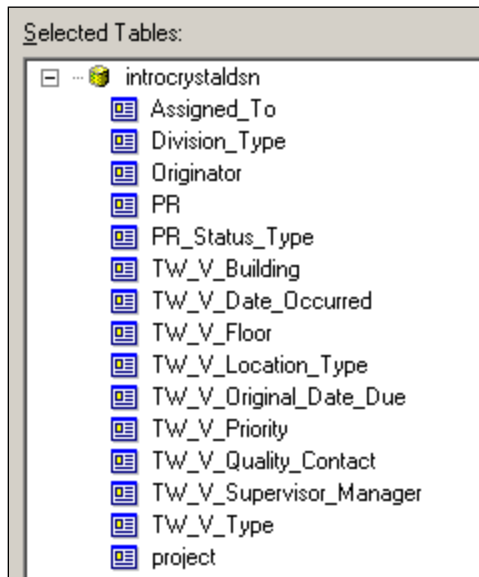


Figure 13-15 Views added to the Selected Tables panel.

5. Click **Links**. Each view links in the same way. Start from the PR table and left-click the ID column. Drag **id** to the view's **pr_id** column.

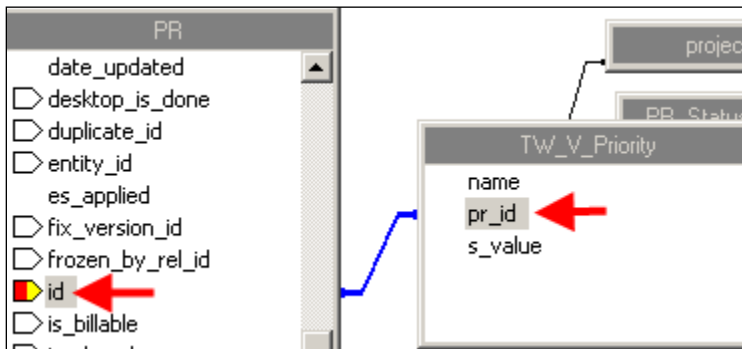


Figure 13-16 Links between PR and Priority View tables.

6. Double-click the link between the two tables. Set the **Join Type** to **Left Outer Join**. This ensures that if a record is not discovered to have a priority value, the other fields display. An incorrect link prevents the rest of the available fields to show up in the record.

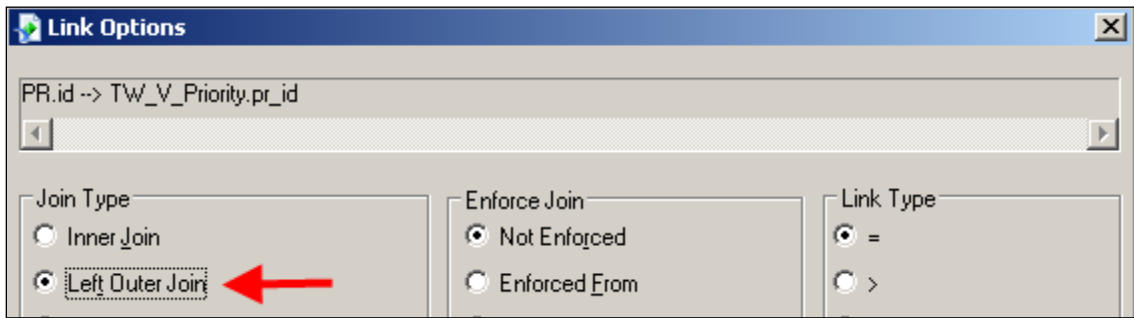


Figure 13-17 Left Outer Join type selected on the Link Options window.

7. Click **OK**. Link the rest of the views in the same way, ensuring each link is set to left outer join. When finished click **OK** to return to design mode.
8. For the rest of the general information section, add a label as a place holder for the file pointer field Related Documents. The field is addressed in the next chapter for multi-fields. Right-click **Group Footer#1** to select **Insert Section Below**. A second group footer is created to allow related documents to grow if necessary without overlapping the fields underneath.

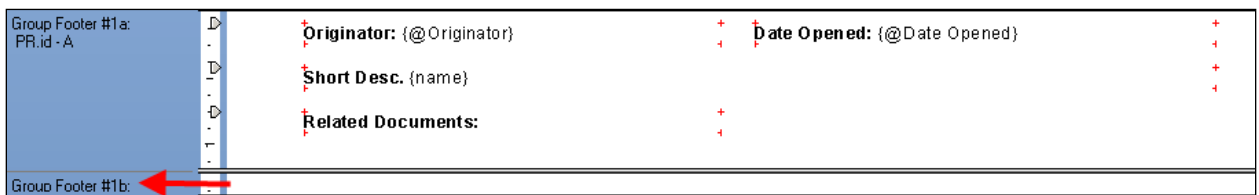


Figure 13-18 Use the Insert Section Below option from Group Footer #1b.

9. In group footer #1b add Date Due by referring to the **PR.date_to_close** column. Set the format for **date_to_close** to not include the time. Only the date appears in TrackWise. Embed the object into the text box.

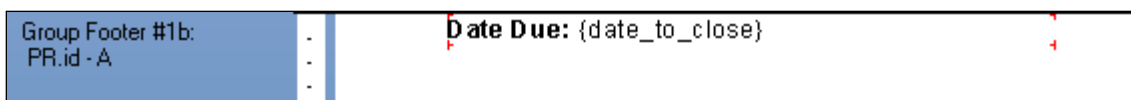


Figure 13-19 Date Due added to design using the PR.date_to_close column.

10. The first user defined field on the report, Original Date Due, is to be added to the right side of Date Due. Locate **TW_V_Original_Date_Due** under Field Explorer. Views contain columns of information as do tables. To identify which column contains the data, right-

click each and select **Browse Data**. Right-click **date_value** to browse and notice the list of dates.

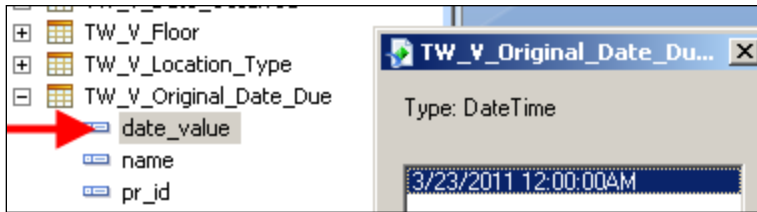


Figure 13-20 Example of value when selecting Browse Data.

11. Place **date_value** into group footer #1b. Right-click **date_value** to change the format to not include the time.



Figure 13-21 Location of Format Field option.

12. Create a new text box with the name of the field. Place **date_value** into the text box.

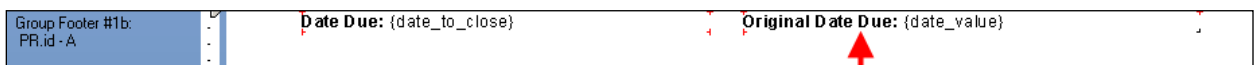


Figure 13-22 Example of date embedded into text object.

13. Save and run the report to confirm the date is here.

14. The next field, **Supervisor/Manager**, is a person field. The column within the view to use is **s_value**. Browse the column to confirm.

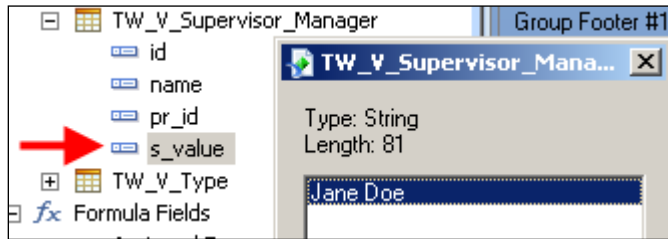


Figure 13-23 Example of value when selecting Browse Data.

15. Create a text box with the name of the field. Place **s_value** into the text box.

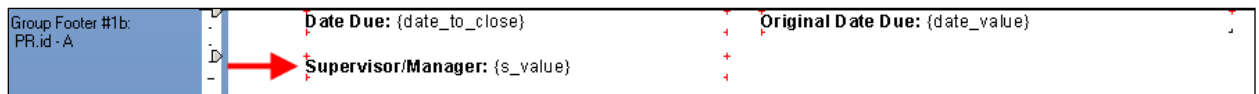


Figure 13-24 s_value embedded in text object.

16. Add **Quality Contact** by adding **s_value** from the view into a text box with the name of the field.

17. Add **Assigned To** based on the existing formula, Assigned To or Blank.

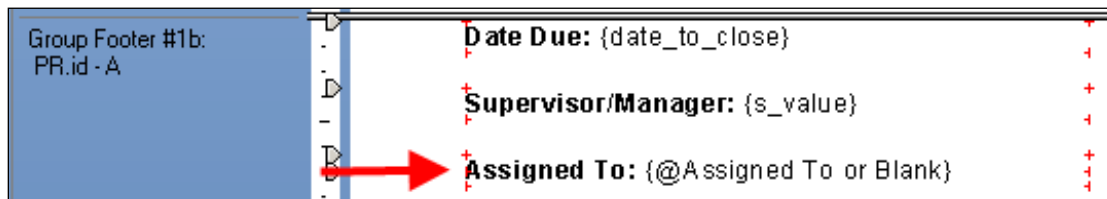


Figure 13-25 Assigned To formula embedded into text object.

18. The following chart displays columns and the associated values they contain.

View column name	Field type
date_value	Date field values.
date_time_value	Date-time value. Use the formula to convert to local time.
dec_value	Decimal values.
s_value	Strings, file attachments and selection values.



1. Add a section name with a text box above originator in group footer 1a with the words **General Information**.
2. Add another section name with a text box under Assigned To in group footer 1c called **Deviation Information**.
3. Add the following fields based on the available views under Field Explorer. Include a label for **Department** on the right side, across from Location Type.
 - Priority
 - Date Occurred (Use the GMT time conversion formula with the date_time column)
 - Type
 - Building
 - Floor
 - Location Type

Deviation Information			
Priority: {s_value}	↑		
Date Occured: {@Date Occured}	↑	Type: {s_value}	↑
Building: {s_value}	↑	Floor: {s_value}	↑
Location Type: {s_value}	↑	Department:	↑

Figure 13-26 Example of design tab with information from Deviation Information section.

4. Save the report and run it. Observe the current values in the report are available.

13.4 Direct Table Linking

Another technique to display user defined fields is direct table linking. It provides a way to use the source tables where this data is from, in order to properly display the information. Although it takes an additional step or two to do, the end result is that the fields are directly connected to the tables in the database, where this information is. It provides less conversion steps later on when redirecting the report to a new database by only referring to five tables instead of the number of views if using the latter method.

Running Totals is used to directly link to the database tables where the information is. It is a three step process that is outlined in this next section.



To start, add the following data tables in field explorer which are used to directly connect to the data tables with. Each table is linked with a left outer join.

1. **PR_addtl_data**

This table contains the following :

- The lookup information to tell the PR table which field(s) are filled out in a record.

- The values for some of these fields such as strings, date, date-time, decimal and number.

It is linked from **PR.ID** to **PR_addtl_data.pr_id**

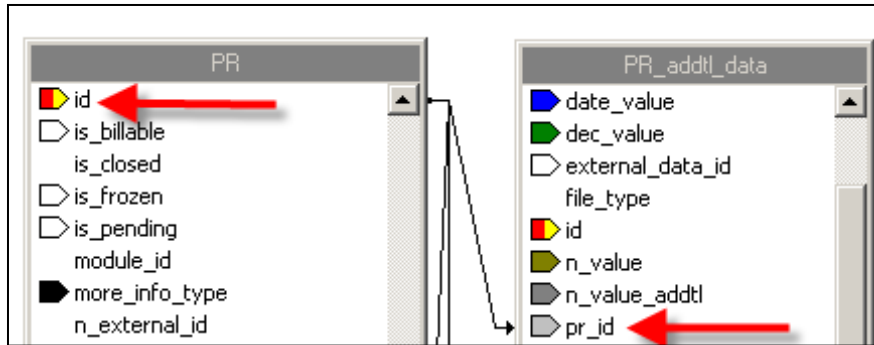


Figure 13-27 Link between PR and PR_addtl_data table.

2. Data_Fields

This table is the lookup for the names of the fields which records have. It is linked from PR_addtl_data.data_field_id where the ID numbers are for these field names to Data_Fields.id

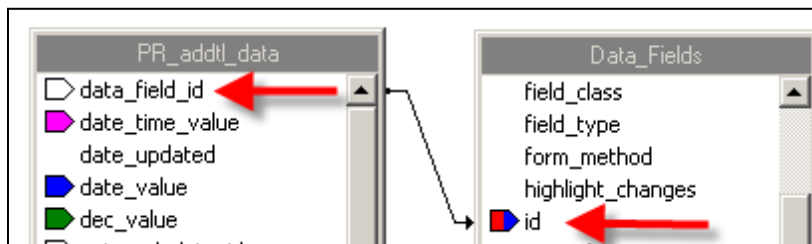


Figure 13-28 Link between PR_addtl_data and Data_Fields table.

- The next three tables contain additional field values which pr_addtl_data does not contain. They each link from pr_addtl_data.n_value to the respected table's ID column.

- **Entity**

This is the lookup table for the names of entities for UDF fields such as Entity and Multi-Entity.

- **Person_Relation**

This is the lookup table for the names of people for UDF fields such as Person and Multi-Person.

- **Addtl_Type**

This is the lookup table for Selection and Multi-Selection values.

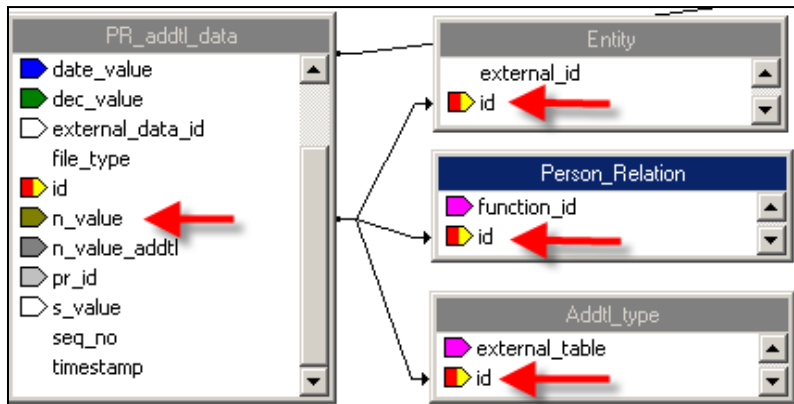


Figure 13-29 Link between PR_addtl_data and Entity, Person_Relation and Addtl_type tables.

4. Make each link left outer. When finished, click **OK**.
5. Insert another break group section for the following fields. Department is a multi-selection field and needs to be at the bottom of the break group.

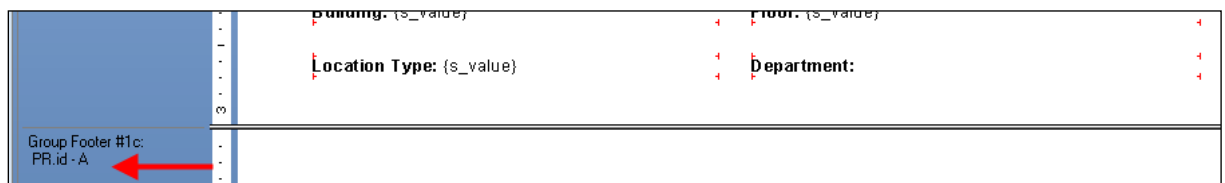


Figure 13-30 Additional section for break group #1 added.

6. The next field to add is the string field called Equipment ID. Refer to the running total option in Field Explorer and right-click. Click **New**.
7. Enter the **Running Total Name** of **Equipment ID**. The next field, Field to Summarize, is looking for the location of string values. All string values are stored in **PR_Addtl_Data.s_value**. Set **Type of Summary** to **maximum**.

Figure 13-31 Summary section for Equipment ID Running Total.

8. The next step is to use a formula to define the exact name of this field. Since it is stored in the **Data_Fields** table, refer to **Evaluate – Use a Formula** and click **X+2**. Use the following formula to pinpoint the exact name of the field. The **Data_Fields** table contains the name column.
9. Click this column name first. Add the **equal** symbol.
10. Right-click the name column and search for the field name **Equipment ID**. Click the field name. It appears in the formula workshop screen.

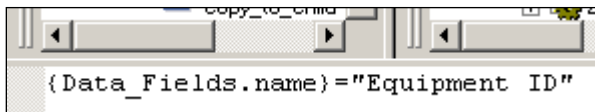


Figure 13-32 Formula for Evaluate section of Equipment ID running total.

11. Click **Save and Close**. The last step is universal. Always clear out the value when the next record comes along, or as referred to as **PR.ID**

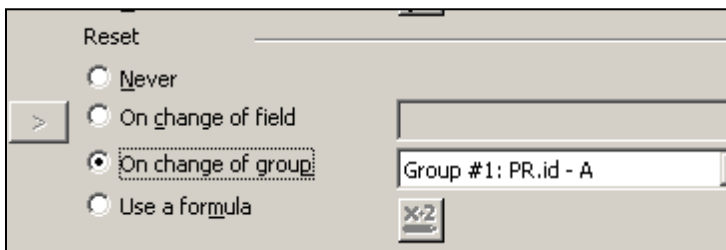


Figure 13-33 Reset section of Equipment ID running total.

12. Click **OK**. Add the running total to the Equipment ID text box. Save the report and preview it.



Figure 13-34 Equipment ID running total embedded into a text object on the report.

13. From the first step above to determine where the values are stored, here is a list of which columns in **PR_Addtl_Data** contain the values.

Column Name	Description
Date_Time_value	Contains date time values. Use a formula after creating the running total for these fields to do the date/time conversion.

Date_value	Date values are stored here. No date time conversion is necessary.
Dec_value	Decimal values are stored here.
S_value	Contains both string field values and file attachments. A formula is used to only show the name of the file for file attachments and not the whole originating path from where the file came from.

Additional Tables with field values:

Table and Column Name	Description
ADDTL_TYPE.NAME	This table and column contains single and multi-selection values.
PERSON_RELATION.NAME	This table and column contains the name of person values.
ENTITY.NAME	This table and column contains the name of entity values.



On Your Own Exercises

1. Add Equipment Description to the report using a running total for direct table linking. Selection values are stored within addtl_type.name.
2. Create a text box and add the running total within.
3. Run the report and refer to the selection value on the report.

Group Footer #1c: PR.id - A	-	Equipment ID: {#Equipment ID}	-	Equipment Description: {#Equipment Description}	-
--------------------------------	---	-------------------------------	---	---	---

Figure 13-35 Example of report with Equipment Description running total and text added.

13.5 Person Address

In addition to displaying the person fields in the report, the address of a person is able to be displayed. The next workbook exercise uses the person field **Quality Contact** and displays this person's contact information.



1. Click **File – Save As** and name the report **SingleReport Person Details.rpt** In Crystal, refer to **Database – Database Expert** and connect to **introcrdsn**. From the table list add **Address** and another copy of **Person_Relation**.



Figure 13-36 Location of Address table.

2. Click **Person_Relation** and press **F2** to rename the table as **Quality_Contact_ID**. The view for Quality Contact does not have the address_id number to send over to the Address table. The Person_Relation table contains this particular ID to use.

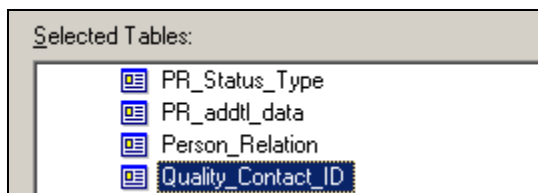


Figure 13-37 Rename the second Person_Relation table to Quality_Contact_ID.

3. Click **Links**. From **TW_V_QUALITY_CONTACT**, link id to **Quality_Contact_ID. id**. This link is the look up for the Quality Contact's person id in order to share it with the address

table. Set the link to left-outer join.

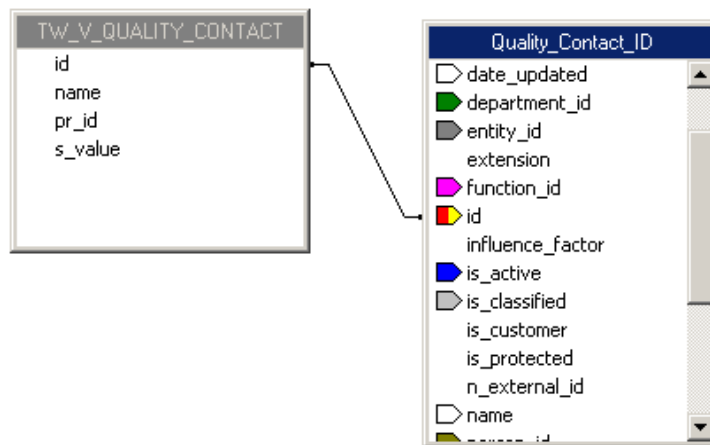


Figure 13-38 Link between the TW_V_Quality_Contact and Quality_Contact_ID table.

4. Continue linking from **Quality_Contact_ID.address_id** to **Address.id**. The id represents details about the contact's address, city, state and zip code. Set the link to left-outer.

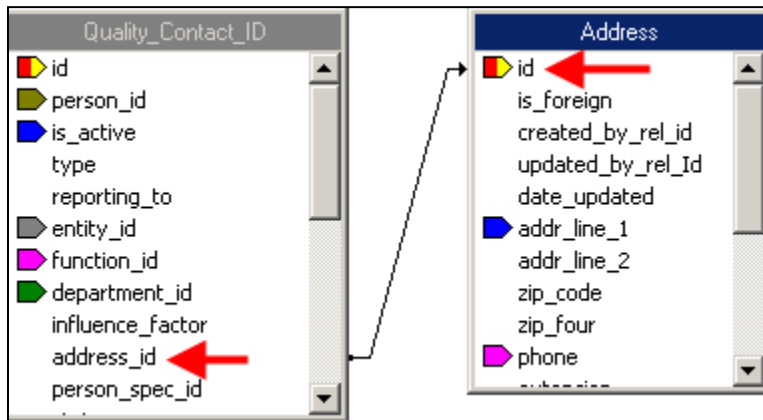


Figure 13-39 Link between the Quality_Contact_ID and Address table.

5. Click **OK** and return to the main report screen. Add the following columns of data from the Address table in Field Explorer to the report under the Quality Contact data field.
 - Addr_line_1
 - City

- State
- zip_code

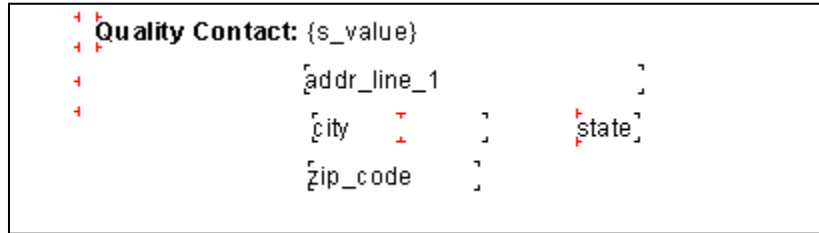


Figure 13-40 Example of report design with Address columns added under Quality Contact.

6. Additional information regarding the contact such as address line 2 (addr_line_2) and email is available for display. To summarize, this requires the view currently in the report, a copy of Person_Relation with an alias and the Address table.

Conclusion:

Single reports contain all of the data fields visible from records. The reports are designed to print one record at a time. The ability to extract the information is based on two techniques using views or direct table linking.

In the next chapter, subreports are a feature used for multi-based fields. This provides a way for specific field types to grow without duplicating report pages when printing.



14. Sub-Reports

14.1 Definition and Explanation

Subreports are stand-alone reports embedded into the main report and logically linked. The main report can have multiple sub-reports; however sub-reports cannot contain their own sub-reports. Subreports are used in TrackWise reports for the following reasons.

- Print multi-fields that include file attachments, multi-selection, multi-text, referenced record and file pointer. These field types make reports print duplicate multiple pages unless subreports are used.
- Print parent and child record data within a report.
- Print related records based on a shared value from the main report.

The following workbook exercise is to print department, a multi-selection field, within the report.

14.2 Displaying Multi-selection Fields with Sub Reports and Views

Open the report from the last chapter, and save it as **SingleReport2.rpt**.



1. In the Design mode, click **Insert - Subreport**.
2. Create the subreport with the report wizard. Name it **Department**.

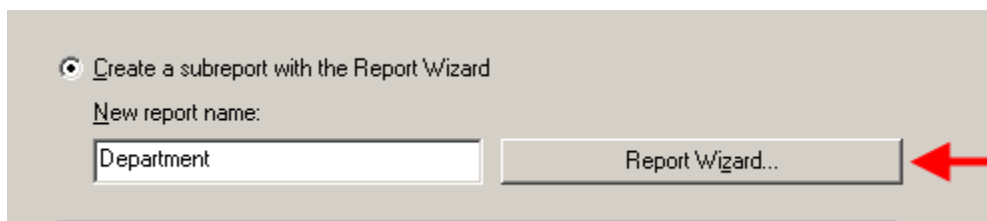


Figure 14-1 Report Wizard button to create a subreport.

3. Click **Report Wizard**. Select the view **TW_V_Department** from the introCrystaldsn data source.

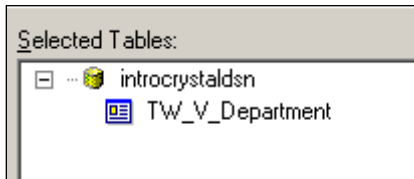


Figure 14-2 View for Department added to the Selected Tables panel.

4. Click **Next**. The same column used for views on the main report is used to display information in the subreport. Select **s_value** for **Fields to Display**. **Browse Data** is used to verify the correct column from the view is used.

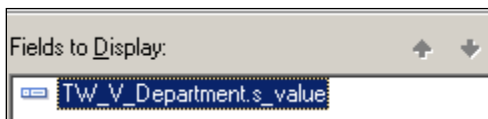


Figure 14-3 s_value column selected as field to display.

5. Click **Finish** and **OK**. On the main report, the cursor is attached to the subreport. Place the subreport under the department text box. Resize the subreport to fit underneath the box.



Figure 14-4 Subreport added under the Department text box and resized.

6. Remove the border by right-clicking on the subreport and click **Format Subreport** then **Border** tab.
7. In the **Line Style** section, select **None** for the Left, Right, Top and Bottom fields. Click **OK** when complete.

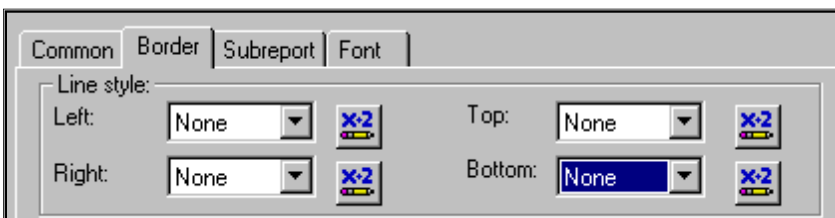


Figure 14-5 Select None to remove border from subreport.

- a. Tip: On the toolbar, you can find an **Outside Border** button. Click the arrow beside it and select the last option symbolizing no borders.

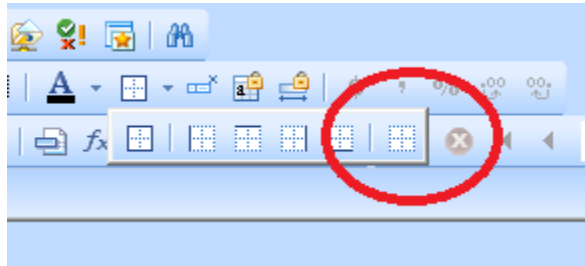


Figure 14-6 Location of Outside Border button on the toolbar.

8. The next step is to address the formatting of the subreport. To do this, click the **Department** tab at the top of the report.
9. On the sub-report screen, remove the **Print Date** and **S_Value** column headers if visible.

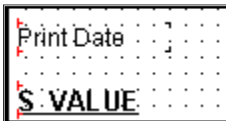


Figure 14-7 Print Date an S_Value column headers for removal.

10. When creating subreports for multi-value fields, to avoid extra space above and below the values in the main report, it is encouraged to suppress the Report Header and Report Footer sections.
11. Collapse or suppress report header and report footer. Leave **Details** visible for **S_VALUE** to display department values.

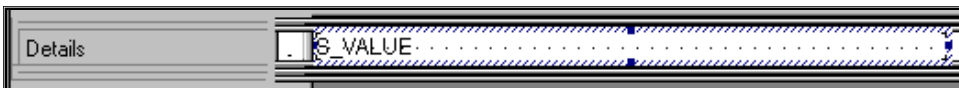


Figure 14-8 S_Value in Details section of subreport.

12. Save the subreport and return to the **Design** tab. The last step is the link between the main report and the subreport. Right-click the subreport and select **Change Subreport Links** from the list. This opens the **Subreport Links** window.
13. From **Available Fields**, select the **PR.ID** field and click the right arrow to move this field into the **Field(s) to link to** . As PR.ID was selected, a new section opened at the bottom.

14. Since the sub-report contains only one table and one compatible field to link to, Crystal Reports makes the correct assumption to link **PR.ID** to **TW_V_Department.PR_ID**.
15. Notice that the **Select data in subreport based on field** option enabled. This creates the link to ensure that the subreport displays only those departments for the current record, represented by PR.ID. All views connect this way, where the PR.ID from the main report passes along the record ID, down to the subreports PR_ID column for the pertinent view.

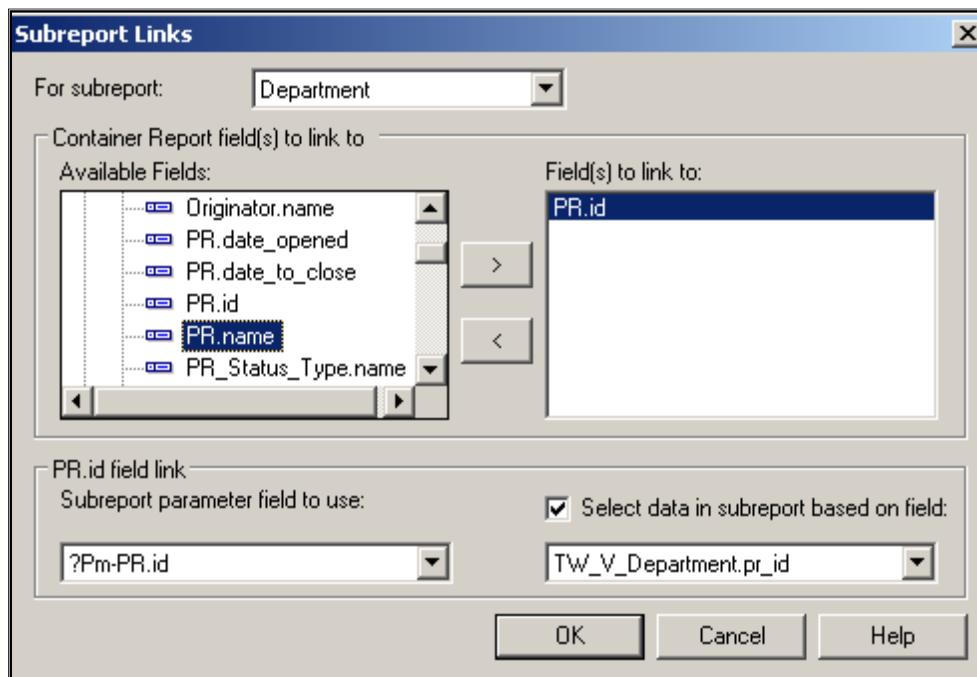


Figure 14-9 Select data in subreport based on field option checked with TW_V_Department.pr_id selected.

15. Copy the report into the **TrackWiseWeb / web app / twreports_1** folder and run the report against your record. Observe how the value shows up on the report. Add additional multi-selection values for Department. Save the record and run the report again. The subreport grows with the additional selection values without overlap of the fields below.

Note: It is always good practice to remove any PR.ID select expert applied to the report prior to loading it on the TrackWise server.

14.3 Displaying FlexFields using Direct Table Link

FlexField information can be displayed by using a view on the main report or by using direct table linking with the actual tables containing the data. The next workbook exercise show the technique by adding three new tables into a subreport with the select expert to define, by name, the flexfield to use.

A subreport is able to be saved on its own and used in other reports. By creating a sub report in the next workbook exercise, it is available for use with other reports that have a need to display Flexfield data. Saving a sub report as a separate report is available by clicking **File – Save Subreport As**.



1. Right-click the latest group footer and insert another section below. This is reserved space for the Description FlexField. Each of the following three flexfields needs to have their own group footer to go across. This ensures that the information is not truncated or a column of information only on the left side of the screen.

2. Insert a new sub-report and name it **Description**. The following are the required tables and the applicable table joins:

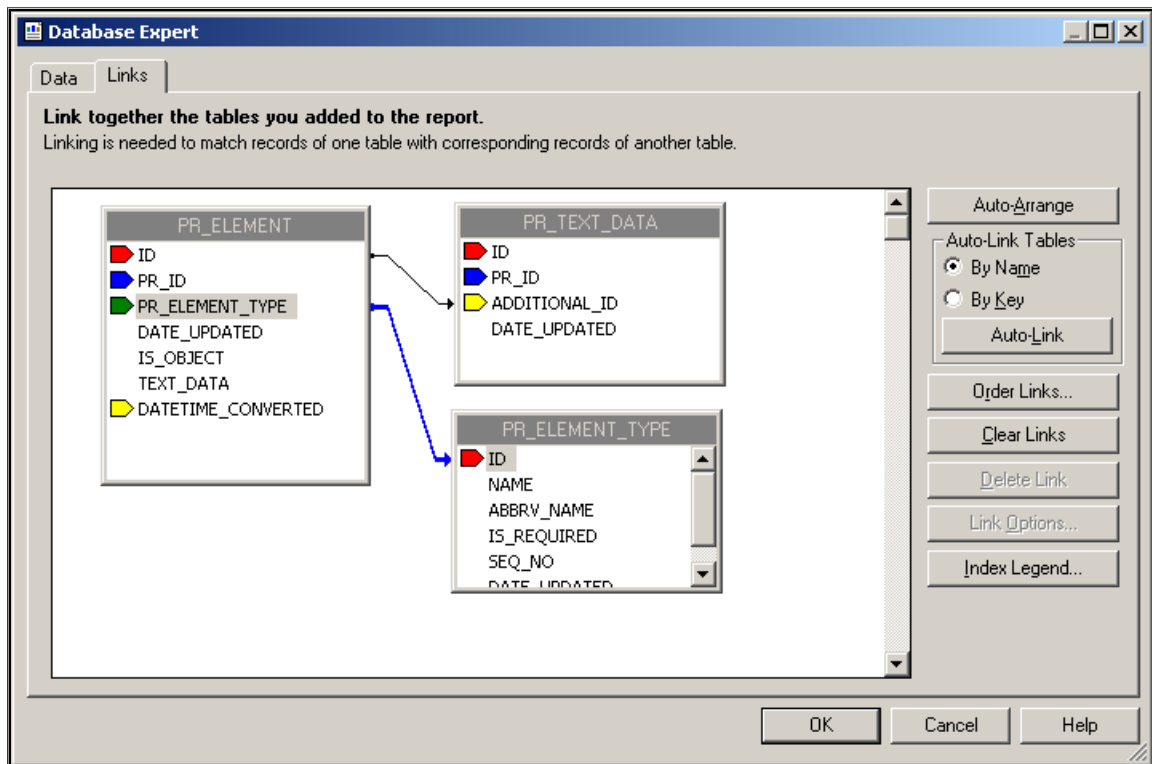


Figure 14-10 Required tables and joins for Flexfield data.

- **PR_ELEMENT** is the lookup table for PRs to see which ones have Flexfield data. Link this table from **PR_ELEMENT.ID** to **PR_TEXT_DATA.ADDITIONAL_ID** as left outer.

Link **PR_ELEMENT.PR_ELEMENT_TYPE** to **PR_ELEMENT_TYPE.ID**

- **PR_TEXT_DATA** contains the actual data for every Flexfield in the database.
 - **PR_ELEMENT_TYPE** is the lookup table for the Flexfield name.
3. The column to be displayed on the sub-report, containing the FlexField information, is **PR_TEXT_DATA.TEXT** which is the only field displayed on the sub-report:



Figure 14-11 PR_TEXT_DATA.TEXT added to the subreport.

4. Click **Finish** in the Report Wizard and enter the subreport.

5. Use the **Select Expert** to select only the FlexField named **Description**. This step is to define which Flexfield in the record is to show its data values.

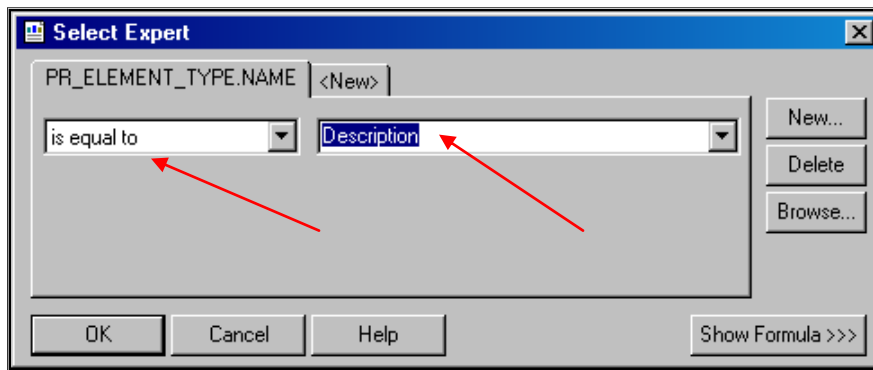


Figure 14-12 PR_ELEMENT_TYPE.NAME selected for Select Expert.

6. Click **OK** and save the report.
7. The final step is to change the sub-report links. This is done from the main report by right-clicking the sub-report and selecting **Change Subreport Link**. On that screen, link **PR.ID** from the main report to **PR_ELEMENT.PR_ID** on the sub-report.

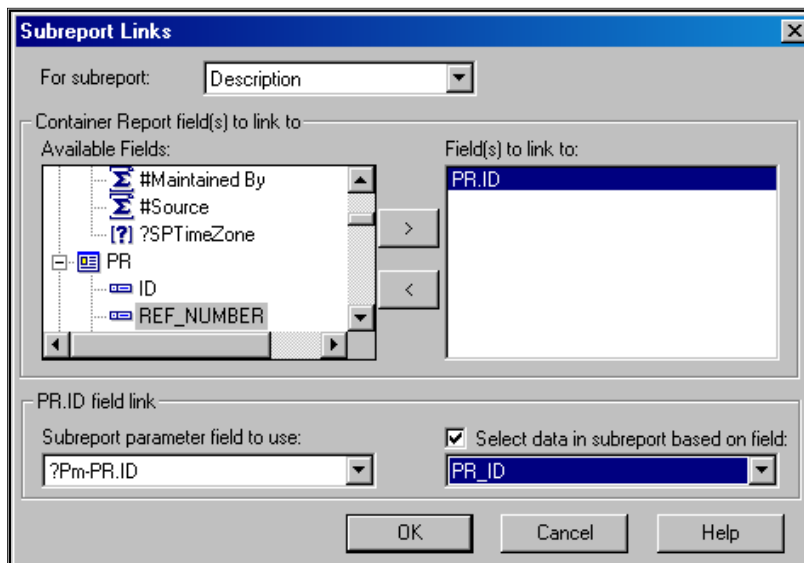


Figure 14-13 Subreport Links window.

8. If **PR_ELEMENT.PR_ID** is not available to link to, go back into the flexfield subreport and add this particular column into a suppressed area of the report. Then, go back to the main screen and try to link again.



1. Insert another group footer below for the next flexfield.
2. Try the same technique, above, for the other flexfield **Immediate Actions**. Set **Select Expert** for **PR_ELEMENT_TYPE.NAME** to equal **Immediate Actions**.
3. Link the main report to the subreport.
4. Insert a group footer for the next Flexfield, **Timeliness Justification**, to add another subreport to the main report.
5. Link the main report to the subreport.

14.4 Grid Fields In Report

Grid Fields are field types that appear as a spreadsheet of information within records. As with multi-selection fields, displaying the contents of a grid field is accomplished with subreports. The report hosts the grid field as a subreport while the subreport contains the views associated with the fields embedded in the grid. In the next exercise, two grid fields are added to the report. They are **Product Information** and **Materials**.

Before incorporating product material detail, log in to TrackWise and open your record, locate this grid field to add a second row of information. Save the record and then refer back to Crystal Reports.

Row #	☐	Lot #	Product Name	Dosage	Form	Expiration Date
1	☐	000000002 Search	Product 2	100 mg	Capsule	10/26/2010 Now
2	☐	000000004 Search	Product 4	12 oz	Cream	3/22/2011 Clear

Figure 14-14 Example for Product Information grid field on the record details screen in TrackWise.



1. In Design mode, click **Insert – Subreport**.

2. Ensure **Create a Subreport with the Report Wizard** is selected and in the **New Report Name** field, enter **Product Information**, named after the Grid. Click **Report Wizard** when complete.

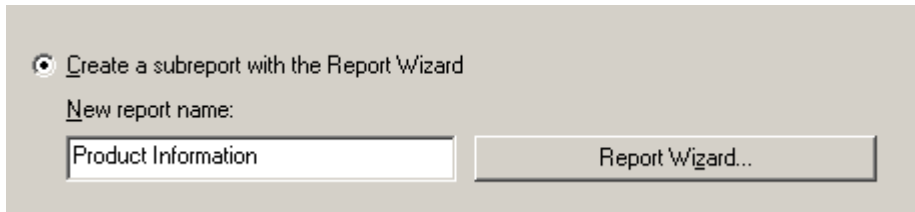


Figure 14-15 Select the Report Wizard to create a subreport.

3. Locate **introCrystaldsn** to open the views folder. Double-click the following views to bring over into the subreport. Fields within a grid use the name of the grid first followed by the name of the field inside the grid. In this example, the fields start with **TW_V_Product_Information_** followed by the name of the field. Refer to the following chart for the correct view to bring over to the subreport.

Field Name	View Name
Lot #	TW_V_Product_Information_Lot
Product Name	TW_V_Product_Information_Produ
Dosage	TW_V_Product_Information_Dosag
Form	TW_V_Product_Information_Form
Expiration Date	TW_V_Product_Information_Expir

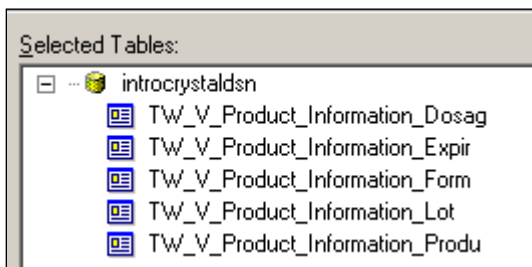


Figure 14-16

4. Click **Next** and refer to the **Links** tab. Arrange the tables for ease of linking. The Grid Fields are linked using a double link method. For each view, connect from pr_id to pr_id of the second view. Do the same for seq_no to seq_no . Make one of them a left outer join which would allow both to become left outer joins. These joins exchange the record number and the order of the values with sequence number.

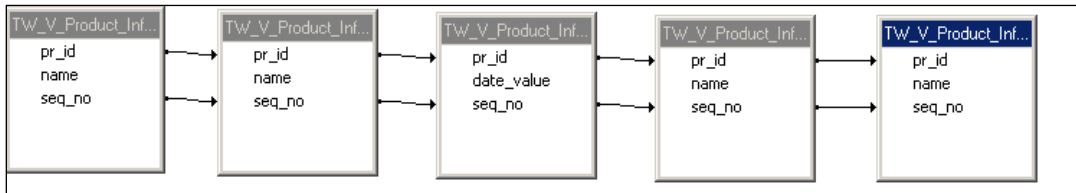


Figure 14-17 Tables and joins for the grid fields.

5. Click **Finish – OK**.
6. Place the subreport in the empty group footer. Double-click the subreport to open the details.
7. Right-click **Report Footer** and select **Suppress (No Drill Down)**. In order to link to the subreport in the next step, drag **pr_id** from any of the views under Field Explorer over to the suppressed **Report Footer** area.

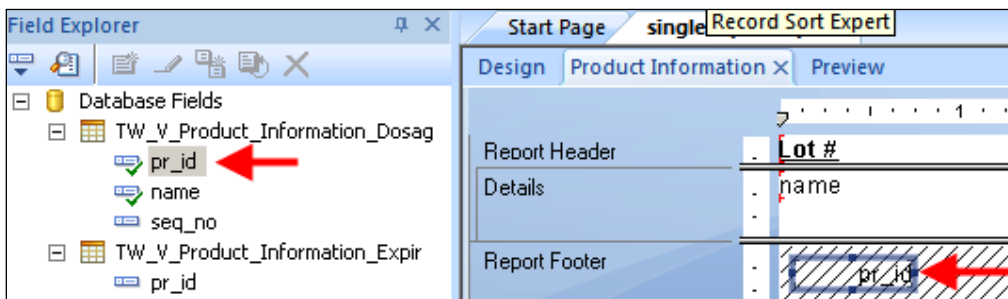


Figure 14-18 pr_id added to the suppressed Report Footer.

8. Arrange the columns of data in the same order as how they appear in the grid by dragging and dropping each of the **name** columns into details. Label each column in report header. For Expiration Date, use **date_value**. Change the format of date_value to not show the date and time, just the date.

Design		Product Information		Preview	
Report Header	Lot #	Product Name	Dosage	Form	Expiration Date
Details	name	name	name	name	date_value

Figure 14-19 Design tab of subreport with name columns added and headers updated with grid field names.

Refer back to Design mode. The next step is to establish the link between the main report and the grid sub-report. Right-click the subreport and select **Change Subreport Links** from the list. Create the following link:

PR.ID – TW_V_PRODUCT_INFORMATION_DOSAG.PR_ID

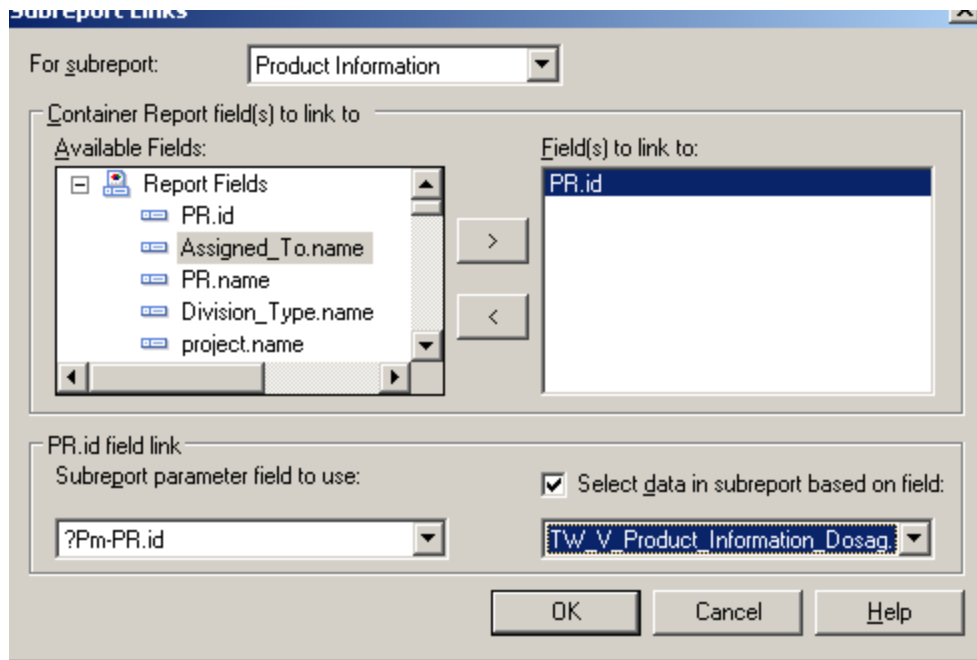


Figure 14-20 Subreport Links window.

9. This link passes from the main report the PR.ID to the subreport view containing the PR_ID column. Each view, which is linked, shares the ID to see if each one contains data within the fields. Click **OK** and save the report. Refresh and refer to the rows of information that appear within the grid for Product Information.

<u>Lot #</u>	<u>Product Name</u>	<u>Dosage</u>	<u>Form</u>	<u>Expiration Date</u>
000000002	Product 2	100 mg	Capsule	October 26, 2010
000000003	Product 3	50 mg	Tablet	March 22, 2011

Figure 14-21 Example of preview of grid data from Product Information.



1. Refer back to your record in TrackWise and add two rows of grid information to Materials. Save the changes to the record.
2. In Crystal, add another subreport called **Materials**. Include the four views for the fields within this grid.

Field Name	View Name
Material Lot #	TW_Materials_Material_Lot
Material Name	TW_Materials_Material_Name
Material Quantity	TW_Materials_Material_Quanti
Material Disposition	TW_Materials_Material_Dispos

3. Link each view accordingly. Suppress the group footer to drag and drop pr_id from any of the four views into here to assist with subreport linking.
4. On the main report right-click the Materials subreport and link from PR.ID to the view's pr_id in the subreport.
5. Save and run the report. The data from Materials appears.

14.5 Displaying Logically Related Records

In the previous example, we used a sub-report to display multi-value fields, one record at a time. However, there could be a situation where records containing the same values are desired to also be shown. For example, if the current record has a Type value of commercial product, the subreport would display every record in the system that contains this same value in the Type field. Please note that record IDs that are displayed may appear from projects which the user does not have access to.

Workbook Exercises

1. Open the report from the previous section and save it as **SingleReport3.rpt**. Insert a new subreport that includes the following tables: **PR** and **TW_V_TYPE**.
2. Join these tables within the subreport **PR.ID - TW_V_TYPE.PR_ID** as left outer join.

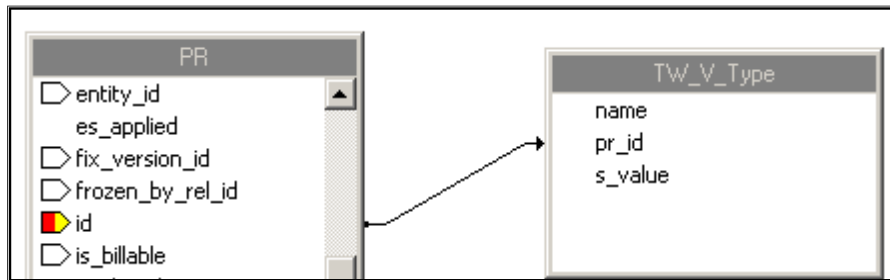


Figure 14-22 Link between PR and TW_V_Type table.

3. Place the **PR.ID** field in the **Fields to Display** window. Click **Finish** then **OK** when complete to return to the main report design screen.

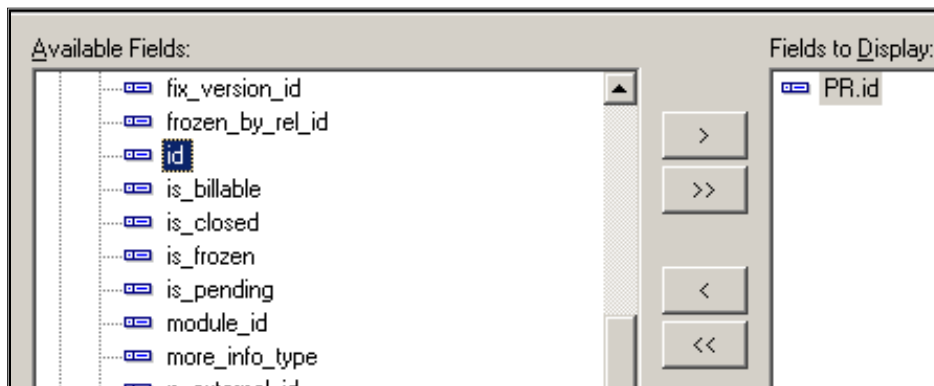


Figure 14-23 PR.id selected as Fields to Display.

4. Click the subreport and modify the formatting. Include text into the subreport to indicate what the original **Type** value is. Change the ID in Details to only show the number and not decimal or comma.

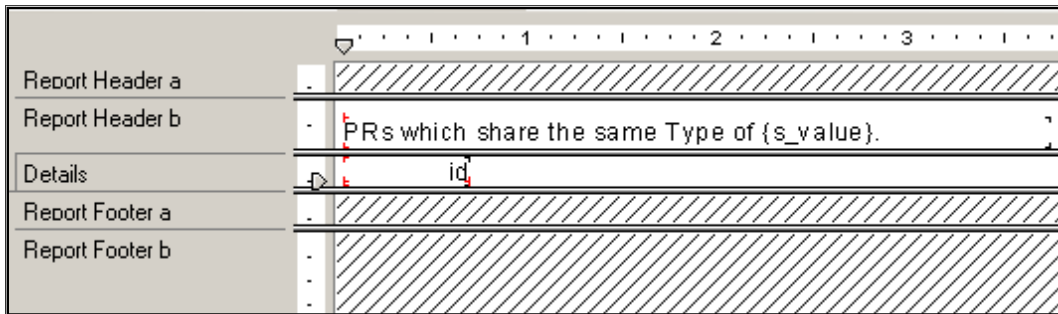


Figure 14-24 Design tab of subreport.

5. Go back to the main report screen. Right-click the subreport and select **Change Subreport Links** from the list.
6. From **Available Fields**, select the **TW_V_TYPE .S_VALUE** field on the main report. Click the right arrow to move this field into the **Field(s) to link to** box.
7. Ensure that in the field below, the **TW_V_TYPE.S_VALUE** field is selected. This link returns all records that share a common Type selection value.

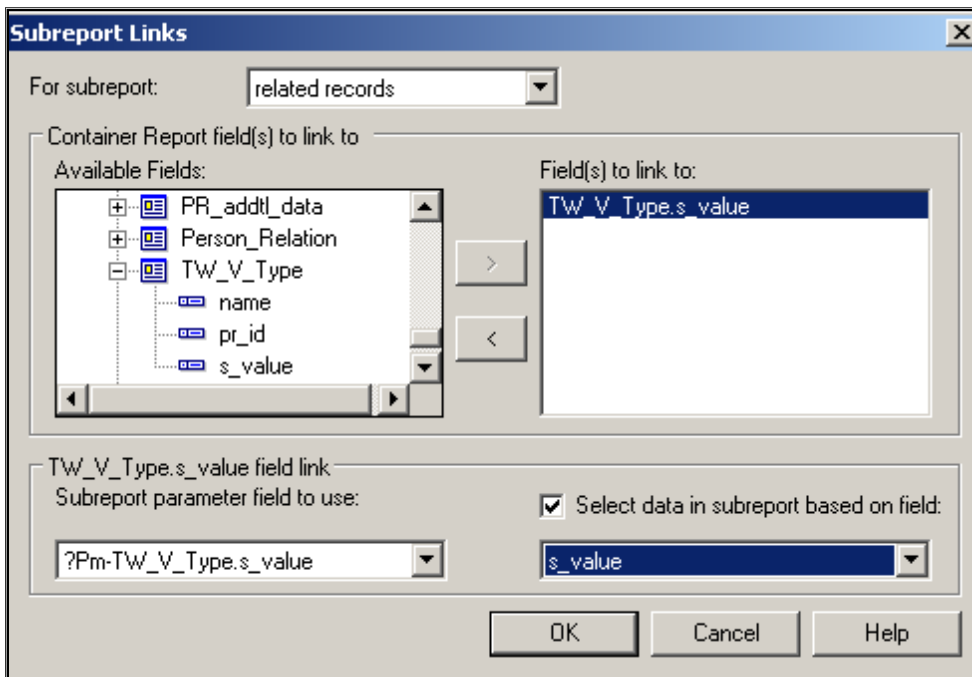


Figure 14-25 Subreport Links window.

8. Click **OK** and save the report. To organize the list of returned records in a table format use the **Section Expert** in the **Related Records** sub report and select **Details**. While **Details** is highlighted, click the box for the option **Format With Multiple Columns**.

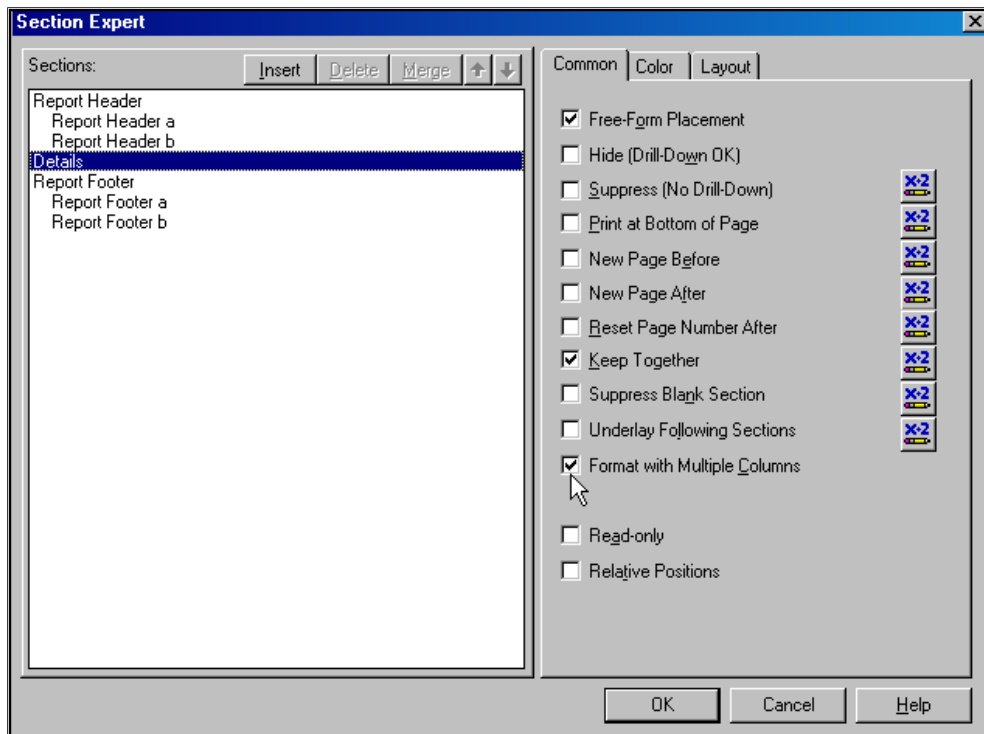


Figure 14-26 Section Expert window with Format with Multiple Columns selected.

9. A layout tab appears on top. Click the **Layout** tab and refer to the following column size settings. This allows the record IDs to properly display rather than just appear as a single ongoing list.

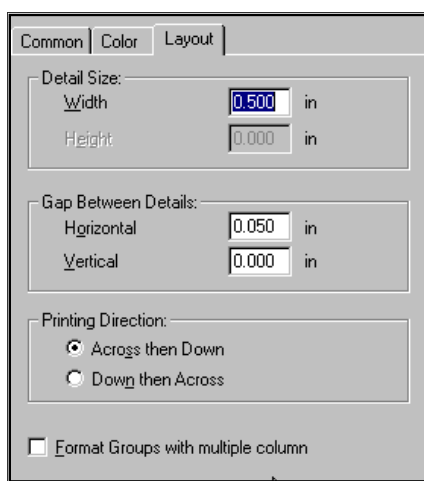


Figure 14-27 Set Detail Size, Gap Between Details, and Printing Direction on the Layout tab.

14.6 File Attachment Fields in Reports

In the previous example, we used a sub-report to display multi-value fields, one record at a time. Here is a formula, to be used within a subreport, to print the name of a file attachment in a record. Start with the file attachments view just like any other multi-based field. The difference is that instead of s_value to be brought over to details, use the following formula to only show the name of the file attachment.



1. Log into TrackWise and refer to your record.
2. Click **Referenced Info / Comments**. Attach a file to the record with the Attached Files field.

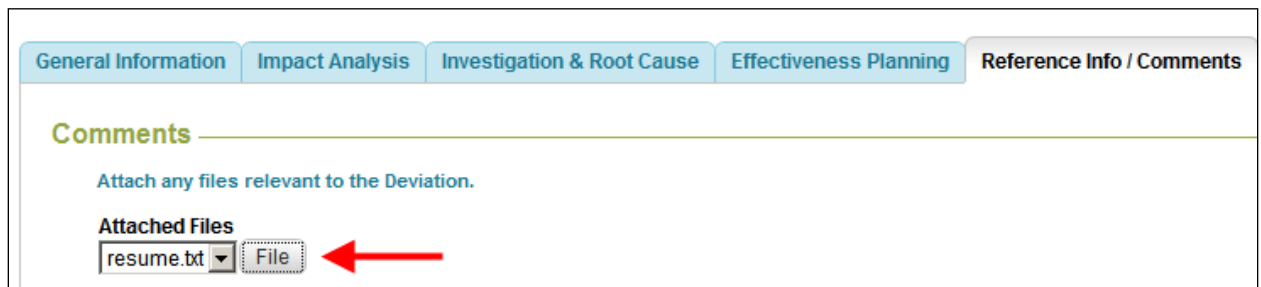


Figure 14-28 File button on the record details in TrackWise.

3. Save the record. Refer back to Crystal Reports. Add a group footer in preparation for the file attachment field. Include the section and field name.



Figure 14-29

4. Insert a subreport. Name it **Attached Files**. Include the view TW_V_ATTACHED_FILES.

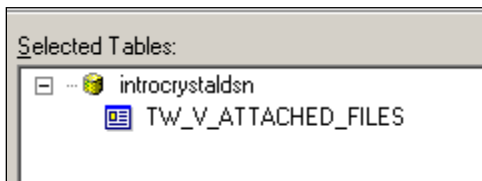


Figure 14-30 TW_V_ATTACHED_FILES added to Selected Tables.

5. Click **Finish** and **OK**. Do not add s_value to display on the report. A formula is used instead to truncate the value from the database to only display the name of the file. Place the subreport underneath the Attached Files text.

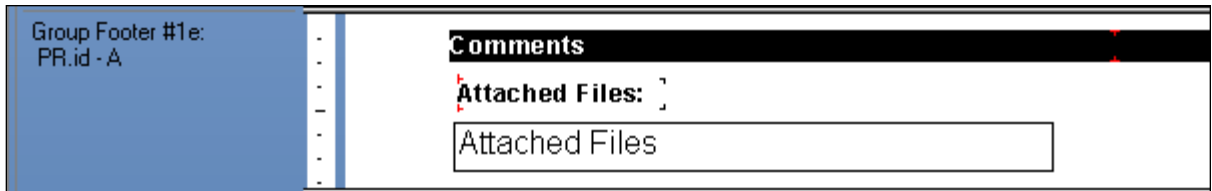


Figure 14-31 Attached Files subreport added to Group Footer #1e.

6. Right-click the subreport and select **Format Subreport**. Remove the border around the subreport.
7. Refer to the detail screen of the subreport. Create the following formula to display the name of the file attachment field.

```
numberVar X :=instrrev ({TW_V_ATTACHED_FILES.s_value},"\\");
```

```
mid({TW_V_ATTACHED_FILES.s_value}, X + 1)
```

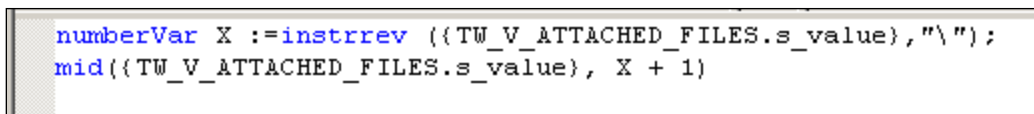


Figure 14-32 Formula to display name of file.

8. Verify the formula for any errors. Click **Save and Close**.
9. Go to the Design screen and right-click the subreport.

10. Select **Change Subreport Links**. Link from **PR.id** to **TW_V_ATTACHED_FILES.pr_id**. Click **OK**.

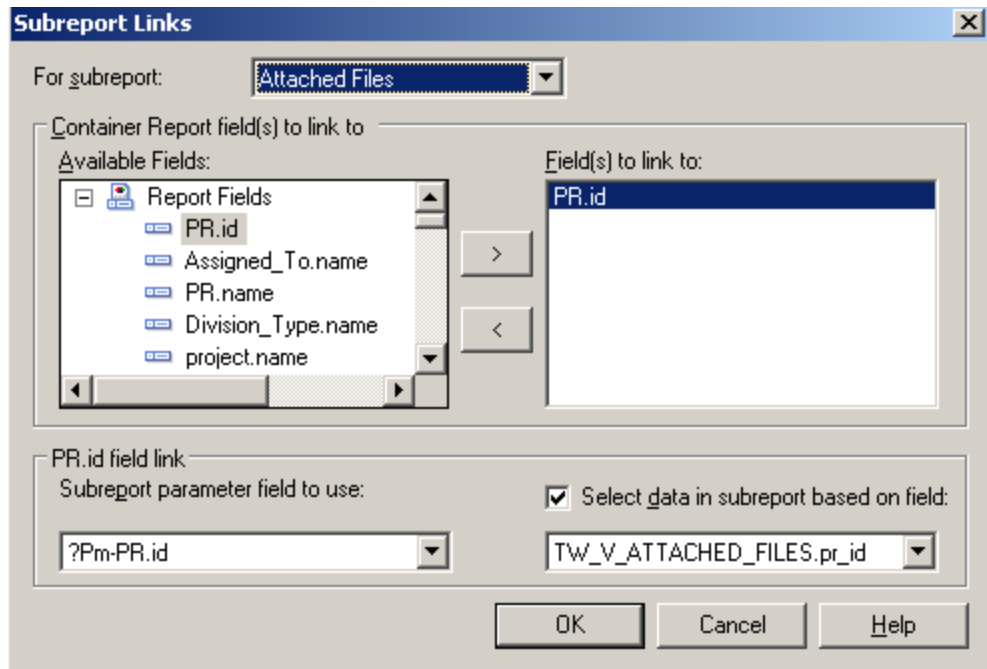


Figure 14-33 Subreport Links.

9. Run the report and refer to the page where the file attachment field is. Notice the name of the file appears.

1156 - Mislabeled medicine in production.

Division / Project: Holmdel, NJ / Deviation **Date Created:** October 27, 2010 2:31 am

PR State: Opened **In State Since:** October 27, 2010 2:31 am

Comments

Attached Files:

resume.txt 

Figure 14-34 Name of file displayed.

14.7 File Pointer Fields in Reports

File pointer fields are considered multi-field related and are used within subreports. Save the report and close out Crystal. Refer to the following steps to add the field to the report.



Workbook Exercises

1. Open Crystal Reports. Refer back to the report.
2. Click **Insert – Subreport**.
3. Name the subreport **Related Documents**. Click **Report Wizard**.
4. Refer to the introCrystalsdn data source. Under the view list, select **TW_V_RELATED_DOCUMENTS**. Click **Next** and select **s_value** to display within the subreport.
5. Click **Finish** then **OK**. Add the subreport underneath the text box. Remove the box around the subreport.
6. In the subreport details screen, suppress one of the report footers and remove the other.
7. Ensure **s_value** reaches the other side of details.

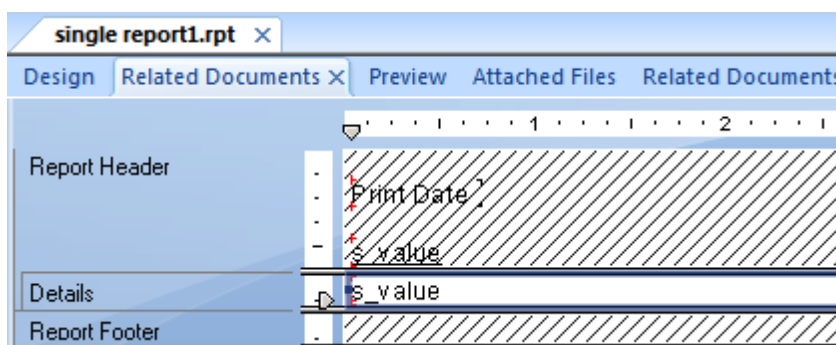


Figure 14-35

8. Refer back to design mode. Right-click **Related Documents** and select **Change Subreport Links**. Connect the subreport from **PR.ID** to **TW_V_RELATED_DOCUMENTS.pr_id**

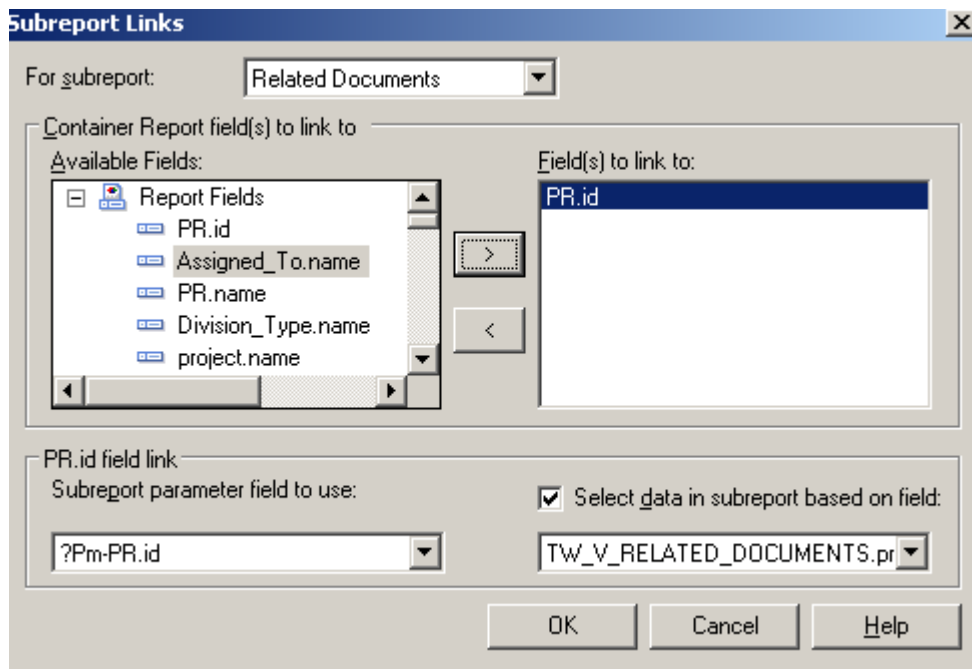


Figure 14-36 Subreport Links

9. Click **OK**. Save and refresh the report. Notice the file pointer refers to a file stored in the specific location.

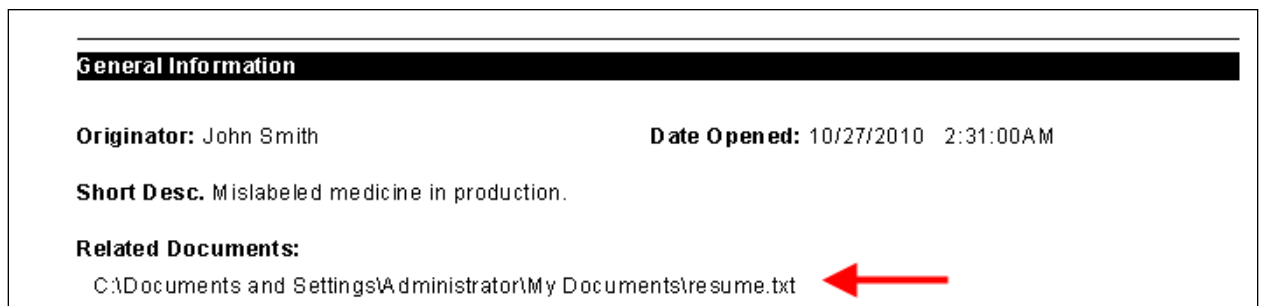


Figure 14-37 Location of file displayed.

14.8 Multilanguage Reports

In the parameter chapter, LanguageName was mentioned as a way to find out the language the user logged in with for TrackWise. It provides the ability to create one report for any number of translations available within TrackWise. Before this parameter was available, reports were created multiple times for each language to use. By providing one report for all users, regardless of dialect logged in with, this consolidates the number of reports required for each translated language.

Refer to the following workbook exercise to be used outside of class if Multilanguage is configured within the TrackWise system.



The prerequisite for the report is to have LanguageName as a parameter in both the main report and the subreport.

1. Open the **template.rpt** and choose **File – Save As** to name it **MultiLanguage.rpt**.
2. Create a sub report for each field name that is translated into another language based on the Multilanguage feature in TrackWise. When prompted in the sub report wizard screen, add the following four tables with left-outer joins. The following table explains what each table is available for.

Table	Description
Data_Fields	This table is part of the formula to verify the name of the field being translated.
Language	This table identifies the language identification number in order to properly translate the name of the field.
Object_Type	This table contains the look up of the item being translated. When translating TrackWise components, this table categorizes them. In this example it is data fields.
Object_Value	This table contains the list of translated terms.

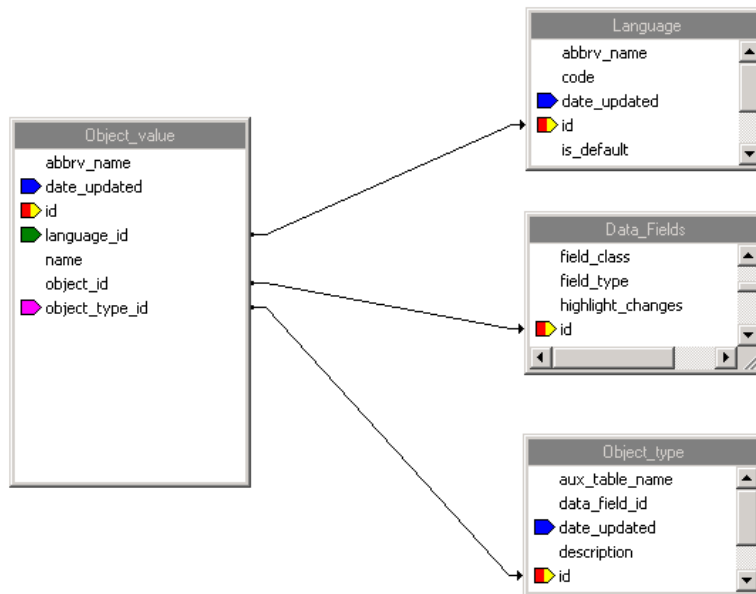


Figure 14-38 Tables and joins for a Multilanguage report.

3. After linking the tables, return to the sub report design screen. Suppress any extra rows and leave Details available.
4. A formula is used to display the name of the field that is translated. Create a new formula called **Label**.
5. Create the following formula. Click **Save and Close** when finished.

if not isNull({Object_value.name}) then {Object_value.name} else "Title"

If Object_Value.name is not blank, use the translated value for the field in question. Otherwise, put the word Title on the screen.

6. Add the formula field onto the report within **Details**.

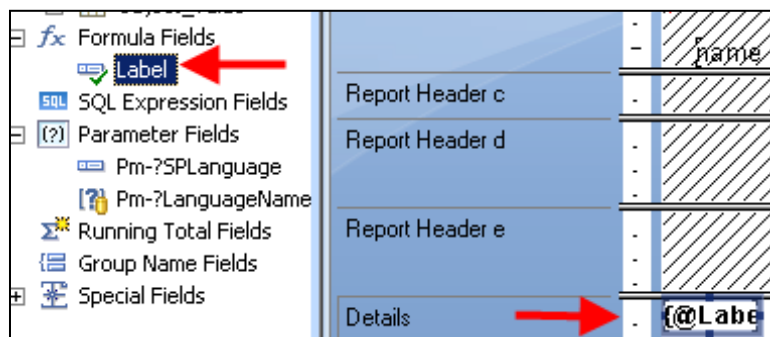


Figure 14-39 Formula added to Details section.

7. To filter out all other fields on the subreport to only show the label for Title, use the select expert. Click **Report – Select Expert – Record**.
8. Add the following two tables and specific filters for each.

Table and Column Name	Select Expert Setting
Data_Fields.name	is equal to Short Desc.
Object_Type.main_table_name	is equal to Data_fields

9. Each of these select expert entries ensures that the sub report only returns the translated term for the data field Title. Repeat this process and replace the Data_Fields.name entry to equal the specific field in question that the sub report is showing. The other setting stays the same.
10. Return to the main report. In this example, the system field Title is translated into Spanish as the word Titulo. The string data within the field is not translated. Only the label on the forms next to each field is translated. Condense the space of the sub report to appear as a text box.

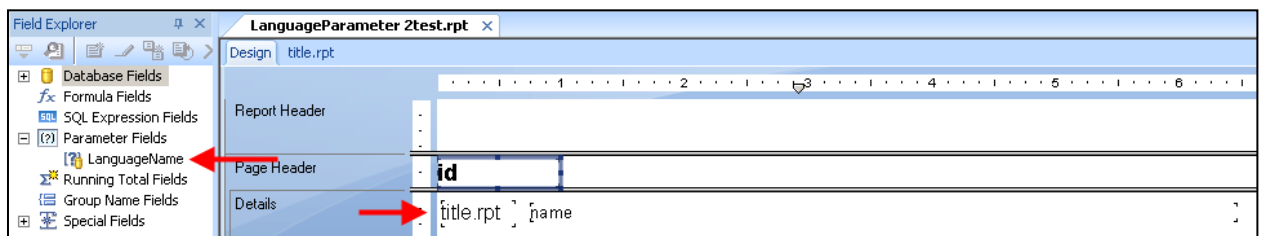


Figure 14-40

11. Based on this screen capture above, add **PR.name** next to the sub report to display the title information.

12. Link the sub report to the main report by referring to the following parameter to table connection.

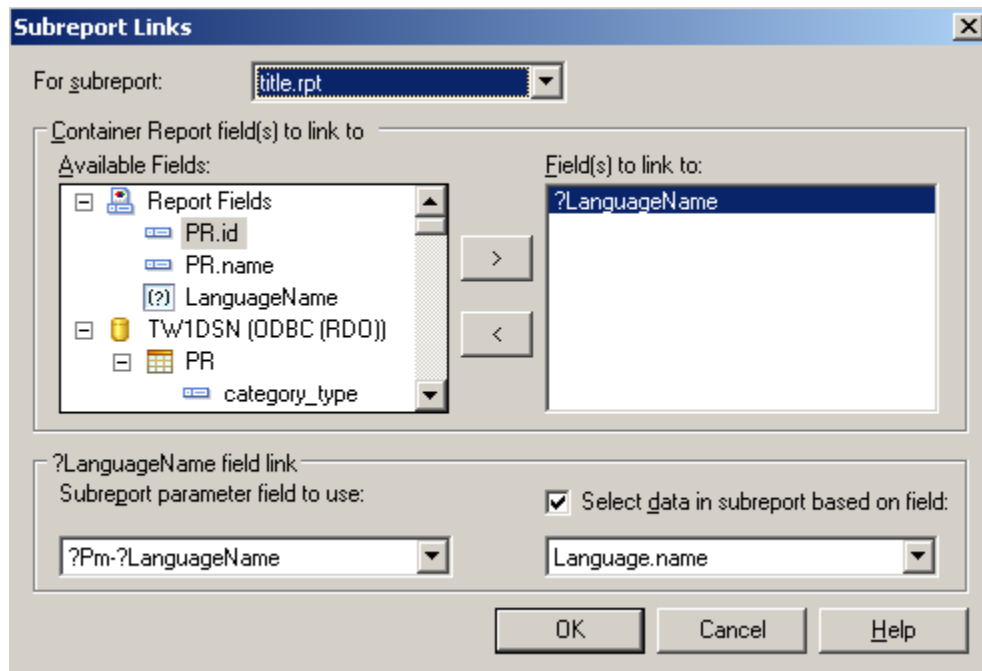


Figure 14-41 Subreport Links window.

13. The parameter on the main report identifies which language is being used. It sends the name down to the Language table which identifies which language to show the static text as.
14. The following is an example of the word Title translated with the actual text next to it.



Figure 14-42 Example of the word Title translated along with actual text.

To complete the report the report designer should repeat the previous steps for all static text labels that are translatable. Another option is to save the sub report on its own as use it as a template to insert other sub reports with. Click **File – Save Subreport As** within the sub report screen and call it **Translated Label Template**. The only part that requires maintenance is the following.

- Delete **Language.name** from the select expert which is based on the original sub report link from the main report to the sub report. The sub report needs to be linked again. Due to the original cached information within the select expert, this needs to be applied again from linking the sub report on the main report.

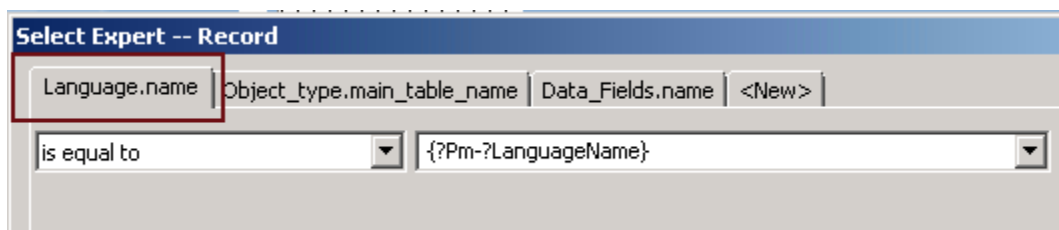


Figure 14-43

- Change **Data_Field.name** to equal whichever field name in question that needs to be translated.

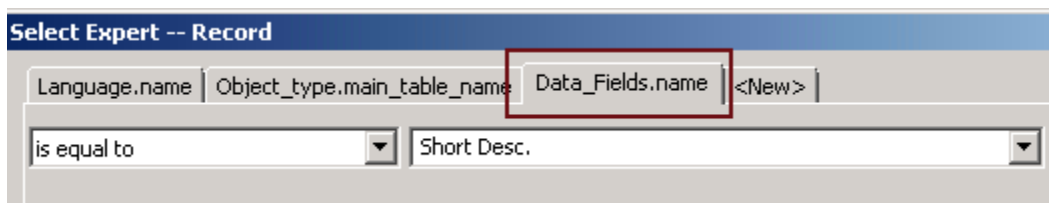
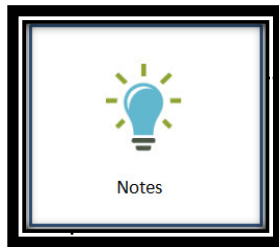


Figure 14-44

Conclusion:

Fields that are multi-based use subreports to prevent duplicate pages from printing when the report generates. Subreports are also used to show records of a similar nature based on a shared value. This feature ignores project and group category rights in TrackWise, but provides a way to see potential trends amongst many records of a similar nature.

The next chapter focuses on audit trail activity history reports. New tables are introduced to acquire this information.



15. Activity History Report

15.1 Definition and Explanation

Activity History reports print the complete audit trail of a single record. Audit trail entries include scheduled activities, performed activities, field changes and notifications. The following workbook exercise provides the steps to display the audit trail history of a single record onto a report.

15.2 Activity History Report Design

Open **template.rpt** and choose **File – Save As** to name it **Activity History.rpt**



Workbook Exercises

1. Click **Database – Database Expert** to add the following tables to print the audit trail activity history.

Table Name	Description
PR_Activity	This table identifies how many activity history entries in a record and some of the characteristics.
PR_Activity_Type	This table identifies the names of the activity history entries.
Activity_Text_Data	This table identifies the activity summary, if available, for an activity history entry.
Person_Relation (Rename this table to Responsible_Name by clicking F2)	This table identifies the name of the person responsible for the action performed within the activity history.

2. Perform the following links:

PR.ID to PR_ACTIVITY.PR_ID [Left Outer]

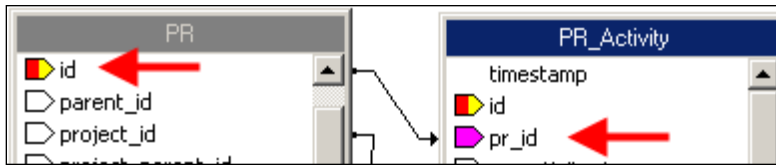


Figure 15-1 Link between PR and PR_Activity table.

PR_ACTIVITY.ID to ACTIVITY_TEXT_DATA.ADDITIONAL_ID [Left Outer]

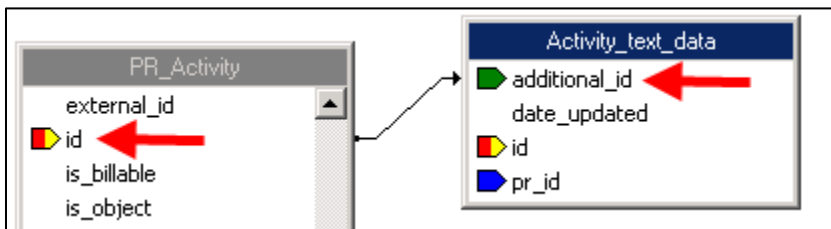


Figure 15-2 Link between PR_Activity and Activity_text_data table.

PR_ACTIVITY.PR_ACTIVITY_TYPE to PR_ACTIVITY_TYPE.ID [Left Outer]

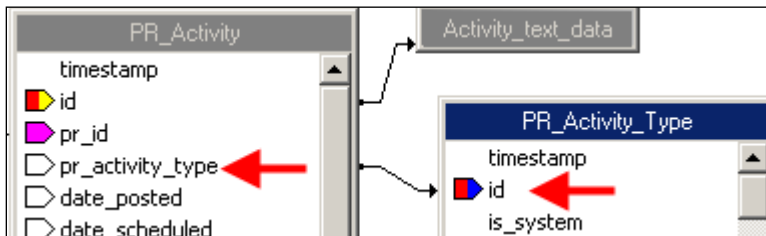


Figure 15-3 Link between PR_Activity and PR_Activity_Type table.

PR_ACTIVITY.RESPONSIBLE_REL_ID to RESPONSIBLE_NAME.ID [Left Outer]

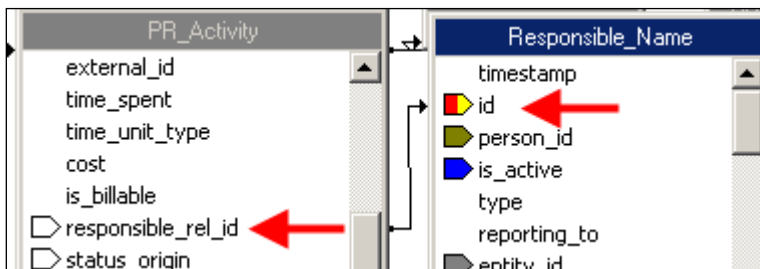


Figure 15-4 Link between PR_Activity and Responsible_Name table.

The screenshot displays a database management interface with the following components:

- PR Table:** Contains fields such as `frozen_by_rel_id`, `id`, `IS_ARCHIVED`, `is_billable`, `is_closed`, `is_frozen`, `is_offline`, `is_pending`, `module_id`, `more_info_type`, `n_external_id`, `name`, `offline_file_id`, `offline_parent_id`, `opened_type`, `originator_dept_id`, `originator_rel_id`, and `parent_id`.
- PR_Activity Table:** Contains fields such as `external_id`, `id`, `is_billable`, `is_object`, `performed_by_name`, `performed_by_role`, `posted_by_name`, `posted_by_rel_id`, `pr_activity_type`, `pr_id`, `pr_state_node_id`, `responsible_name`, `responsible_rel_id`, and `responsible_role_type`.
- Activity_text_data Table:** Contains fields `additional_id`, `date_updated`, `id`, and `pr_id`.
- PR_Activity_Type Table:** Contains fields `date_updated`, `id`, and `is_active`.
- Responsible_Name Table:** Contains fields `function_id`, `id`, and `influence_factor`.

Relationships are indicated by arrows: `PR` to `PR_Activity`, `PR_Activity` to `Activity_text_data`, `PR_Activity` to `PR_Activity_Type`, and `PR_Activity` to `Responsible_Name`.

3. In the report header, add a text box for the title of the report. Include the following fields based on the available tables from field explorer.

- Report Header

Activity History Report

PR # {id}

Date Opened: {@Date Opened}

Division / Project : {name} / {name}

State : {name}

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4. Suppress page header. Use the details section to add the following fields from the newly added tables. Refer to the following chart.

Table and Column Name	Description
PR_ACTIVITY_TYPE.NAME	This column provides the name of the activity history entry.
Responsible_Name.name	This column provides the name of the person who performed the action as noted in the activity history.
PR_ACTIVITY.DATE_PERFORMED	Create a formula to convert this date time field to local time. GetLocalTimeFromGMT({?SPTimeZone},{PR_Activity.date_performed})
ACTIVITY_TEXT_DATA.TEXT	This column provides the activity summary, often found with field changes in the audit trail and with performed activities.

5. Add a line underneath the text data field from Activity Summary. That way, the next entry is distinguishable from the others. Make sure the line is check with the option **Move to Bottom of Section When Printing**.

Details	@ Activity Name Activity Summary text	Responsible Name : {name}	Date Performed : {@Date Performed}
Report Footer			

Figure 15-7 Include line under the text data field from Activity Summary.

6. Save the report. Set Select Expert to retrieve PR.ID 1040. Run the report.

Activity History Report

PR # 1,040

Date Opened: August 9, 2012 4:00 am

Division / Project: Holmdel, NJ / Audit - Internal

State:

Open

Responsible Name: Smith, John

Date Performed: August 9, 2012 4:00 am

Activity Summary:

08/09/2012 12:00 AM (GMT-04:00): Opened by: John Smith, PID-000002.

Assign

Responsible Name: Smith, John

Date Performed: August 9, 2012 4:00 am

Activity Summary:

08/09/2012 12:00 AM (GMT-04:00): Assigned to: Harrel Bramley by: John Smith, PID-000002.

Admin Event

Responsible Name: Smith, John

Date Performed: August 9, 2012 4:00 am

Activity Summary:

08/09/2012 12:00 AM (GMT-04:00): PR State was set to: 'Pending Response' by: John Smith, PID-000002.

Figure 15-8 Example of preview of activity history report.

15.3 Sorting Activities by Date Performed

The activity history entries can be sorted to appear in a particular order when the report generates the data. The following exercise allows the most recent change in the record to show up in the beginning of the report by sorting date performed in descending order.



Workbook Exercises

1. Click **Report – Record Sort Expert**.
2. Based on the formula created in step four from page the previous section to convert the date time field Date Performed, select **@Date Performed** and add it to **Sort Fields**.

3. Select **Descending** from the **Sort Direction** section.

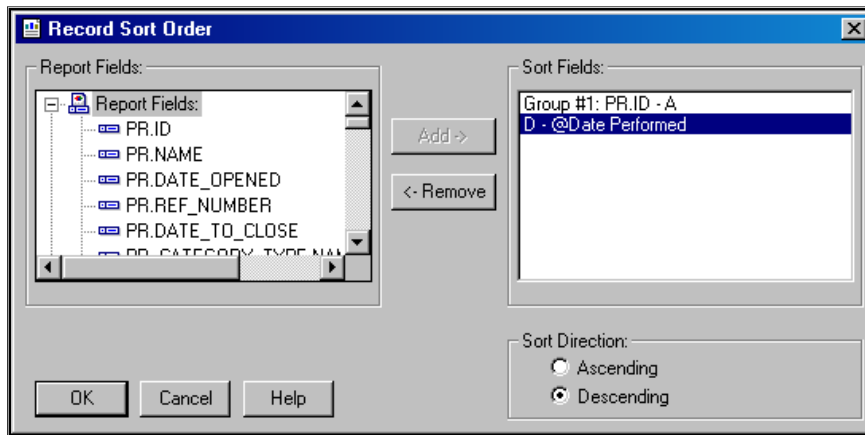


Figure 15-9 Sorted by Date Performed in descending sort direction.

4. Click **OK**. Save and run the report. Notice the most recent event appears on the first page. In TrackWise, update **Main Contact** and **Other Auditors** in **PR#1040**.
5. Save the changes in TrackWise and refresh the report. Notice the recent changes appear on the top.

Activity History Report		PR # 1040	Date Opened : January 27, 2009 12:23 pm
Other Auditors Changed Activity Summary : 03/24/2011 01:46 PM (GMT-4:00) Other Auditors was modified by: John Smith, PID-000002. Changed from: [None] Changed to: Harrel Bramley Jane Doe Clarke Joannee Meaning: Initial Entry Meaning Comments: Responsible Name : Smith, John Date Performed : March 24, 2011 1:46 pm		Division / Project : Holmdel, NJ / Audit - Internal State : Pending Response	
Main Contact Changed Activity Summary : 03/24/2011 01:46 PM (GMT-4:00) Main Contact was modified by: John Smith, PID-000002. Changed from: [None] Changed to: Chad Johnson Meaning: Initial Entry Meaning Comments: Responsible Name : Smith, John Date Performed : March 24, 2011 1:46 pm			

Figure 15-10 Preview of report based on date performed in descending sort order.

15.4 Appending E-Sig Applied to Activity Name

When looking at the activity history screen in TrackWise, some of the performed activities display a pen and paper symbol to indicate an e-signature was applied. The following exercise uses a formula to print the words “e-signature applied” next to the activity name if this is the case. If there is not an e-signature, then the name only appears in the report.



Workbook Exercises

1. Create a new formula to display the activity name.
2. In the text section of the formula, type the following.

if ({PR_ACTIVITY.ES_APPLIED} = 1) then {PR_ACTIVITY_TYPE.NAME} + ' (e-signature applied)' else {PR_ACTIVITY_TYPE.NAME}
3. Check the formula for errors. The column es_applied is set to one when an e-signature is applied. It includes the name of the activity with an accompanying message to indicate an applied e-signature. If es_applied is set to zero, no e-signature was applied so just the activity name appears.
4. When finished click **Save and Close**.

```
if ({PR_Activity.es_applied}=1) then {PR_Activity_Type.name}+ ' (e-signature applied)'  
else  
{PR_Activity_Type.name}
```

Figure 15-11 Formula to add text of e-signature applied with activities that involved an e-signature.

5. Replace the **PR_ACTIVITY_TYPE.NAME** field on the report with the formula. Change the select expert to filter for PR#1132.

6. Save and run the report. Notice the names of the performed activities display the words (e-signature applied) next to the name.

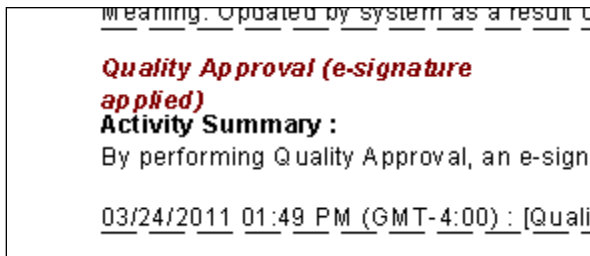
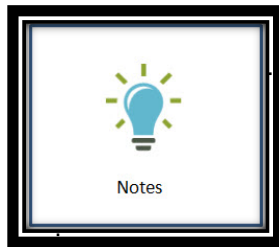


Figure 15-12 Example of preview report with e-signature formula.

Conclusion:

Activity history reports print field changes, performed activities and notifications from a record's audit trail. Through the use of four tables, this information is accessible from a report to print the complete history of changes. The report includes additional fields such as status, division and project values. The report is designed to run against one record at a time.

In the next chapter, an additional use of subreports is used to display parent and child record information.



16. Parent-Child Report

16.1 Definition and Explanation

Reporting on a family of records, from within a single report, provides a way to check the status and content of each record. Records within a family tree structure rely on the condition and status of one another to determine which records can progress and which need additional information. The following exercise uses subreports to extract data pertaining to children records and parent.

16.2 Displaying Children Information on the Parent Record

Open **SingleReport3.rpt** from the last section and save it as **SingleReport4.rpt**. Insert a new section below the last group footer for the subreport.



Workbook Exercises

1. Click **Insert – Subreport**. When prompted, choose the existing report template.rpt to establish the child record report with.

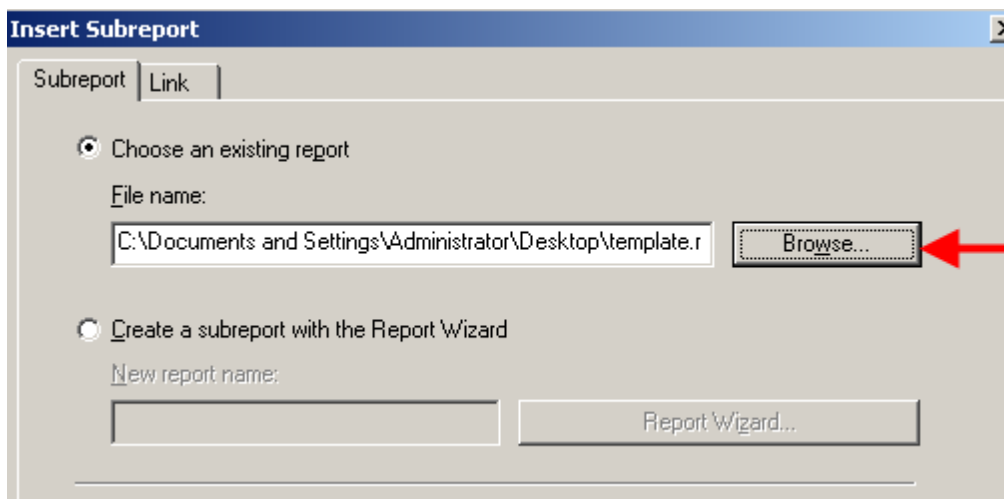


Figure 16-1 Select Browse to locate existing report.

2. Click **OK**. Drop the report into the new group footer. Maximize the space by expanding the subreport to reach the end of the report.
3. Right-click the subreport and select **Format Subreport**. Click the **Subreport** tab. Change the subreport name to **Children Records**. Click **OK**.



Figure 16-2 Subreport name changed to Children Records.

4. Double-click the subreport and refer to the design screen. In the first report header, add a text box to provide the name of the report.

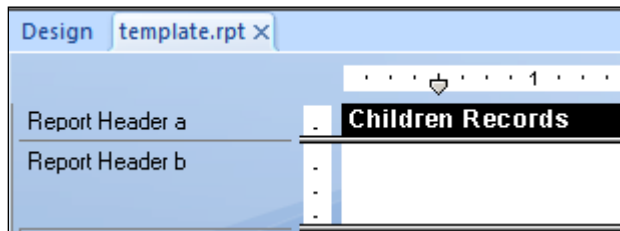


Figure 16-3 Text added to subreport for the name of the report.

5. When adding subreports using an existing report, the subreport contains an extra set of report headers and report footers. Be mindful and to maintain consistency, remove the extra sections.
6. Use the available tables and formulas from field explorer to build the child subreport with. Incorporate basic fields such as PR number, state and project to show an overview of what the children records contain. Consider this type of report as a mini multi-report to display several available children records. Refer to the following table as an example.

Report Header b	PR #	Project	Short Desc.	Assigned To	State	Date Due	Originator
Details	PR.id	Project.name	PR.name	@Assigned To or Blank	PR_Status_Type.name	PR.date_to_close	@Originator

7. Add **PR.root_parent_id** into a suppressed section of the report. This assists with the linking step to ensure the children records properly appear in the subreport.

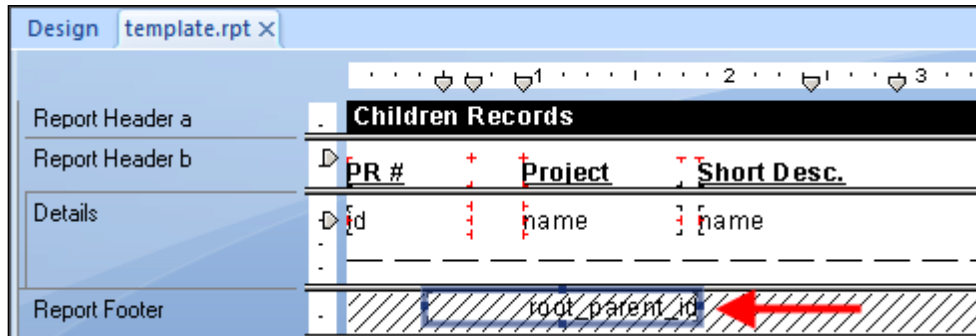


Figure 16-4 PR.root_parent_id added to the suppressed Report Footer.

8. When finished, refer back to design mode of the main report.

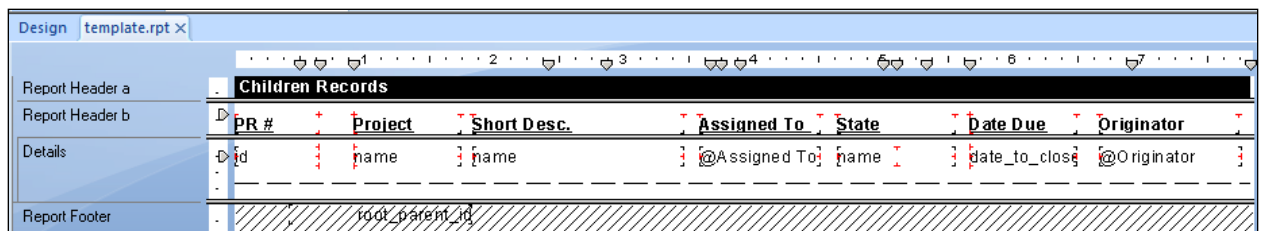


Figure 16-5 Completed subreport.

9. The next step is to link the main report to the subreport. Right-click the subreport and select **Change Subreport Links**. From **Available Fields**, select **PR.ROOT_PARENT_ID**. Click the right arrow to move this field into the **Field(s) to link to**.

Ensure that in the field below, **Select data in subreport based on field**, the table **PR.root_parent_id** field is selected. This link brings back only records related to the one on the main report.

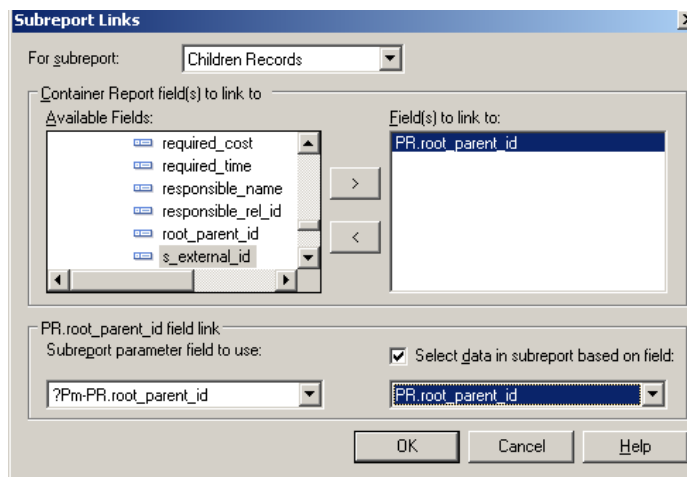


Figure 16-6 Subreport Links window.

10. Click **OK**.
11. In TrackWise, go into the details of PR#1156 and create two children records. The first is an Investigation and the other a Corrective Action record. Fill out the following fields in each.
 - Short Desc
 - Assigned To
 - Date Due
12. Save each record and run the report in Crystal. The data from both children records now appear within the report. To save the subreport for future use, go back into the subreport details. Click **File – Save Subreport As**. Call this subreport **Parent Child Subreport Template.rpt** It is used in the next workbook exercise.

13. The root parent also appears in the subreport with the children. The reason is because the parent record also contains a root parent number. Refer back to design mode of the children records subreport. Click **Report - Select Expert – Record** and add the following filter. This prevents the Root Parent ID from appearing twice on the report as the subreport is only meant to show children records.

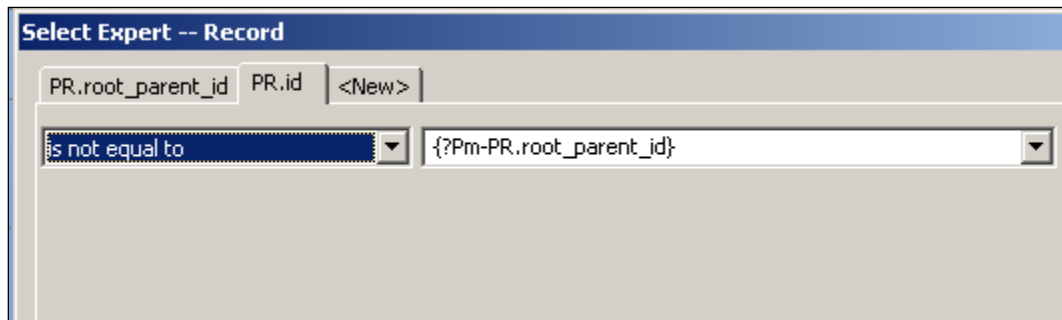


Figure 16-7 Select Expert – Record with PR.id.

14. Click **OK** and run the report again. Notice the root parent is absent as intended from the subreport. By running a single report against a parent record, the children records appear with examples of available data.

16.3 A complete Parent Child Report

The following exercise shows how to build a complete parent child report. This report shows the parent record on top of the report and then all generations of children below. The children records can be sorted by record ID, grouped by Project or any other field attribute, as previously described. One of the advantages of running this report is that no matter what record the report is generated upon, it always has a consistent look.

In order to understand the logic of this report, it is important to understand the link between PR.ID, PR.Parent_ID and PR.Root_Parent_ID, attributes that every TrackWise record contains. The following table shows a family of records based on PR#1164 which is a Training Employee record. The children records are training documents.

PR # ▲	Parent ID	Project	Division	Short Desc.
☒ 1164 		Training - Employee	Holmdel, NJ	Ingrid Latos: Employee Record
☐ 1172 	1164	Training	Holmdel, NJ	AD-002 TrackWise from an IT Perspective
☐ 1173 	1164	Training	Holmdel, NJ	QA-009 Policy Document
☐ 1174 	1164	Training	Holmdel, NJ	SOP-001 Document

Figure 16-8 Desktop display of the 1164 family of records.

ID	PARENT_ID	ROOT_PARENT_ID
1164	---	1164
1165	1164	1164
1166	1164	1164
1167	1164	1164

As seen in the table above, the PR.ID is the beginning of the chain and all related records share a common attribute, ROOT_PARENT_ID. This notion is the basic concept necessary in understanding the parent child relationship between records. The idea in this report is to have two sub-reports. One subreport to display information about the root parent record. The second subreport shows iterations of children records related to the root parent record.



Workbook Exercises

1. Create a new report and save it as **ParentChild1.rpt**.
2. Add the **PR table** to the report. The PR table is used to anchor the following two subreports on this main report and determines which record is the root parent and which are the related records. Include a title in the report header.
3. Suppress **Details**.

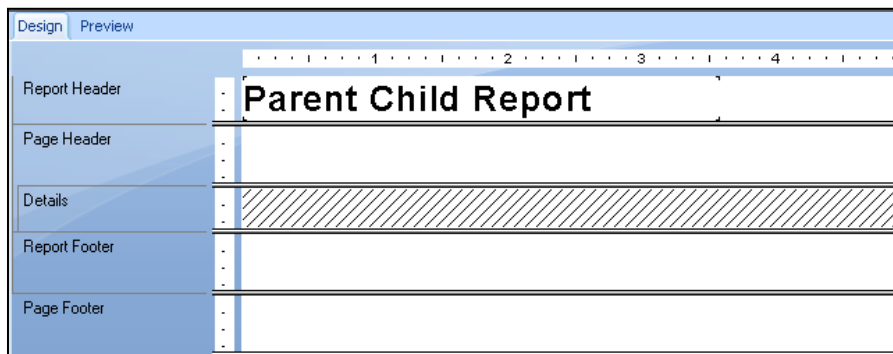


Figure 16-9 Suppressed Details section.

4. Create a new sub-report by using the existing **Parent Child Subreport .rpt**. Place the subreport into the page header. Right-click the subreport and select **format subreport**.
5. Click the **Subreport** tab. Change the subreport name to **Parent Subreport**. This identifies the functionality of this subreport.

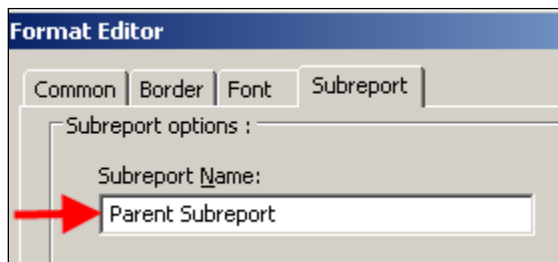


Figure 16-10 Subreport renamed to Parent Subreport.

6. Click **OK**. Refer back to design mode and double-click the subreport. In the subreport design screen, clear out all existing select expert rules. When using a template, start without any pre-established filters to avoid report generating issues.

7. Change the name in the report header to say **Parent Record**.

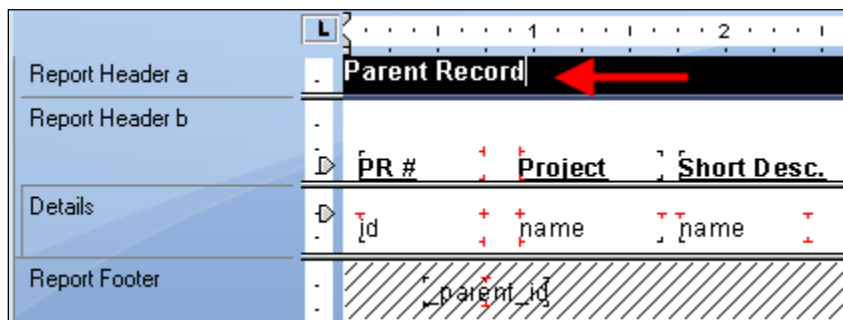


Figure 16-11 Report Header a modified to display Parent Record.

8. Refer back to the main report design screen. Right-click the subreport and click **Change Subreport Links**. For this parent subreport, link from **PR.ROOT_PARENT_ID** to **PR.ID**. No matter which record is reported upon, the root parent is sent to this first subreport in order to print the information.

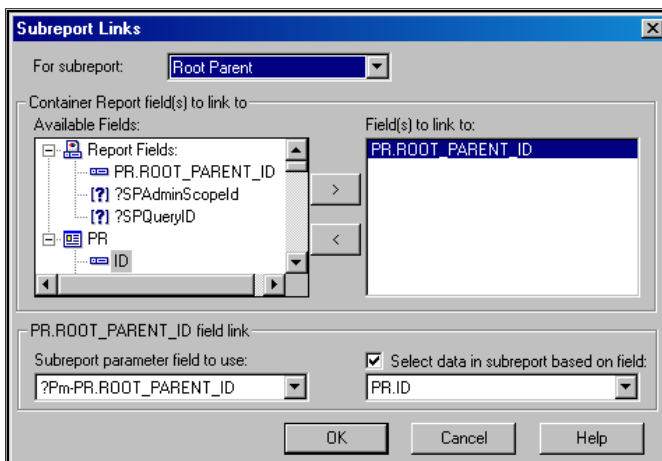


Figure 16-12 Subreport Links window.

9. Click **Report – Select Expert – Record** to filter for the **PR.ID** to equal **1164**. Save and run the report in Crystal. The root parent employee record appears.
10. Insert a second sub-report which shows all children records. Use **Parent Child Subreport Template.rpt** to design the child subreport with. Place the sub-report in the **Report Footer** section of the main report.
11. Delete any existing select expert rules inherited from the template subreport. Verify the report header displays **Children Records**.

12. Click **Report – Select Expert – Record**. Include the following filter to not allow the root parent to show up in this child subreport.

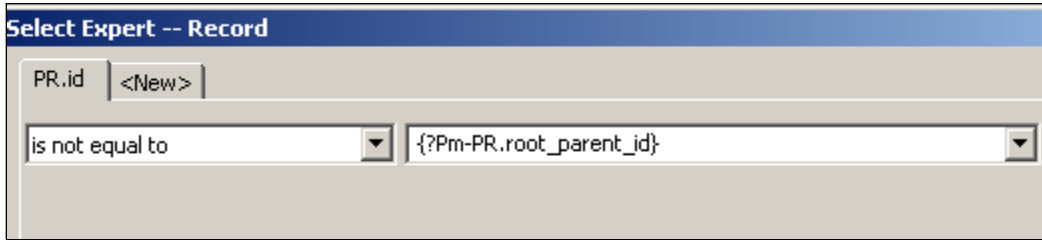


Figure 16-13 Select Expert – Record with PR.id filter.

13. Finally, link the main report to the subreport with the following link. On the design screen right-click the child subreport and select **Change Subreport Links**. From the main report, link from **PR.root_parent_id** to the subreport **PR.root_parent_id**. All of the children records return based on this number. With the select expert, the root parent is not included as intended.

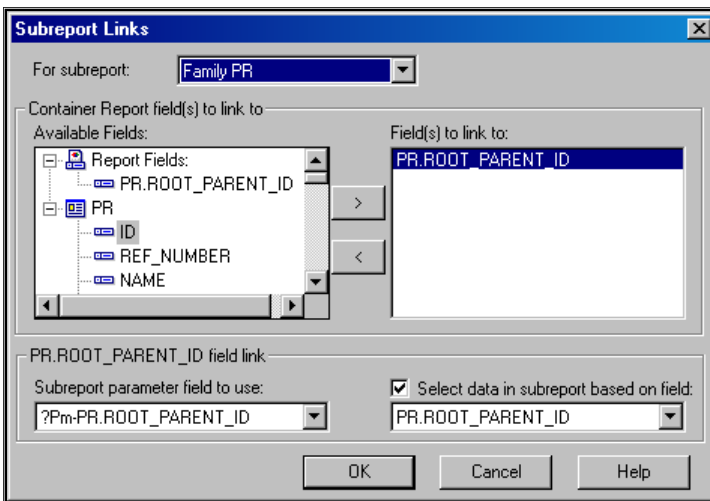


Figure 16-14 Subreport Links window.

14. The above example shows all children and grandchildren records because they all share the same root parent ID. An alternative is to only show children or grandchildren. Refer to the following chart for the different links available. Include the select expert to show a specific project where project.name equals a specific value.

Field(s) to link to	Select data in subreport based on field	Sub report type
PR.ID	PR.parent_id	Child Subreport
PR.ID	PR.root_parent_id	Grandchild Subreport

15. If PR.root_parent.id is not available from the subreport, click **Cancel** and refer back to the design screen of the child subreport. Drag root_parent_id from the PR table to the suppressed report footer of the report and try again from the main report.

Report Header b	PR #	Short Desc.
Details	[id]	name
Report Footer	root_parent_id	

Figure 16-15 PR.root_parent.id added to suppressed Report Footer.

16. Copy the report template into the single report folder and run the report against different records within the same family of records and observe the results. Any single record, regardless of parent or child type, displays the complete family tree of records. This type of report is similar to the Family button within TrackWise records with regards to the information shown.

Conclusion:

Parent child reports incorporate related records into one single report. The first example incorporated one subreport to show the status and details of the available children records. The second example used two subreports to show a preview of the parent record with another to show the children. The links for both subreports were different based on what kind of information was intended to appear.

In the next chapter, the student creates a new single report based on an existing project record.



17. Single Report Review

17.1 Definition and Explanation

The following workbook exercise is designed to create a new single report for another project. The previous chapters covered the best practice and approach for creating single reports. Use views or direct table linking to add the fields onto the report. Start by opening template.rpt and then click **File – Save As** to save a copy of the template for the next report.



1. Log into TrackWise. Set the Scope to **All Projects**. Create a new record by clicking **New PR**.

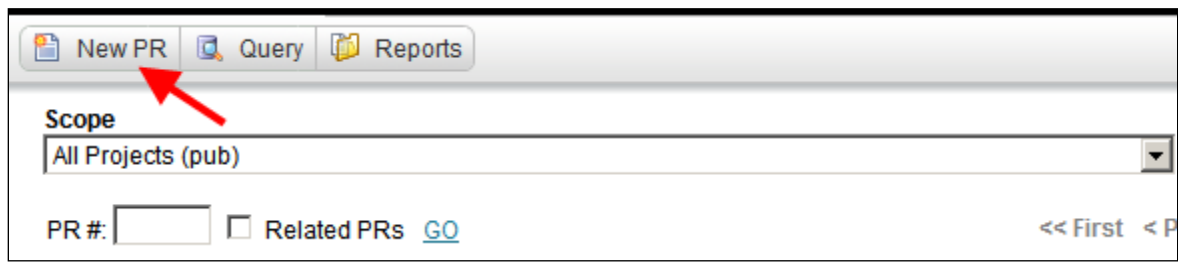
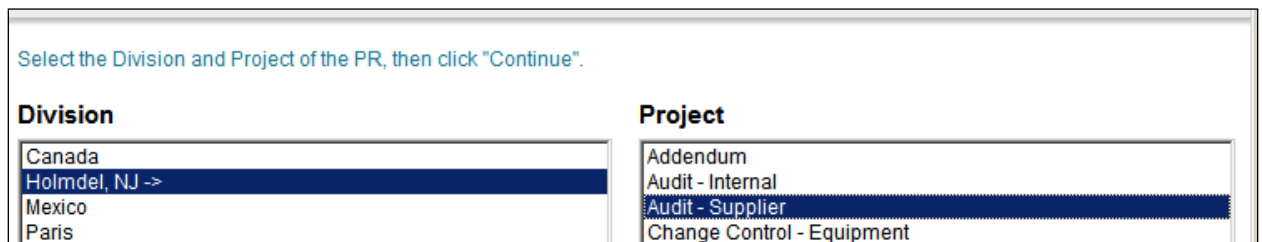


Figure 17-1 New PR button on the Desktop.

2. Under Division click **Holmdel, NJ** and the Project **Audit – Supplier**. Click **Continue**.



Division	Project
Canada	Addendum
Holmdel, NJ ->	Audit - Internal
Mexico	Audit - Supplier
Paris	Change Control - Equipment

Figure 17-2 Holmdel, NJ Division and Audit-Supplier Project selected.

3. Complete the following fields in order to design the report from.

Short Desc.	
Assigned To	Other Auditors
Date Due	
Type	
Auditee	Main Contact
Description	
Purpose	Scope

General Information

Originator
[John Smith](#)

Date Opened
03/25/2011 09:28 AM  [Now](#)

Short Desc. 
Auditing Corporate Inc.

Assign the Lead Auditor for the Audit record. The Lead Auditor will manage the audit process.

Assign any additional Auditors that will assist.

Assigned To
 [Info](#)

Other Auditors
 [Edit](#)

Set the Date Due to the expected finish date of the Audit.

Date Due
 

Audit Information

Type

General

Auditee

Corporate Inc

Info

Main Contact

Larry Johnson

Info

Description

Audit Corporate Inc. to ensure they are still meeting our standards.

Purpose and Scope of Audit

Purpose

Annual audit of supplier.

Scope

Auditing manufacturing.

Figure 17-3 Example of data entered on the records details screen.

- Click **Save**. Write down the PR# generated by TrackWise. This is the same ID to use with select expert in Crystal while designing the report.

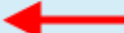
PR #	Parent ID	Project ▲	Division	Short Desc.
 1168		Audit - Supplier	Holmdel, NJ	Auditing Corporate Inc.

Figure 17-4 Location of PR#.

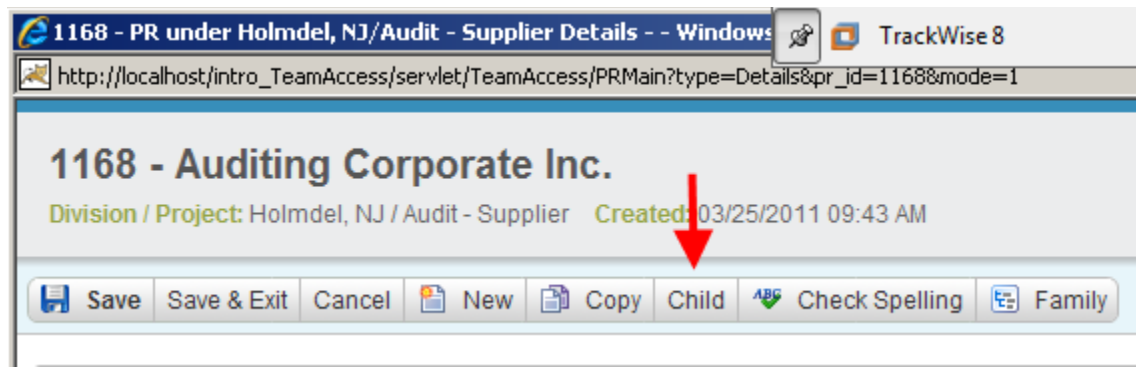
- Refer back to Crystal Reports. Open **template.rpt**.
- Click **File – Save As** and name the report **Audit Supplier**. Set the select expert for PR.ID to equal the newly generated record ID.
- If using views, create several detail sections for multi-based fields that need to grow but not to cover the fields underneath. If direct table linking is to be used, create several break group footers based on pr.id for the running totals.

8. Refer back to the audit record for the order and layout of the fields. For this example add the fields under the general information tab. When finished, save the report.

RHa RHb Da Db Dc Dd RF	1168 - Auditing Corporate Inc. Division / Project: Holmdel, NJ / Audit - Supplier Created: March 25, 2011 9:43 am General Information Originator: John Smith Date Opened: March 25, 2011 9:43 am Short Desc: Auditing Corporate Inc. Assigned To: Ingrid Latos Other Auditors: Corey Dillon Jane Doe Tiki Barber Date Due: March 31, 2011 Audit Information Type: General Auditee: Corporate Inc Main Contact: Larry Johnson Description: Audit Corporate Inc. to ensure they are still meeting our standards. Purpose and Scope of Audit Purpose: Annual audit of supplier. Scope: Auditing manufacturing.
--	--

Figure 17-5 Example of preview report based on the audit record layout of fields.

9. Create a child Observation record from the record used in this example. In the record details, click **Child**.



1168 - PR under Holmdel, NJ/Audit - Supplier Details - - Windows TrackWise 8

http://localhost/intro_TeamAccess/servlet/TeamAccess/PRMain?type=Details&pr_id=1168&mode=1

1168 - Auditing Corporate Inc.

Division / Project: Holmdel, NJ / Audit - Supplier Created: 03/25/2011 09:43 AM

Save Save & Exit Cancel New Copy Child Check Spelling Family

Figure 17-6 Location of Child button in the parent Audit-Supplier record.

10. This record serves as data for the subreport in the main audit record report. Complete the following fields and then click **Save** and **Save & Exit** when finished.

- Short Desc.
- Type
- Assigned To

PR # ▲	Parent ID	Project	Division
 1168 		Audit - Supplier	Holmdel, NJ
 1169 		Observation - Supplier	Holmdel, NJ

Figure 17-7 Desktop display parent and child record.

11. Refer back to Crystal Report. Insert another section below from details or group footer. Click **Insert – Subreport**. Use **template.rpt** to create a new report for observation record data. Clear out all select expert settings before adding the fields. Refer to the following chart for report requirements. Use the view **TW_V_TYPE** for the Type field.

PR#	Short Desc.	Type	Assigned To
PR.ID	PR.name	TW_V_TYPE.s_value	@Assigned To or Blank formula

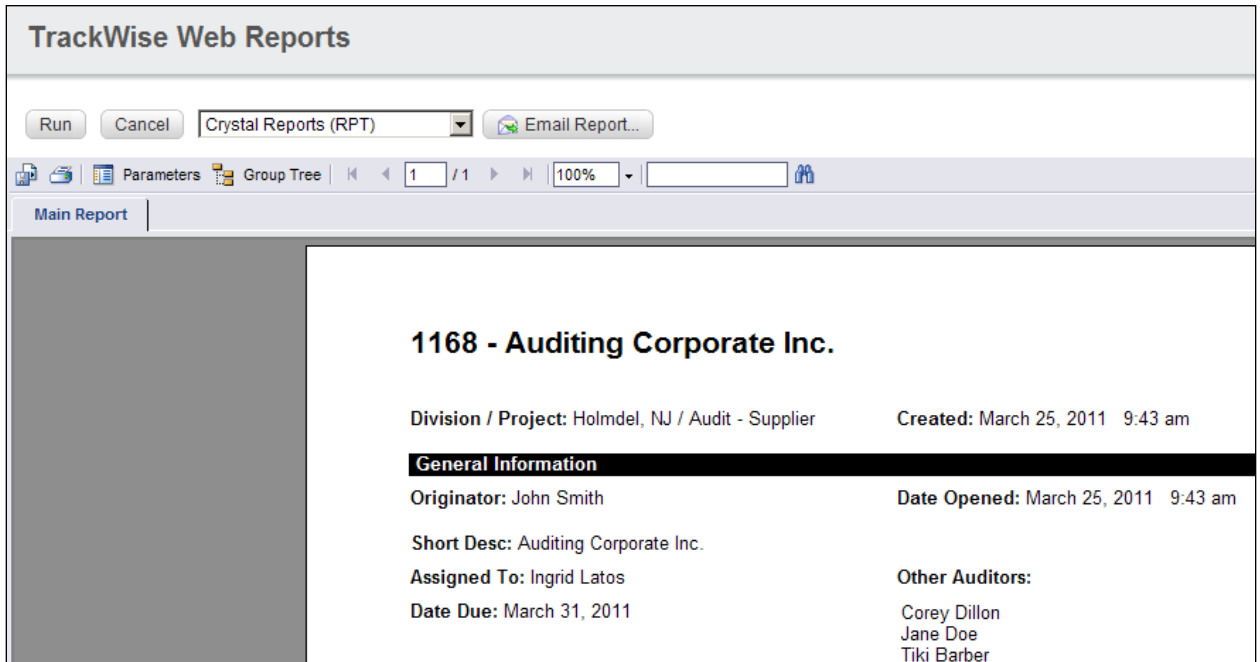
12. When finished, link the main report to the subreport based on the PR.root_parent_id and include the filter to prevent the root parent from appearing.

13. Save and run the report. Notice the child observation record appears.

Observation Records			
PR # 1169	Short Desc: Supplier does not have a disaster recovery plan in place.	Type Process Validation	Assigned To Harrel Bramley

Figure 17-8 Example of preview for child record subreport.

14. Copy the report over to the single reports **TWReports_1** folder. Run the report in TrackWise on any audit supplier record.



TrackWise Web Reports

Run Cancel Crystal Reports (RPT) Email Report...

Parameters Group Tree 1 / 1 100%

Main Report

1168 - Auditing Corporate Inc.

Division / Project: Holmdel, NJ / Audit - Supplier Created: March 25, 2011 9:43 am

General Information

Originator: John Smith Date Opened: March 25, 2011 9:43 am

Short Desc: Auditing Corporate Inc.

Assigned To: Ingrid Latos Other Auditors:

Date Due: March 31, 2011

Corey Dillon
Jane Doe
Tiki Barber

Figure 17-9 Example of report generated in TrackWise.

18. Converting Reports

18.1 Definition and Explanation

With different TrackWise databases available such as Development and Production, refer to the following steps to convert the report from one database to another, if the report is to be moved to a different environment after being created. This is done after the report has been verified to run correctly, but it is going to be used in a different TrackWise environment such as Production.

18.2 Converting Database Driver for Existing Reports

1. In one's environment outside of training, open the intended report and click **Database - Set Datasource Location**.

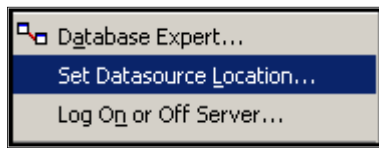


Figure 18-1

2. If there is no current connection listed in the **Replace with** section, establish a new connection by expanding the **Create New Connection** folder and selecting the **ODBC (RDO)** folder.
3. Select the former database DSN in the **Current Data Source** section and select the current database DSN in the **Replace with** section.
4. Click **Update**. Click **Close**. This allows the current tables and views on the report to work with the newer data source.
5. Select **Refresh**. New Database data is displayed.
6. This technique is also used when editing a report in Crystal and the current data source is not the exact DSN name intended to be used. If more than one Crystal designer is working on the same report, the DSN name from one person's ODBC DSN screen may be different than the other. These steps convert the name over as intended.

19. Coordinator Auto Reports

19.1 Definition and Explanation

Auto Reports allows Coordinator to generate Crystal Reports in PDF format based on a predefined schedule. These PDFs are emailed to recipients for reporting purposes. All the rules for report design and formatting still apply for Coordinator-based auto reports. The reports appear as multi-reports to display many records at once. The following two parameters are used within auto reports to show the scope and query content of the coordinator.

19.2 SPAdminScopeId

The Crystal parameter SPAdminScopeId [Number] is used to create a sub-report containing the Divisions and Projects in the admin scope for the given Admin Query. Using the template report, create a new report called **AutoReport.rpt**



Workbook Exercises

1. Define a new parameter in the main report: **SPAdminScopeId** [number]
2. Create a new subreport and add the **AQ_SCOPE** table.
3. Choose to display the **AQ_SCOPE.RPT_SCOPE** field in the **Details** section, and perform any necessary formatting.

4. On the main report, right-click the subreport and select **Change Subreport Link** from **?SPAdminScopeID** to **AQ_SCOPE.ID** field on the subreport.

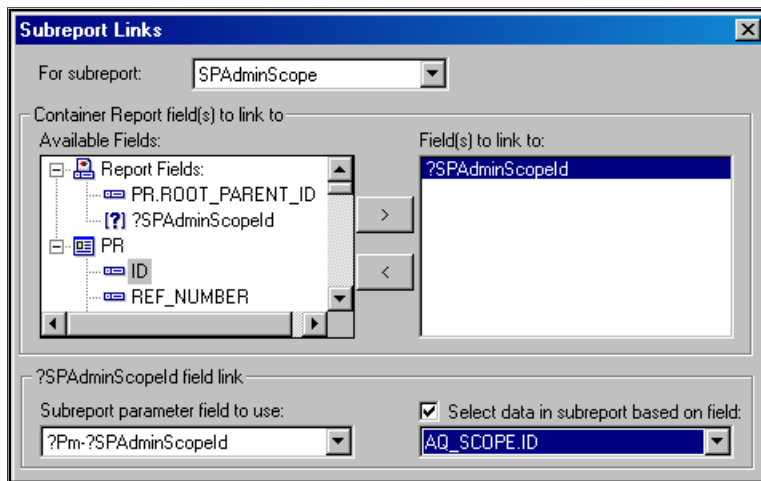


Figure 19-1 Subreport Links window.

5. Save the report when finished.

19.3 SPQueryId

The Crystal parameter **SPQueryId** [Number] is used to create a subreport containing the coordinator query description.



1. Define a new parameter in the main report: **SPQueryId[number]**
2. Create a new subreport and add the **PR_QUERY** and **ADMIN_QUERY** tables. Link these tables from **PR_QUERY.ID - ADMIN_QUERY.PR_QUERY_ID**
3. Choose to display the **PR_QUERY.DESCRPTION** field in the **Details** section and perform any necessary formatting.

4. On the main report, right-click this subreport and select **Change Subreport Link** so the **?SPQueryID** parameter links to the **ADMIN_QUERY.ID** field on the subreport. Save the report when finished.

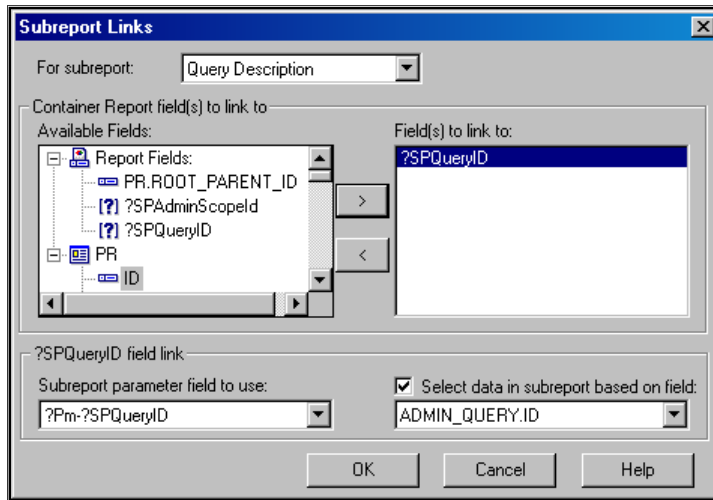


Figure 19-2 Subreport Links window.

Conclusion:

This is tested within one's own TrackWise environment using the coordinator to fetch records and distribute auto reports. In addition to these two subreports, the rest of the report can display additional information about the returned records such as PR number, short description, assigned to, project and any additional fields necessary.



20. Troubleshooting Reports

20.1 Definition and Explanation

In the development and test execution of these reports, one may encounter problems or issues with the template or data being returned. The following are the most frequently reported issues with Crystal Reports.

20.2 Records are missing from the report

The first component that might contribute to records missing from the report is incorrect table joins such as **Equal** and **Left-outer**. Refer back to the chapter on when to use each different join. Furthermore, ensure your tables are correctly joined through the use of the necessary columns.

The other component that might contribute to missing records is an included Select Expert condition not required in TrackWise. During report development, it is very common to introduce a temporary condition. However, it is not uncommon that before promoting a report to production, this condition is not removed. Therefore, if a select expert condition is set to only fetch records where the PR.ID = 1024, when this report is run against any record (except 1024), the report could potentially fail. Ensure that all unnecessary select expert filters are removed.

20.3 Report not working after promoting to Production

Report development typically occurs against the data in a development TrackWise database. However, when promoting a report to a different environment, the report may not work and even return errors.

The first possible solution is that one may have defined new User Defined fields in the development database and created the applicable views. However, when promoting this report into the production environment, the new fields exist but views were not created yet. Use the **Update Schema** utility to build the views.

Another potential issue is that the field names in the development and production database do not match. An example is the data field, **Lot Number** was created in the development environment and used to create the report, was created as **Lot #** in the production database.

Although the user understands the difference, Crystal does not. In this case, the report template must be changed in the production environment.

21. Creating Views With Web Administrator

21.1 Definition and Explanation

Views are created from within Web Administrator. Contact your TrackWise Administrator to create the views by synchronizing the newer data fields. This is implemented **under System Configuration – Data Fields**. The following workbook exercise includes the steps necessary to create the views with.



Workbook Exercises

1. Click **Start – Programs – TrackWiseWeb – intro_WebAdmin**
2. Log in with admin for username and password.
3. Click **System Configuration – Data Fields**. This screen displays all of the data fields in TrackWise.
4. Click **Synchronize Views**. This feature creates views for data fields that need one.



Figure 21-1 Synchronize Views button in Web Administrator.

5. A prompt appears to confirm the creation of views for data fields without one and remove invalid views. There is also an option to rename existing views when an administrator renames a data field and the original view pertains to the former name. This can impact reports still using the older named view. They need to be removed from the report and the new ones added if necessary.

6. Click **OK**. The views are created.

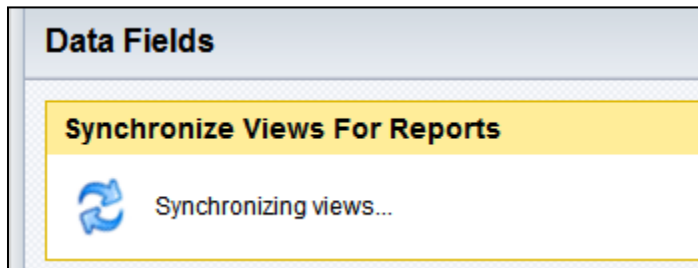


Figure 21-2 Status message displayed after clicking Synchronize views.

7. Click any of the data fields listed on the screen. Refer to the details section where the view appears. The view name assists Crystal Report designers with the actual view name for the corresponding field.

Field Name - Customized	Actual Start Date
Field Type	Date
Field Category	PR (User Defined)
View Name	TW_V_ACTUAL_START_DATE
Seq. No	12

Figure 21-3 View Name created and associated with the data field.

8. Refer back to the main data fields screen and click **Report**. The report displays the names of the data fields and the views that are available. This is printable and used when designing reports.

TrackWise Data Fields Configuration Report				
Data Fields				
Process Group: Initial Group				
Field Name - Customized	Associated with Initial Group	Field Category	Field Type	View For Reports
(A)NDA #	Yes	PR (User Defined)	String	TW_V_A_NDA
Account Manager	Yes	Entity	Person	
Actions Taken	Yes	PR (FlexField)	FlexField	TW_V_ACTIONS_TAKEN
Actual Last Audit Day	Yes	PR (User Defined)	Date/Time	TW_V_ACTUAL_LAST_AUDIT_DAY
Actual Start Date	Yes	PR (User Defined)	Date	TW_V_ACTUAL_START_DATE
Additional Investigators	Yes	PR (User Defined)	Multi Person	TW_V_ADDITIONAL_INVESTIGATORS

Figure 21-4

22. Report Validation and Explanation

22.1 Definition and Explanation

In a regulated environment, completed reports never get promoted directly into the production environment. They need to go through validation based on testing and verification. The following explains the documentation process for the report template.

22.2 Report Template Documentation

There are three components that need to be captured in order to best document the report.

1. The first component is the report design screen itself. The following is a screen capture of the report layout.

Report Header a	{id} - {name}
Report Header b	Division / Project: {name} / {name} Created: {Date Created}
Page Header	
Details a	General Information Originator: {Originator} Date Opened: {Date Opened} Short Desc: {name} Assigned To: {Assigned To or Blank} Other Auditors: Date Due: {date_to_close} Other Auditors
Details b	Audit Information Type: {s_value} Auditee: {s_value} Main Contact: {s_value} Description: Description
Details c	Purpose and Scope of Audit Purpose: Purpose
Details d	Scope: Scope
Details e	Observation
Report Footer	
Page Footer	Page N of M

Figure 22-1 Report design layout.

2. The next component to be documented is the SQL statement being generated by the report. This can be captured by clicking **Database – Show SQL Query**.

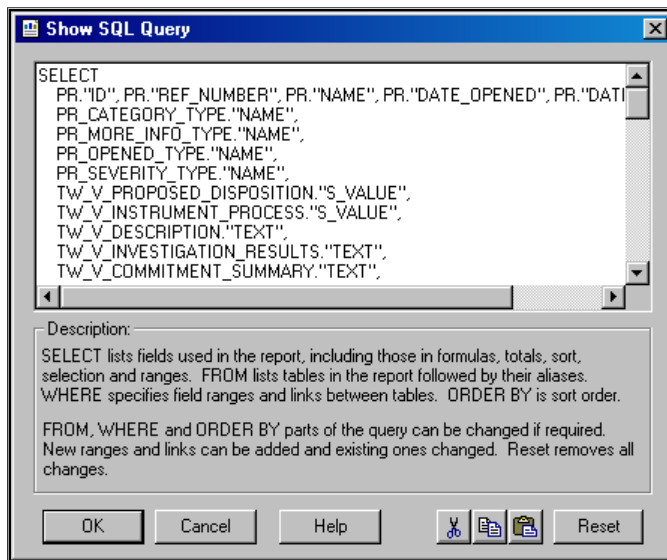


Figure 22-2 Show SQL Query window.

3. The third component to be captured is the fields and formulas attributes on the report. Click **File – Export – Export Report**.

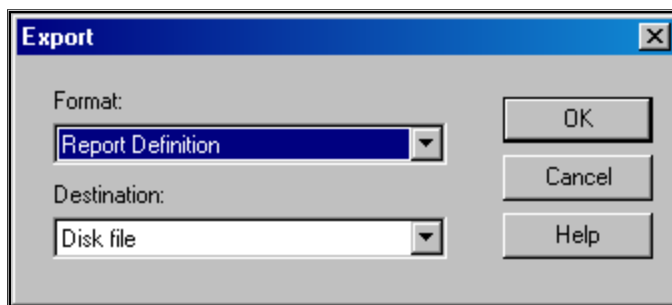


Figure 22-3 Export Report window.

4. Click **OK** and choose the location with filename to save this file.
5. The following is an example from a report definition file:

Title:

String, Visible, Left Alignment, Top Alignment,
Keep Together

{PR.NAME}

String, Visible, Left Alignment, Top Alignment,

Keep Together, Using System Default Formatting, Word Wrap

Date Opened:

String, Visible, Left Alignment, Top Alignment,

23. Database Tables

The following list contains database tables used to generate reports in class.

Table Name	Column Name	Notes
Division_Type	Name	Contains the Division name values.
Project	Name	Contains the Project name values.
Person_Relation	Name	Contains Person name values.
PR	ID Name Date_Opened Date_Created Date_To_Close	Record ID Number Record Short Descr. / Title Use a formula to convert Use a formula to convert Date Due of a record
PR_Status_Type	Name	Contains the State name.
PR_Activity	Date_Scheduled Date_Performed Responsible_Rel_id	Use a formula to covert. Use a formula to covert. The person responsible for the activity.
PR_Activity_Type	Name	Name of activity entry in the audit trail history.
Activity_Text_Data	Text	Activity Summary information.
PR_Addtl_Data	Dec_Value Date_Value Date_Time_Value S_Value Number	Decimal Values Date Values Date Time (use formula) String and File Attachments Number
Data_Fields	Name	Names of data fields
Addtl_Type	Name	Selection Values
Entity	Name	Entity Values

24. Notes

